Tongji Calculus, 7th Edition · Chapter 1

Functions and Limits

Detailed Solutions

100 classic-method adaptations and high-quality synthesis problems

All mathematics compiled by XeTeX / LaTeX · 4:3 study edition

How to Use This Workbook

The material progresses from basic to challenging. IDs Q001–Q100 match across both languages and both document types. Every formula comes from explicit LaTeX source and is compiled by the XeTeX mathematics engine.

Open-text method adaptations

20 / 100

Classic-method variants

37 / 100

Original synthesis / diagnosis

43 / 100

Source-lineage policy

Problems are redesigned around classic methods found in open textbooks, public courses, and standard calculus teaching. Copyrighted books are used only for scope and method alignment; their wording and solutions are not copied. See SOURCES.md.

1. On the first attempt, use only the exercise workbook and show complete reasoning.
2. During correction, record the cause of each error instead of copying the final answer.
3. Redo errors after 48 hours and interleave topics one week later.

Basic | Single choice | Foundation | Open-text method adaptation

Q001 · Identify a Valid Mapping

Problem

Let $X = \{-1, 0, 1\}$ and $Y = \{0, 1\}$. Which correspondence defines a function from X to Y?

A. For every $x \in X$, define $f(x) = x^2$

B. Define $f(-1) = 1$ and $f(1) = 1$, but do not define $f(0)$

C. Define $f(-1) = 1$ and $f(0) = 0$, while assigning both $f(1) = 0$ and $f(1) = 1$

D. For every $x \in X$, define $f(x) = x + 1$

Answer

A

Knowledge points

- every domain element must have an image
- each input must have a unique output
- every output must belong to the codomain

Reading and method choice

Check the three defining requirements one by one: every element of X must be covered, its image must be unique, and every image must lie in Y. Failure of any one condition invalidates the correspondence.

Detailed derivation

1. For $A, (-1)^2 = 1, 0^2 = 0$, and $1^2 = 1$, so all three inputs have exactly one image.
2. All outputs in A are either 0 or 1, hence they belong to Y; therefore A is valid.
3. B leaves $f(0)$ undefined, so it does not cover the whole domain X.
4. C gives the same input 1 two outputs, while D gives $f(1) = 2 \notin Y$; therefore C and D are invalid.

Common pitfalls

- checking only whether a formula can be evaluated and ignoring the prescribed codomain
- allowing one input to have two different outputs

Verification

Substitution in A gives the ordered-pair set $\{(-1,1),(0,0),(1,1)\}$. Every first component appears exactly once, and every second component lies in Y.

Method takeaway

A mapping is a function only when it is defined everywhere on the domain, single-valued, and codomain-respecting.

Extension

If the codomain in D were changed to $\{0,1,2\}$, then D would define a function from X to that new codomain.

Source lineage: Open-text method adaptation · mapping / definition of a function · reference IDs: openstax-v1-1.1-functions. See SOURCES.md for scope and adaptation boundaries; this note does not claim verbatim reproduction.

Basic **Fill in the blank** **Foundation** **Open-text method adaptation**

Q002 · Common Domain of a Radical and a Logarithm

Problem

The natural domain of $f(x) = \sqrt{2-x} + \ln(x+1)$ is ____.

Answer

$(-1, 2]$

Knowledge points

- an even root requires a nonnegative radicand
- the argument of a natural logarithm must be strictly positive
- simultaneous restrictions are intersected

Reading and method choice

Because both expressions occur in the same function, x must satisfy both restrictions. The radical may include its zero endpoint, whereas the logarithm may not.

Detailed derivation

1. For $\sqrt{2-x}$ to be defined, require $2 - x \geq 0$, hence $x \leq 2$.
2. For $\ln(x+1)$ to be defined, require $x + 1 > 0$, hence $x > -1$.
3. Intersect $(-1, +\infty)$ with $(-\infty, 2]$.
4. The intersection is $(-1, 2]$; -1 is excluded and 2 is included.

Common pitfalls

- replacing the logarithmic condition $x + 1 > 0$ with $x + 1 \geq 0$
- listing the two restrictions without taking their intersection

Verification

At $x = 2$ the value $\sqrt{0} + \ln 3$ is defined, while $x = -1$ would produce $\ln 0$, confirming the stated endpoint types.

Method takeaway

For a natural domain, collect every restriction, intersect them, and distinguish strict from non-strict inequalities.

Extension

If $\ln(x+1)$ were replaced by $\frac{1}{\ln(x+1)}$, one would also require $\ln(x+1) \neq 0$, excluding $x = 0$.

Source lineage: Open-text method adaptation · domain / radical · reference IDs: openstax-v1-1.1-functions. See SOURCES.md for scope and adaptation boundaries; this note does not claim verbatim reproduction.

Basic True/false with justification Foundation Classic-method variant

Q003 · A Necessary Condition for Composition

Problem

True or false: If $(f \circ g)(x_0)$ is defined, then $g(x_0)$ must belong to the domain of f.

Answer

True.

Knowledge points

- definition of a composite function
- input condition for the outer function

Reading and method choice

$(f \circ g)(x_0)$ means $f(g(x_0))$. For this expression to exist, $g(x_0)$ must first exist and must then be an admissible input for f.

Detailed derivation

1. By definition, $(f \circ g)(x_0) = f(g(x_0))$.
2. Because the composite value is defined, x_0 must lie in the domain of g, so $g(x_0)$ exists.
3. For $f(g(x_0))$ to be evaluated, the input $g(x_0)$ must lie in the domain of f.
4. Thus the stated condition is necessary, and the statement is true.

Common pitfalls

- requiring x_0 itself to lie in both domains instead of checking the inner output $g(x_0)$
- reversing the order of composition

Verification

If $g(x_0)$ were outside the domain of f, then $f(g(x_0))$ would be undefined, contradicting the hypothesis.

Method takeaway

For a composition, inspect the output of the inner function because that output becomes the input of the outer function.

Extension

For the converse, one must also require x_0 to lie in the domain of g; together, these two conditions are sufficient.

Source lineage: Classic-method variant · composition / domain condition · reference IDs: openstax-v1-1.1-functions. See SOURCES.md for scope and adaptation boundaries; this note does not claim verbatim reproduction.

Basic | Single choice | Foundation | Classic-method variant

Q004 · Recognize the Rigorous Definition of a Sequence Limit

Problem

Which ε-N statement correctly expresses that the sequence $\{a_n\}$ converges to A?

A. For every $\varepsilon > 0$, there exists a positive integer N such that $|a_n - A| < \varepsilon$ whenever $n > N$

B. There exists a positive integer N such that, for every $\varepsilon > 0, |a_n - A| < \varepsilon$ whenever $n > N$

C. For every $\varepsilon > 0$ and every positive integer $n, |a_n - A| < \varepsilon$

D. There exists $\varepsilon > 0$ such that, for every positive integer N, one can find $n > N$ with $|a_n - A| < \varepsilon$

Answer

A

Knowledge points

- ε-N definition of sequence convergence
- order of quantifiers
- control of the tail

Reading and method choice

The definition allows N to depend on the requested accuracy ε. Once N is chosen, every term with $n > N$ must lie in the ε-neighborhood of A.

Detailed derivation

1. First, an arbitrary accuracy $\varepsilon > 0$ is prescribed; this is the initial universal quantifier.
2. One may then choose a positive integer $N = N(\varepsilon)$ depending on that ε.
3. The requirement concerns every $n > N$, not merely one selected index: all tail terms satisfy $|a_n - A| < \varepsilon$.
4. A has exactly this quantifier order and tail condition. B and C are too strong, while D allows only scattered close terms.

Common pitfalls

- choosing N before ε and forcing one N to work for all accuracies
- replacing all sufficiently large indices by the existence of one large index

Verification

The quantifier form of A is $\forall \varepsilon > 0, \exists N \in N^{+}, \forall n > N : |a_n - A| < \varepsilon$, which is the standard definition.

Method takeaway

Remember the order: prescribe ε, choose N, then control every term in the tail.

Extension

Replacing $|a_n - A| < \varepsilon$ by $a_n \in (A - \varepsilon, A + \varepsilon)$ gives an equivalent neighborhood formulation.

Source lineage: Classic-method variant · sequence limit / ε-N definition · reference IDs: openstax-v2-5.1-sequences. See SOURCES.md for scope and adaptation boundaries; this note does not claim verbatim reproduction.

Basic | Fill in the blank | Foundation | Classic-method variant

Q005 · Choose N for a Prescribed Accuracy

Problem

For $a_n = \frac{(-1)^n}{n}$, the limit is ____. When $\varepsilon = 0.01$ and the definition uses $n > N$, one may take $N =$ ____.

Answer

0; 100.

Knowledge points

- an alternating sign disappears in an absolute-value estimate
- ε-N definition
- strict inequality

Reading and method choice

Although $(-1)^n$ alternates in sign, the distance from 0 is always $\frac{1}{n}$. To make this distance strictly smaller than 0.01, it is enough to require $n > 100$.

Detailed derivation

1. Compute the distance from the candidate limit $0:|a_n - 0| = \frac{|(-1)^n|}{n} = \frac{1}{n}$.
2. The condition $\frac{1}{n} < 0.01 = \frac{1}{100}$ is equivalent to $n > 100$.
3. Because the definition uses $n > N$, choosing $N = 100$ makes every $n > N$ satisfy the strict inequality.
4. The same reasoning works for every $\varepsilon > 0$ by choosing $N \geq \frac{1}{\varepsilon}$, so $a_n \to 0$; the blanks are 0 and 100.

Common pitfalls

- mistaking alternating signs for divergence even when magnitudes go to zero
- thinking N must be 101 even though the convention is $n > N$

Verification

At $n = 101, |a_{101}| = \frac{1}{101} < 0.01$. At $n = 100$ the value equals 0.01, but $n = 100$ is not part of the tail $n > N$.

Method takeaway

Convergence controls distance; absolute values remove the alternating sign.

Extension

If the convention were $n \geq N$, one could choose $N = 101$ to preserve the strict inequality.

Source lineage: Classic-method variant · sequence limit / ε-N estimate · reference IDs: openstax-v2-5.1-sequences. See SOURCES.md for scope and adaptation boundaries; this note does not claim verbatim reproduction.

Basic True/false with justification Foundation Classic-method variant

Q006 · Does Boundedness Guarantee Convergence?

Problem

True or false: Every bounded sequence converges.

Answer

False. The sequence $a_n = (-1)^n$ is bounded but divergent.

Knowledge points

- every convergent sequence is bounded
- the converse is false
- counterexample using even and odd subsequences

Reading and method choice

The valid theorem says convergence implies boundedness; it cannot be reversed. A sequence that keeps oscillating between two values gives a direct counterexample.

Detailed derivation

1. Let $a_n = (-1)^n$. Then $|a_n| = 1$ for every n, so the sequence is bounded.
2. Its even subsequence satisfies $a_{2k} = 1$ and therefore converges to 1.
3. Its odd subsequence satisfies $a_{2k-1} = -1$ and therefore converges to -1.
4. If the original sequence converged, all subsequences would have the same limit. Since these two limits differ, the original sequence diverges and the statement is false.

Common pitfalls

- reversing a necessary consequence into a sufficient condition
- claiming oscillation without a verifiable argument for divergence

Verification

All terms lie in $[-1, 1]$, confirming boundedness, but no candidate limit can make both positive and negative tail terms arbitrarily close to it.

Method takeaway

Convergence implies boundedness, but boundedness controls size only and does not prevent perpetual oscillation.

Extension

Adding monotonicity changes the conclusion: every monotone bounded sequence converges, an existence criterion developed later in the chapter.

Source lineage: Classic-method variant · bounded sequence / convergent sequence · reference IDs: openstax-v2-5.1-sequences. See SOURCES.md for scope and adaptation boundaries; this note does not claim verbatim reproduction.

Basic | Single choice | Foundation | Open-text method adaptation

Q007 · Does a Point Value Affect a Function Limit?

Problem

Define $f(1) = 7$, and for $x \neq 1$ let $f(x) = \frac{x^2-1}{x-1}$. What is $\lim_{x\to1} f(x)$?

A. 0

B. 1

C. 2

D. 7

Answer

C

Knowledge points

- a limit examines a deleted neighborhood
- factor and cancel a common factor
- a point value may differ from the limit

Reading and method choice

As $x \to 1$, only values with x close to but different from 1 are relevant. Therefore the formula for $x \neq 1$ is used, and the assigned value $f(1) = 7$ does not enter the limit.

Detailed derivation

1. For $x \neq 1$, factor $x^2 - 1 = (x-1)(x+1)$.
2. Within the deleted neighborhood, cancel $x - 1$ to obtain $f(x) = x + 1$.
3. Therefore, as $x \to 1, f(x) = x + 1 \to 2$.
4. Although $f(1) = 7$, changing one point does not alter the nearby trend, so the correct choice is C.

Common pitfalls

- substituting the assigned value $f(1) = 7$ merely because x approaches 1
- claiming the original quotient equals $x + 1$ at $x = 1$, even though cancellation is valid only for $x \neq 1$

Verification

At $x = 0.99$ and $x = 1.01$, the values are 1.99 and 2.01, both close to 2 rather than 7.

Method takeaway

A finite-point limit is determined by values in a deleted neighborhood and is independent of the function value at the point itself.

Extension

If $f(1)$ were changed to 2, the function would be continuous at 1. Any other point value would leave the limit equal to 2.

Source lineage: Open-text method adaptation · function limit / removable-hole expression · reference IDs: openstax-v1-2.2-function-limits. See SOURCES.md for scope and adaptation boundaries; this note does not claim verbatim reproduction.

Basic | True/false with justification | Foundation | Classic-method variant

Q008 · Are Two Existing One-Sided Limits Sufficient?

Problem

True or false: If $\lim_{x\to a^-} f(x)$ and $\lim_{x\to a^+} f(x)$ both exist and are finite, then $\lim_{x\to a} f(x)$ must exist.

Answer

False. The two one-sided limits must also be equal.

Knowledge points

- criterion for a two-sided limit
- one-sided limits
- counterexample

Reading and method choice

Existence of the two one-sided limits gives a stable trend on each side separately. A single two-sided limit exists only when both trends approach the same value.

Detailed derivation

1. A two-sided limit exists exactly when both one-sided limits exist and are equal.
2. The statement assumes existence and finiteness but omits equality.
3. For a counterexample, let $f(x) = 0$ for $x < a$ and $f(x) = 1$ for $x \geq a$. The left-hand limit is 0 and the right-hand limit is 1.
4. Both one-sided limits exist and are finite, but they are unequal, so the two-sided limit does not exist; the statement is false.

Common pitfalls

- computing both one-sided limits without comparing them
- thinking that the value assigned at a can repair unequal one-sided limits

Verification

Approaching through $x < a$ gives only the value 0, while approaching through $x > a$ gives only 1; no single candidate limit can fit both sides.

Method takeaway

A two-sided limit exists precisely when the left and right limits exist and equal the same finite value.

Extension

Redefining $f(a)$ cannot change either one-sided limit in this counterexample and therefore cannot create a two-sided limit.

Source lineage: Classic-method variant · one-sided limits / two-sided limit · reference IDs: openstax-v1-2.2-function-limits. See SOURCES.md for scope and adaptation boundaries; this note does not claim verbatim reproduction.

Basic | Single choice | Foundation | Classic-method variant

Q009 · Recognizing an Infinitesimal

Problem

As $x \to 0$, which of the following functions is an infinitesimal?

A. $2 + x$

B. $\frac{x^2}{1+|x|}$

C. $\frac{1}{x^2}$

D. $1 + \sin x$

Answer

B

Knowledge points

- Definition of an infinitesimal: a function with limit 0
- Term-by-term limit checking
- Difference between boundedness and being infinitesimal

Reading and method choice

To decide whether an expression is infinitesimal, check whether its limit is 0 in the specified process. Merely containing x is not enough.

Detailed derivation

1. For A, $\lim_{x\to 0}(2 + x) = 2$, so it is not infinitesimal.
2. For $B, 0 \le \frac{x^2}{1+|x|} \le x^2$ and $x^2 \to 0$; the squeeze theorem gives limit 0.
3. For $C, \frac{1}{x^2} \to +\infty$ as $x \to 0$, so it is infinite rather than infinitesimal.
4. For D, $\lim_{x\to 0}(1 + \sin x) = 1$, so it is not infinitesimal.
5. Therefore B is the unique correct choice.

Common pitfalls

- Calling every expression containing x infinitesimal
- Confusing boundedness with convergence to 0

- Ignoring the specified process $x \to 0$

Verification

At $x = 0.1$ and $x = 0.01, B$ is about 0.00909 and 0.000099, respectively, while the other expressions approach $2, +\infty$, and 1.

Method takeaway

Being infinitesimal is a limit property tied to a specified process, not a visual or absolute-size judgment.

Extension

As $x \to a, f(x)$ is infinitesimal if and only if $\lim_{x \to a} f(x) = 0$.

Source lineage: Classic-method variant · infinitesimal / definition · reference IDs: openstax-v1-2.2-function-limits, openstax-v1-2.5-precise-limit. See SOURCES.md for scope and adaptation boundaries; this note does not claim verbatim reproduction.

Basic | True/false with justification | Foundation | Classic-method variant

Q010 · Is an Infinitesimal a Small Number?

Problem

True or false, with justification: An infinitesimal is a nonzero number whose absolute value is very small.

Answer

False. An infinitesimal is a variable or function whose limit is 0 in a specified process, not a fixed nonzero real number; the constant 0 also qualifies.

Knowledge points

- Process-dependent nature of infinitesimals
- Variables versus constants
- Limits of constant functions

Reading and method choice

The statement replaces a limit property by an informal claim about the size of a fixed number. Any fixed positive number, however small, has itself as its constant-function limit.

Detailed derivation

1. By definition, $\alpha(x)$ is infinitesimal as $x \to a$ when $\lim_{x\to a} \alpha(x) = 0$.
2. For any fixed $c \neq 0$, even if $|c|$ is tiny, the constant function $\alpha(x) = c$ has limit $c \neq 0$.
3. Thus a fixed nonzero small number is not an infinitesimal; the limiting process must be stated.
4. The constant function $\alpha(x) = 0$ has limit 0, so zero itself satisfies the definition.
5. Both parts of the original assertion are therefore inaccurate, so it is false.

Common pitfalls

- Confusing informal smallness with convergence to 0
- Assuming an infinitesimal can never equal 0
- Omitting the limiting process

Verification

For $c = 10^{-100}$, $\lim_{x\to 0} c = c \neq 0$; for $\alpha(x) = x$, $\lim_{x\to 0} \alpha(x) = 0$. Both can be numerically small, but only the latter tends to 0.

Method takeaway

An infinitesimal describes a trend of variation, not a fixed magnitude.

Extension

The same function can behave differently in different processes: x is infinitesimal as $x \to 0$ but not as $x \to +\infty$.

Source lineage: Classic-method variant · infinitesimal / concept distinction · reference IDs: openstax-v1-2.2-function-limits, openstax-v1-2.5-precise-limit. See SOURCES.md for scope and adaptation boundaries; this note does not claim verbatim reproduction.

Basic | Single choice | Foundation | Classic-method variant

Q011 · When Can Limits Be Divided Directly?

Problem

Which conclusion follows directly from the quotient limit law?

A. $f(x) \to 2$ and $g(x) \to 3$, so $\frac{f(x)}{g(x)} \to \frac{2}{3}$

B. $f(x) \to 2$ and $g(x) \to 0$, so $\frac{f(x)}{g(x)} \to 0$

C. $f(x) \to 0$ and $g(x) \to 0$, so $\frac{f(x)}{g(x)} \to 1$

D. Neither $f(x)$ nor $g(x)$ has a limit, so $\frac{f(x)}{g(x)}$ must have no limit

Answer

A

Knowledge points

- Quotient limit law
- Nonzero denominator-limit condition
- Indeterminate forms

Reading and method choice

The quotient law requires finite limits for numerator and denominator and a nonzero denominator limit. Only A satisfies all conditions.

Detailed derivation

1. In A, lim $g = 3 \neq 0$, so the quotient law gives lim$(\frac{f}{g}) = \frac{2}{3}$.
2. In B, the denominator limit is 0, so a key hypothesis of the law is missing.
3. C is the indeterminate form $\frac{0}{0}$; the two zero limits do not determine the quotient.
4. D is false because if $f(x) = g(x) \neq 0$, their quotient can be identically 1 even while both functions oscillate.
5. Thus A is the unique direct consequence.

Common pitfalls

- Treating $\frac{0}{0}$ as an ordinary fraction
- Forgetting to check that the denominator limit is nonzero
- Assuming two nonexistent limits cannot cancel in a quotient

Verification

A matches every theorem hypothesis. Counterexamples such as $f = x, g = x^2$ address B and C, while $f = g = 2 + \sin(\frac{1}{x})$ addresses D.

Method takeaway

Before citing the quotient law, explicitly verify $\lim g \neq 0$.

Extension

The denominator may vanish at isolated points, provided the quotient is defined in a deleted neighborhood and the denominator limit is nonzero.

Source lineage: Classic-method variant · limit laws / quotient law · reference IDs: openstax-v1-2.3-limit-laws. See SOURCES.md for scope and adaptation boundaries; this note does not claim verbatim reproduction.

Basic Calculation Foundation Classic-method variant

Q012 · Direct Substitution for a Polynomial

Problem

Evaluate $\lim_{x\to 2}(3x^3 - 2x^2 + x - 5)$.

Answer

13

Knowledge points

- Continuity of polynomials
- Sum, difference, and product limit laws
- Direct substitution

Reading and method choice

A polynomial is built from constants and powers by finitely many algebraic operations, so direct substitution is valid at every real number.

Detailed derivation

1. From $\lim_{x\to 2} x = 2$, obtain $\lim x^2 = 4$ and $\lim x^3 = 8$.
2. Multiply by constants: $\lim 3x^3 = 24$, $\lim(-2x^2) = -8$, and $\lim x = 2$.
3. Include the constant term -5.
4. Combine: $24 - 8 + 2 - 5 = 13$.
5. Therefore the limit is 13.

Common pitfalls

- Dropping a minus sign
- Computing 2^3 as 6
- Overcomplicating a polynomial that permits substitution

Verification

$P(2) = 3 \cdot 8 - 2 \cdot 4 + 2 - 5 = 13$, matching the limit-law computation.

Method takeaway

At finite points, substitute directly into polynomials first.

Extension

Every polynomial P satisfies $\lim_{x \to a} P(x) = P(a)$.

Source lineage: Classic-method variant · polynomial limit / direct substitution · reference IDs: openstax-v1-2.3-limit-laws. See SOURCES.md for scope and adaptation boundaries; this note does not claim verbatim reproduction.

Basic Calculation Foundation Open-text method adaptation

Q013 · Removing a Zero Factor with a Difference of Cubes

Problem

Evaluate $\lim_{x\to 2} \frac{x^3-8}{x-2}$ without derivatives.

Answer

12

Knowledge points

- Difference-of-cubes identity
- Cancellation in a deleted neighborhood
- Limits versus point values

Reading and method choice

Substitution gives $\frac{0}{0}$, which only says the quotient law is not directly applicable. Factor $x^3-8=(x-2)(x^2+2x+4)$ and cancel for $x\neq 2$.

Detailed derivation

1. Direct substitution gives $\frac{0}{0}$, signaling that algebraic transformation is needed.
2. Use $x^3-8=(x-2)(x^2+2x+4)$.
3. For $x\neq 2$, the quotient equals x^2+2x+4; cancellation is valid because a limit uses a deleted neighborhood.
4. Substitute $x=2$ into the polynomial to obtain $4+4+4=12$.
5. Hence the original limit is 12.

Common pitfalls

- Treating $\frac{0}{0}$ as the answer
- Failing to state $x\neq 2$ during cancellation
- Assuming an undefined point prevents a limit

Verification

At $x=2.001$, the quotient equals $x^2+2x+4\approx 12.006001$, close to 12.

Method takeaway

The form $\frac{0}{0}$ signals simplification; common zero factors may be canceled in a deleted neighborhood.

Extension

Similarly, $\lim_{x\to a} \frac{x^n - a^n}{x-a} = na^{n-1}$ follows from factorization, without derivatives.

Source lineage: Open-text method adaptation · factorization / removable factor · reference IDs: openstax-v1-2.3-limit-laws. See SOURCES.md for scope and adaptation boundaries; this note does not claim verbatim reproduction.

Basic | Single choice | Foundation | Open-text method adaptation

Q014 · Recognizing the Complete Squeeze-Theorem Hypotheses

Problem

Which set of conditions is sufficient to conclude $f(x) \to L$ by the squeeze theorem?

A. $g(x) \le f(x) \le h(x)$, with $g(x) \to L$ and $h(x) \to L$

B. $g(x) \le f(x)$, with $g(x) \to L$

C. $f(x) \le h(x)$, with $h(x) \to L$

D. $g(x) \le f(x) \le h(x)$, but $g(x) \to L_1$ and $h(x) \to L_2$ with $L_1 < L_2$

Answer

A

Knowledge points

- Squeeze theorem
- Two-sided control
- Common limiting value

Reading and method choice

The squeeze theorem needs both bounds to converge to the same number in the same limiting process. One-sided control or different endpoint limits cannot determine f uniquely.

Detailed derivation

1. A gives $g \le f \le h$ and both bounds tend to L, exactly the squeeze theorem.
2. B gives only a lower bound, so f may remain far above L.
3. C gives only an upper bound, so f may remain far below L.
4. D traps f between two distinct limiting levels but does not force a unique limit.
5. Therefore A is correct.

Common pitfalls

- Using squeeze with only one bound
- Ignoring that the two bound limits must agree

- Failing to ensure the inequalities hold in one neighborhood

Verification

For B take $g = 0, f = 1$; for C take $h = 0, f = -1$; for D take $g = 0, h = 2, f = 1$. None determines the stated L.

Method takeaway

Squeeze requires two-sided eventual inequalities and a common limit for both bounds.

Extension

The sequence version of the squeeze theorem has the identical logical structure.

Source lineage: Open-text method adaptation · squeeze theorem / theorem hypotheses · reference IDs: openstax-v1-2.3-limit-laws. See SOURCES.md for scope and adaptation boundaries; this note does not claim verbatim reproduction.

Basic Calculation Foundation Open-text method adaptation

Q015 · Suppressing Oscillation with a Vanishing Factor

Problem

Evaluate $\lim_{x\to 0} x\,\sin(\frac{1}{x})$.

Answer

0

Knowledge points

- Boundedness of sine
- Absolute-value squeeze
- Oscillation with a vanishing factor

Reading and method choice

Although $\sin(\frac{1}{x})$ oscillates near 0, its absolute value never exceeds 1. Multiplication by x reduces the amplitude to at most $|x|$.

Detailed derivation

1. For every $x \neq 0$, $|\sin(\frac{1}{x})| \leq 1$.
2. Hence $|x\sin(\frac{1}{x})| = |x||\sin(\frac{1}{x})| \leq |x|$.
3. Equivalently, $-|x| \leq x\sin(\frac{1}{x}) \leq |x|$.
4. As $x \to 0$, both $-|x|$ and $|x|$ tend to 0.
5. The squeeze theorem gives the limit 0.

Common pitfalls

- Trying to find a limit of $\sin(\frac{1}{x})$ itself
- Saying bounded times zero without a uniform bound
- Forgetting absolute values in the squeeze

Verification

For every x, $|x\sin(\frac{1}{x})| \leq |x|$; in particular, $|x| < 10^{-3}$ forces the product's absolute value below 10^{-3}.

Method takeaway

A uniformly bounded oscillatory factor can be controlled by a factor tending to 0.

Extension

If $|g(x)| \le M$ and $\alpha(x) \to 0$, then $\alpha(x)g(x) \to 0$.

Source lineage: Open-text method adaptation · squeeze theorem / oscillatory function · reference IDs: openstax-v1-2.3-limit-laws. See SOURCES.md for scope and adaptation boundaries; this note does not claim verbatim reproduction.

Basic | Fill in the blank | Foundation | Classic-method variant

Q016 · Scaling the First Fundamental Limit

Problem

$\lim_{x\to 0} \frac{\sin(5x)}{3x} = \_\_\_\_$.

Answer

$\frac{5}{3}$

Knowledge points

- $\lim_{u\to 0} \frac{\sin u}{u} = 1$
- Creating the standard form
- Constant factors

Reading and method choice

Rewrite the denominator using the same argument 5x as the sine, preserving the constant ratio.

Detailed derivation

1. Write $\frac{\sin(5x)}{3x} = (\frac{5}{3}) \cdot [\frac{\sin(5x)}{5x}]$.
2. Let $u = 5x$; then $u \to 0$ as $x \to 0$.
3. By the first fundamental limit, $\frac{\sin u}{u} \to 1$.
4. Thus the original limit is $(\frac{5}{3}) \cdot 1 = \frac{5}{3}$.
5. The blank is $\frac{5}{3}$.

Common pitfalls

- Writing 1 without accounting for scaling
- Dropping the factor $\frac{5}{3}$
- Replacing $\sin(5x)$ by $5 \sin x$

Verification

At $x = 0.001$, $\frac{\sin(0.005)}{0.003} \approx 1.66666$, close to $\frac{5}{3}$.

Method takeaway

For $\frac{\sin(kx)}{x}$, create $\frac{\sin(kx)}{kx}$ and retain the factor k.

Extension

In general, $\lim_{x\to 0} \frac{\sin(ax)}{bx} = \frac{a}{b}$ when $b \neq 0$.

Source lineage: Classic-method variant · first fundamental limit / variable scaling · reference IDs: openstax-v1-2.3-limit-laws, mit-18.01sc-session-8-trig-limits. See SOURCES.md for scope and adaptation boundaries; this note does not claim verbatim reproduction.

Basic | Single choice | Foundation | Classic-method variant

Q017 · Recognizing Equivalent Infinitesimals

Problem

As $x \to 0$, which equivalence of infinitesimals is correct?

A. $\ln(1+x) \sim x$

B. $1-\cos x \sim x$

C. $\sin x \sim x^2$

D. $e^x - 1 \sim 1$

Answer

A

Knowledge points

- Definition of equivalent infinitesimals
- Standard equivalences
- Comparison of orders

Reading and method choice

The test for $\alpha \sim \beta$ is $\frac{\alpha}{\beta} \to 1$. Apply the fundamental limits to each option.

Detailed derivation

1. For A, $\frac{\ln(1+x)}{x} \to 1$, so $\ln(1+x) \sim x$.
2. For B, $\frac{1-\cos x}{x} = [\frac{1-\cos x}{x^2}]x \to (\frac{1}{2}) \cdot 0 = 0$, not 1.
3. For C, $\frac{\sin x}{x^2} = \frac{[\frac{\sin x}{x}]\cdot 1}{x}$ does not tend to 1.
4. For D, $e^x - 1 \to 0$ while 1 does not, so they cannot be equivalent infinitesimals.
5. Only A is correct.

Common pitfalls

- Assuming any two infinitesimals are equivalent
- Checking only that both tend to 0

- Ignoring their different orders

Verification

The ratios in A, B, and C tend to 1, 0, and an unbounded quantity, respectively; D does not even compare two infinitesimals.

Method takeaway

Equivalence means more than both tending to 0 : their ratio must tend to 1.

Extension

Other standard relations are $\sin x \sim x$, $\tan x \sim x$, $e^x - 1 \sim x$, and $1 - \cos x \sim \frac{x^2}{2}$.

Source lineage: Classic-method variant · equivalent infinitesimals / single choice · reference IDs: openstax-v1-2.3-limit-laws. See SOURCES.md for scope and adaptation boundaries; this note does not claim verbatim reproduction.

Basic | Single choice | Foundation | Open-text method adaptation

Q018 · The Essential Condition for Continuity

Problem

Suppose f is defined in a neighborhood of a, including at a. Which statement is equivalent to saying that f is continuous at a?

A. $\lim_{x\to a} f(x) = f(a)$

B. $f(a)$ exists

C. The left-hand limit $\lim_{x\to a^-} f(x)$ exists

D. f is bounded in some neighborhood of a

Answer

A

Knowledge points

- definition of continuity at a point
- relation between a function value and a limit
- two-sided and one-sided limits

Reading and method choice

Continuity requires more than the existence of a function value or one one-sided limit. The two-sided limit as x approaches a must exist and must equal the actual value of the function at a.

Detailed derivation

1. By definition, f is continuous at a if and only if $\lim_{x\to a} f(x) = f(a)$.
2. Choice B is insufficient. For example, let $f(0) = 1$ and $f(x) = 0$ for $x \neq 0$; then $f(0)$ exists, but f is not continuous at 0.
3. Choice C supplies only the left-hand limit. Continuity requires both one-sided limits to exist, to be equal, and to have common value $f(a)$.
4. Choice D is also insufficient. The function $\sin(\frac{1}{x})$ is bounded in every punctured neighborhood of 0, yet no assigned value at 0 makes it continuous there.
5. Thus only choice A states the complete necessary-and-sufficient condition.

Common pitfalls

- confusing being defined with being continuous
- checking only one one-sided limit
- mistaking local boundedness for existence of a limit

Verification

Choice A can be unpacked into three checks: the two-sided limit exists, it is finite, and it equals $f(a)$; these are precisely the requirements in the definition.

Method takeaway

At a point, always check that the function value exists, the limit exists, and the two are equal.

Extension

At an endpoint of the domain, continuity is usually defined with the appropriate one-sided limit.

Source lineage: Open-text method adaptation · definition of continuity / function limit · reference IDs: openstax-v1-2.4-continuity, mit-18.01sc-session-5-discontinuity. See SOURCES.md for scope and adaptation boundaries; this note does not claim verbatim reproduction.

Basic | Fill in the blank | Foundation | Open-text method adaptation

Q019 · Defining a Missing Value to Obtain Continuity

Problem

Let $f(x) = \frac{x^2-4}{x-2}$ for $x \neq 2$, and define $f(2) = c$. To make f continuous at $x = 2$, one must take $c =$ ____.

Answer

$c = 4$

Knowledge points

- removable discontinuity caused by a common factor
- continuous extension
- assigning the limiting value

Reading and method choice

The original quotient is undefined at $x = 2$, but the common factor $x - 2$ can be canceled in a punctured neighborhood. The only value that gives a continuous extension is the limit at that point.

Detailed derivation

1. Factor the numerator: $x^2 - 4 = (x - 2)(x + 2)$.
2. For $x \neq 2$, $f(x) = \frac{(x-2)(x+2)}{x-2} = x + 2$.
3. Hence $\lim_{x\to 2} f(x) = \lim_{x\to 2}(x + 2) = 4$.
4. Continuity at 2 requires $f(2) = \lim_{x\to 2} f(x)$.
5. Since $f(2) = c$, the unique choice is $c = 4$.

Common pitfalls

- forgetting that cancellation is valid only for $x \neq 2$
- stopping after direct substitution gives $\frac{0}{0}$
- giving the simplified expression instead of the required numerical value

Verification

With $c = 4$, the resulting function agrees with $x + 2$ at every real number, so it is continuous at 2 and everywhere else.

Method takeaway

At a removable discontinuity, define the missing value to be the limit after the removable factor is canceled.

Extension

If $c \neq 4$, then $x = 2$ remains a removable discontinuity because the limit exists but differs from the assigned value.

Source lineage: Open-text method adaptation · removable discontinuity / continuous extension · reference IDs: openstax-v1-2.4-continuity, mit-18.01sc-session-5-discontinuity. See SOURCES.md for scope and adaptation boundaries; this note does not claim verbatim reproduction.

Basic Single choice Foundation Classic-method variant

Q020 · Finding an Interval of Continuity from the Domain

Problem

Which set is both the domain and the interval of continuity of $F(x) = \frac{\ln(4-x^2)}{\sqrt{x+1}}$?

A. $(-2, 2)$

B. $(-1, 2)$

C. $[-1, 2)$

D. $(-1, +\infty)$

Answer

B

Knowledge points

- domain of a logarithm
- square-root and denominator restrictions
- continuity of elementary functions on their domains

Reading and method choice

A composite elementary function is continuous wherever it is defined. We therefore impose both positivity of the logarithm's argument and strict positivity of the square root that appears in the denominator.

Detailed derivation

1. The logarithm requires $4 - x^2 > 0$, which is equivalent to $-2 < x < 2$.
2. The square root requires $x + 1 \geq 0$, but because it is in the denominator, zero must also be excluded. Hence $x + 1 > 0$, or $x > -1$.
3. Intersecting the two conditions gives $(-2, 2) \cap (-1, +\infty) = (-1, 2)$.
4. On this open interval, $4 - x^2$, $\ln(4 - x^2)$, $\sqrt{x+1}$, and their quotient are continuous by the algebra and composition theorems.
5. Thus both the domain and interval of continuity are $(-1, 2)$, so B is correct.

Common pitfalls

- using only $x + 1 \geq 0$ and forgetting that a denominator cannot vanish
- replacing the strict logarithm condition by $4 - x^2 \geq 0$
- failing to intersect all domain conditions

Verification

At $x = -1$ the denominator is zero, and at $x = 2$ the logarithm's argument is zero, so neither endpoint is included. Every $-1 < x < 2$ satisfies both restrictions.

Method takeaway

The continuity intervals of an elementary expression are the interval components of its domain, found by intersecting every layer of restrictions.

Extension

If the square root were multiplied in the numerator instead, $x = -1$ could be included, subject still to the logarithm condition.

Source lineage: Classic-method variant · elementary function / interval of continuity · reference IDs: openstax-v1-2.4-continuity, mit-18.01sc-session-4-limits-continuity. See SOURCES.md for scope and adaptation boundaries; this note does not claim verbatim reproduction.

Basic **Multiple choice** **Foundation** **Classic-method variant**

Q021 · Algebra Rules for Continuous Functions

Problem

Which statements about continuity are correct?

A. Every polynomial is continuous on $\mathbb{R}$

B. A rational function is continuous at every point where its denominator is nonzero

C. A composition of continuous functions is continuous wherever the composition is meaningful

D. The quotient of two continuous functions must remain continuous at a point where the denominator function is zero

Answer

A, B, C

Knowledge points

- sums, products, and quotients of continuous functions
- continuity of compositions
- continuity of elementary functions

Reading and method choice

A polynomial is obtained from constant and identity functions by finitely many sums and products. A rational function additionally requires a nonzero denominator. A composition requires continuity of both layers and a meaningful expression.

Detailed derivation

1. Polynomials are built by finite sums and products of continuous functions, so they are continuous on all of $\mathbb{R}$; A is correct.
2. A rational function is a quotient of polynomials, and the quotient theorem applies wherever its denominator is nonzero; B is correct.
3. If the inner function is continuous at the point, the outer function is continuous at the corresponding image, and the composition is defined, then the composition is continuous; C is correct.
4. The quotient theorem explicitly requires a nonzero denominator at the point. If the denominator is zero, the expression may not even be defined; D is false.
5. Therefore the correct selections are A, B, and C.

Common pitfalls

- omitting the nonzero-denominator condition
- checking only the outer function and not whether the composition is defined
- assuming every elementary expression is continuous on all real numbers

Verification

A direct counterexample to D is $f(x) = 1$ and $g(x) = x$: both are continuous at 0, but $\frac{f(x)}{g(x)} = \frac{1}{x}$ is undefined and discontinuous there.

Method takeaway

Algebra rules for continuity have hypotheses; domain checks are essential for quotients and compositions.

Extension

Compositions involving roots, logarithms, and inverse trigonometric functions must also obey their respective domain restrictions.

Source lineage: Classic-method variant · algebra of continuous functions / composition · reference IDs: openstax-v1-2.4-continuity, mit-18.01sc-session-4-limits-continuity. See SOURCES.md for scope and adaptation boundaries; this note does not claim verbatim reproduction.

Basic | True/false with justification | Foundation | Open-text method adaptation

Q022 · Are Extrema Guaranteed on an Open Interval?

Problem

True or false: If f is continuous on an open interval (a, b), then it must attain both a maximum and a minimum in (a, b).

Answer

False

Knowledge points

- extreme value theorem
- closed-interval hypothesis
- counterexample

Reading and method choice

The extreme value theorem requires continuity on a closed interval $[a, b]$. On an open interval, function values may approach endpoint bounds without ever attaining them.

Detailed derivation

1. The standard extreme value theorem assumes that f is continuous on the closed interval $[a, b]$.
2. The hypothesis here gives continuity only on (a, b), so the endpoints are absent and the theorem does not apply.
3. Take the counterexample $f(x) = x$ on $(0, 1)$. As a polynomial, it is continuous throughout this open interval.
4. For every $x \in (0, 1)$, the point $\frac{x+1}{2} \in (0, 1)$ has a larger function value, so no maximum exists.
5. Likewise, $\frac{x}{2} \in (0, 1)$ has a smaller function value, so no minimum exists.
6. Therefore continuity on an open interval does not guarantee attained extrema, and the statement is false.

Common pitfalls

- remembering only that continuous functions have extrema and omitting the closed-interval condition
- confusing a supremum with a maximum
- assuming boundedness guarantees attainment

Verification

For $f(x) = x$ on $(0, 1)$, the supremum is 1 and the infimum is 0, but neither endpoint belongs to the domain, so neither bound is attained.

Method takeaway

Whenever using the extreme value theorem, explicitly check both continuity and the closed-interval hypothesis.

Extension

Even a continuous and bounded function on an open interval may fail to attain either its maximum or its minimum.

Source lineage: Open-text method adaptation · extreme value theorem / closed interval · reference IDs: openstax-v1-2.4-continuity, openstax-v1-4.3-extrema. See SOURCES.md for scope and adaptation boundaries; this note does not claim verbatim reproduction.

Standard Calculation Methods Open-text method adaptation

Q023 · A Composite Function and Its Domain

Problem

Let $f(x) = \sqrt{x+2}$ and $g(u) = \frac{1}{u-1}$. Find the formula and natural domain of $(g \circ f)(x)$.

Answer

$(g \circ f)(x) = \frac{1}{\sqrt{x+2}-1}$, **with domain** $[-2, -1) \cup (-1, +\infty)$.

Knowledge points

- order of composition
- the inner function must be defined
- the inner output must lie in the outer domain

Reading and method choice

Substitution alone is not enough. The radical must exist, and after substitution the denominator of the outer function must remain nonzero.

Detailed derivation

1. The order $g \circ f$ means first compute $f(x) = \sqrt{x+2}$, then use it as the input of g.
2. Substitution gives $(g \circ f)(x) = g(f(x)) = \frac{1}{\sqrt{x+2}-1}$.
3. The inner radical requires $x + 2 \geq 0$, so $x \geq -2$.
4. The input of g cannot equal 1, so $\sqrt{x+2} \neq 1$, or $x \neq -1$; combining the restrictions gives $[-2, -1) \cup (-1, +\infty)$.

Common pitfalls

- reversing $g \circ f$ into $f \circ g$
- keeping only $x \geq -2$ and missing the zero denominator at $x = -1$

Verification

At $x = -2$ the denominator is -1, so the function is defined. At $x = -1$ the denominator is 0, so the point must be removed. Every other $x \geq -2$ satisfies both restrictions.

Method takeaway

A composite domain requires two checks: the inner function must be evaluable, and its output must be admissible for the outer function.

Extension

One may also find $f \circ g$; its restrictions differ from those of $g \circ f$, illustrating that composition is generally not commutative.

Source lineage: Open-text method adaptation · composite function / domain of a composition · reference IDs: openstax-v1-1.1-functions. See SOURCES.md for scope and adaptation boundaries; this note does not claim verbatim reproduction.

Standard | Multiple choice | Methods | Classic-method variant

Q024 · Analyze Global Properties of a Function

Problem

Let $h(x) = \frac{|x|}{1+x^2}$ on R. Which statements are correct?

A. h is even

B. For every $x \in \mathbb{R}, 0 \leq h(x) \leq \frac{1}{2}$

C. h is periodic

D. h is strictly increasing on $[0, +\infty)$

Answer

A, B

Knowledge points

- test for an even function
- algebraic bounds
- counterexamples to periodicity and monotonicity

Reading and method choice

Parity follows by comparing $h(-x)$ and $h(x)$. Boundedness follows from $(|x| - 1)^2 \geq 0$. Periodicity and monotonicity can be rejected by consequences or explicit values that contradict their definitions.

Detailed derivation

1. $h(-x) = \frac{|-x|}{1+(-x)^2} = \frac{|x|}{1+x^2} = h(x)$, so A is correct.
2. From $(|x| - 1)^2 \geq 0$ we obtain $x^2 + 1 \geq 2|x|$, hence $0 \leq \frac{|x|}{1+x^2} \leq \frac{1}{2}; B$ is correct.
3. As $|x| \to +\infty, h(x) \to 0$, while $h(1) = \frac{1}{2}$. A nonzero periodic repetition is incompatible with this decay, so C is false.
4. Since $h(1) = \frac{1}{2}$ but $h(2) = \frac{2}{5} < \frac{1}{2}, h$ is not strictly increasing on $[0, +\infty); D$ is false.

Common pitfalls

- stopping after noticing the absolute value and checking only parity
- asserting boundedness or monotonicity from a sketch without algebraic evidence

Verification

Equality in $x^2 + 1 \geq 2|x|$ occurs at $|x| = 1$, so the upper bound $\frac{1}{2}$ is attained. The values $h(1) = \frac{1}{2}$ and $h(2) = \frac{2}{5}$ directly disprove D.

Method takeaway

Test each global property independently by its definition or a rigorous inequality.

Extension

The monotonicity of h on $[0, 1]$ and $[1, +\infty)$ can be studied algebraically through differences, without derivatives.

Source lineage: Classic-method variant · function properties / parity · reference IDs: openstax-v1-1.1-functions. See SOURCES.md for scope and adaptation boundaries; this note does not claim verbatim reproduction.

Standard Calculation Methods Open-text method adaptation

Q025 · An Inverse Function with an Interval Restriction

Problem

The domain of $f(x) = 2x - 3$ is $[1, 4]$. Find the range of f, the formula for its inverse, and the domain and range of the inverse.

Answer

The range of f is $[-1, 5]$. The inverse is $f^{-1}(y) = \frac{y+3}{2}$, with domain $[-1, 5]$ and range $[1, 4]$.

Knowledge points

- a strictly monotone function is invertible on its interval
- the original range becomes the inverse domain
- solve after interchanging input and output roles

Reading and method choice

The linear function is strictly increasing on the prescribed interval, so it is one-to-one there. Determine the range from endpoint values, then solve $y = 2x - 3$ for x.

Detailed derivation

1. As x increases from 1 to $4, 2x - 3$ increases strictly, so f is one-to-one on $[1, 4]$.
2. The endpoint values are $f(1) = -1$ and $f(4) = 5$, hence the range of f is $[-1, 5]$.
3. Solving $y = 2x - 3$ gives $x = \frac{y+3}{2}$, so $f^{-1}(y) = \frac{y+3}{2}$.
4. The inverse swaps domain and range: Dom$(f^{-1}) = [-1, 5]$ and Range$(f^{-1}) = [1, 4]$.

Common pitfalls

- keeping $[1, 4]$ as the domain of the inverse after finding its formula
- confusing $f^{-1}(y)$ with the reciprocal $\frac{1}{2y-3}$

Verification

For $x \in [1, 4]$, $f^{-1}(f(x)) = \frac{(2x-3)+3}{2} = x$. For $y \in [-1, 5]$, $f(f^{-1}(y)) = \frac{2\cdot(y+3)}{2} - 3 = y$.

Method takeaway

Finding an inverse requires swapping not only the variables but also the original domain and range.

Extension

If the original domain were all of R, the same inverse formula would hold, but both its domain and range would be R.

Source lineage: Open-text method adaptation · inverse function / range · reference IDs: openstax-v1-1.1-functions, openstax-v1-1.4-inverse-functions. See SOURCES.md for scope and adaptation boundaries; this note does not claim verbatim reproduction.

Standard Single choice Methods Classic-method variant

Q026 · Find the Least Positive Period of a Sum

Problem

What is the least positive period of $p(x) = \sin(2x) + \cos(4x)$?

A. $\frac{\pi}{4}$

B. $\frac{\pi}{2}$

C. π

D. 2π

Answer

C

Knowledge points

- periods of sine and cosine
- common periods
- least positive period

Reading and method choice

The component periods show that π is a period of the sum, but a common period does not by itself prove minimality, and in general one may not assume that every period of a sum is a period of each summand. For a rigorous proof, let $T > 0$ be an arbitrary period of the sum and use values at two special points to constrain $\sin(2T)$ and $\cos(2T)$.

Detailed derivation

1. For $\sin(2x), 2T_1 = 2\pi$ gives the least positive period $T_1 = \pi$.
2. For $\cos(4x), 4T_2 = 2\pi$ gives the least positive period $T_2 = \frac{\pi}{2}$.
3. π is a period of both terms, so $p(x + \pi) = p(x)$; hence π is indeed a positive period of p.
4. Let $T > 0$ be any period of p. From $p(T) = p(0) = 1$, set $s = \sin(2T)$ and use $\cos(4T) = 1 - 2s^2$. Then $s + 1 - 2s^2 = 1$, so $s \in 0, \frac{1}{2}$.
5. Also $p(\frac{\pi}{4}) = 0$, so $p(\frac{\pi}{4} + T) = 0$. With $c = \cos(2T)$, we have $p(\frac{\pi}{4} + T) = c - \cos(4T) = c - (2c^2 - 1) = 0$, hence $c \in 1, -\frac{1}{2}$.
6. Combining these candidates with $s^2 + c^2 = 1$ leaves only $s = 0$ and $c = 1$. Thus $\sin(2T) = 0$ and $\cos(2T) = 1$, so $2T = 2k\pi$ and $T = k\pi$.

7. Every positive period is therefore a positive integer multiple of π, while π itself is a period. Hence the least positive period is π.

Common pitfalls

- taking the smaller component period automatically
- assuming that every period of a sum must separately be a period of each summand
- ruling out only $\frac{\pi}{2}$ while leaving other positive numbers below π untreated

Verification

Direct substitution verifies $p(x+\pi) = p(x)$. The converse argument proves that every period T must have $T = k\pi$, ruling out not only $\frac{\pi}{2}$ but every other positive number below π.

Method takeaway

A common period proves only that it is a period; minimality requires a rigorous constraint on an arbitrary candidate period.

Extension

For $\sin(4x) + \cos(4x)$, both components have least positive period $\frac{\pi}{2}$, so the sum also has least positive period $\frac{\pi}{2}$.

Source lineage: Classic-method variant · periodic function / least positive period · reference IDs: openstax-v1-1.1-functions. See SOURCES.md for scope and adaptation boundaries; this note does not claim verbatim reproduction.

Standard Calculation Methods Classic-method variant

Q027 · Limit of a Rational Sequence with Equal Degrees

Problem

Evaluate $\lim_{n\to\infty} \frac{5n^2-3n+1}{2n^2+n-4}$.

Answer

$\frac{5}{2}$.

Knowledge points

- divide by the highest power
- limit laws for sequences
- nonzero denominator limit

Reading and method choice

Both numerator and denominator are quadratic. Dividing by n^2 converts all lower-degree terms into multiples of $\frac{1}{n}$ or $\frac{1}{n^2}$, which vanish, leaving the ratio of leading coefficients.

Detailed derivation

1. For $n \geq 1$, divide both numerator and denominator by n^2.
2. The expression becomes $\frac{5-\frac{3}{n}+\frac{1}{n^2}}{2+\frac{1}{n}-\frac{4}{n^2}}$.
3. As $n \to \infty$, both $\frac{1}{n}$ and $\frac{1}{n^2}$ tend to 0, so the numerator tends to 5 and the denominator to 2.
4. Since the denominator limit is $2 \neq 0$, the quotient law gives the limit $\frac{5}{2}$.

Common pitfalls

- writing only 'compare leading terms' without showing the valid normalization
- forgetting that the quotient law requires a nonzero denominator limit

Verification

At $n = 100$ the ratio is $\frac{49701}{20096} \approx 2.4732$, already close to 2.5; the lower-order corrections continue to shrink.

Method takeaway

For equal-degree polynomial ratios at infinity, the limit is the ratio of leading coefficients.

Extension

If the numerator degree were lower than the denominator degree, the same normalization would give limit 0.

Source lineage: Classic-method variant · rational sequence limit / leading terms · reference IDs: openstax-v2-5.1-sequences. See SOURCES.md for scope and adaptation boundaries; this note does not claim verbatim reproduction.

Standard Proof Methods Classic-method variant

Q028 · Prove Convergence of a Rational Sequence from the Definition

Problem

Use the ε-N definition to prove $\lim_{n\to\infty} \frac{3n-2}{n+1} = 3$.

Answer

Given $\varepsilon > 0$, choose $N = [\frac{5}{\varepsilon}]$. If $n > N$, then $|\frac{3n-2}{n+1} - 3| = \frac{5}{n+1} < \varepsilon$, so the limit is 3.

Knowledge points

- structure of an ε-N proof
- working backward to choose N
- strict inequality

Reading and method choice

First simplify the distance between the sequence and the target 3. Then solve the desired inequality $\frac{5}{n+1} < \varepsilon$ for a sufficient lower bound on n.

Detailed derivation

1. Let $\varepsilon > 0$ be arbitrary and compute $|\frac{3n-2}{n+1} - 3|$.
2. Combining the terms gives $|\frac{3n-2-3n-3}{n+1}| = \frac{5}{n+1}$.
3. To obtain $\frac{5}{n+1} < \varepsilon$, it is enough that $n + 1 > \frac{5}{\varepsilon}$. Choose the convenient safe value $N = [\frac{5}{\varepsilon}]$.
4. If $n > N$, then $n > [\frac{5}{\varepsilon}] \geq \frac{5}{\varepsilon}$, so $n + 1 > \frac{5}{\varepsilon}$ and consequently $\frac{5}{n+1} < \varepsilon$.
5. Since this works for every $\varepsilon > 0$, the ε-N definition gives $\frac{3n-2}{n+1} \to 3$.

Common pitfalls

- stating a value of N without verifying that it implies the desired inequality
- allowing N to depend on n instead of only on ε

Verification

For $\varepsilon = 0.1$, take $N = 50$. If $n > 50$, then $\frac{5}{n+1} \leq \frac{5}{51} < 0.1$, confirming the estimate.

Method takeaway

A definition proof follows the chain: simplify the error, solve backward for n, choose $N(\varepsilon)$, and verify forward.

Extension

For the general sequence $\frac{an+b}{n+c} \to a$, simplify the error to $\frac{|b-ac|}{|n+c|}$ and use the same pattern.

Source lineage: Classic-method variant · ε-N proof / rational sequence · reference IDs: openstax-v2-5.1-sequences. See SOURCES.md for scope and adaptation boundaries; this note does not claim verbatim reproduction.

Standard | Fill in the blank | Methods | Classic-method variant

Q029 · Determine a Parameter from One-Sided Limits

Problem

Let $f(x) = \begin{cases} 2x + a, & x < 1, \\ x^2 + 1, & x > 1. \end{cases}$ For $\lim_{x\to 1} f(x)$ to exist, a must equal ____, and the limit is then ____.

Answer

0; 2.

Knowledge points

- one-sided limits of a piecewise function
- existence criterion for a two-sided limit
- parameter equation

Reading and method choice

Use the left piece as $x \to 1^-$ and the right piece as $x \to 1^+$. The two-sided limit exists when the two results agree; whether $f(1)$ is defined is irrelevant here.

Detailed derivation

1. The left-hand limit is $\lim_{x\to 1^-}(2x + a) = 2 + a$.
2. The right-hand limit is $\lim_{x\to 1^+}(x^2 + 1) = 2$.
3. Set the one-sided limits equal: $2 + a = 2$, which gives $a = 0$.
4. With $a = 0$, both one-sided limits equal 2, so $\lim_{x\to 1} f(x) = 2$.

Common pitfalls

- substituting $x = 1$ into the wrong branch or demanding that $f(1)$ be specified
- finding a but omitting the common limit value

Verification

When $a = 0$, the left expression 2x tends to 2 and the right expression $x^2 + 1$ tends to 2. If $a \neq 0$, the two limits differ by a.

Method takeaway

At a piecewise junction, a limit parameter is determined by equality of the left and right limits, independently of the point value.

Extension

If $f(1) = b$ is added and continuity at 1 is required, one must also set $b = 2$.

Source lineage: Classic-method variant · piecewise-function limit / parameter · reference IDs: openstax-v1-2.2-function-limits. See SOURCES.md for scope and adaptation boundaries; this note does not claim verbatim reproduction.

Standard | Calculation | Methods | Classic-method variant

Q030 · Use One-Sided Limits to Show Nonexistence

Problem

Evaluate $\lim_{x \to 0} \frac{|x|}{x}$. If it does not exist, explain why.

Answer

The limit does not exist; the left-hand limit is -1 and the right-hand limit is 1.

Knowledge points

- piecewise representation of absolute value
- one-sided limits
- criterion for a two-sided limit

Reading and method choice

$|x|$ has different formulas on the two sides of 0. Simplify separately on each side and compare the one-sided limits; direct substitution at $x = 0$ is not appropriate.

Detailed derivation

1. For $x < 0, |x| = -x$, so $\frac{|x|}{x} = -1$.
2. Therefore $\lim_{x \to 0^-} \frac{|x|}{x} = -1$.
3. For $x > 0, |x| = x$, so $\frac{|x|}{x} = 1$ and $\lim_{x \to 0^+} \frac{|x|}{x} = 1$.
4. The one-sided limits -1 and 1 are unequal, so $\lim_{x \to 0} \frac{|x|}{x}$ does not exist.

Common pitfalls

- canceling $\frac{|x|}{x}$ to 1 and ignoring negative x
- claiming nonexistence only because the denominator is zero at $x = 0$, without examining nearby values

Verification

Along $x_n = \frac{1}{n}$ the function values are always 1, while along $y_n = -\frac{1}{n}$ they are always -1. The two approaches give different results.

Method takeaway

Limits involving absolute values often require sign-based one-sided analysis; unequal one-sided limits rule out a two-sided limit.

Extension

The right-hand limit exists and equals 1, and the left-hand limit exists and equals -1, so the behavior at 0 is a jump.

Source lineage: Classic-method variant · absolute-value function / one-sided limits · reference IDs: openstax-v1-2.2-function-limits. See SOURCES.md for scope and adaptation boundaries; this note does not claim verbatim reproduction.

Standard Calculation Methods Classic-method variant

Q031 · Handle the Sign of a Radical as x Approaches Negative Infinity

Problem

Evaluate $\lim_{x\to-\infty} \frac{2x-1}{\sqrt{x^2+1}}$.

Answer

-2.

Knowledge points

- limits at infinity
- $\sqrt{x^2} = |x|$
- sign on the negative direction

Reading and method choice

The crucial identity is $\sqrt{x^2} = |x|$, not x. As $x \to -\infty$, $|x| = -x$, and this determines the sign of the limit.

Detailed derivation

1. Divide numerator and denominator by $|x|$ to obtain $\frac{\frac{2x-1}{|x|}}{\sqrt{1+\frac{1}{x^2}}}$.
2. As $x \to -\infty, x < 0$, so $|x| = -x$ and therefore $\frac{x}{|x|} = -1$.
3. Hence $\frac{2x-1}{|x|} = \frac{2x}{|x|} - \frac{1}{|x|} \to -2 - 0 = -2$.
4. Also $\sqrt{1 + \frac{1}{x^2}} \to 1$, so the quotient tends to $-\frac{2}{1} = -2$.

Common pitfalls

- replacing $\sqrt{x^2}$ by x and mishandling the negative direction
- comparing leading coefficients while ignoring the absolute value introduced by the square root

Verification

At $x = -100$, the value is $\frac{-201}{\sqrt{10001}} \approx -2.0099$, close to -2. A result of $+2$ would immediately contradict the numerical sign.

Method takeaway

Whenever $\sqrt{x^2}$ appears in an infinite limit, write $|x|$ first and then use the direction to decide whether $|x|$ equals x or $-x$.

Extension

For the same expression as $x \to +\infty$, the limit is 2, showing why the two infinite directions must be treated separately.

Source lineage: Classic-method variant · limit at infinity / radical · reference IDs: openstax-v1-2.2-function-limits. See SOURCES.md for scope and adaptation boundaries; this note does not claim verbatim reproduction.

Standard Calculation Methods Open-text method adaptation

Q032 · Cancel the Common Factor Causing Zero Division

Problem

Evaluate $\lim_{x\to 3} \frac{x^2-9}{x-3}$.

Answer

6.

Knowledge points

- difference of squares
- cancellation in a deleted neighborhood
- limit versus point value

Reading and method choice

Direct substitution gives $\frac{0}{0}$, but the numerator contains the same factor as the denominator. A limit uses x close to 3 with $x \neq 3$, so the common factor can be canceled first.

Detailed derivation

1. Factor the numerator using $x^2 - 9 = (x-3)(x+3)$.
2. For $x \neq 3$, $\frac{x^2-9}{x-3} = x + 3$.
3. The deleted neighborhood used by the limit satisfies $x \neq 3$, so this simplification is valid for the limit calculation.
4. Therefore $\lim_{x\to 3} \frac{x^2-9}{x-3} = \lim_{x\to 3}(x+3) = 6$.

Common pitfalls

- treating the direct-substitution form $\frac{0}{0}$ as proof that the limit does not exist
- claiming after cancellation that the original quotient is defined at $x = 3$

Verification

At $x = 2.99$ and $x = 3.01$, the original quotient equals 5.99 and 6.01, approaching 6 from both sides.

Method takeaway

The form $\frac{0}{0}$ is not an answer; it signals that algebraic simplification may reveal the limit. A deleted neighborhood permits cancellation of the vanishing factor.

Extension

More generally, $\lim_{x\to a} \frac{x^2-a^2}{x-a} = 2a$ by the same difference-of-squares factorization.

Source lineage: Open-text method adaptation · function-limit calculation / factorization · reference IDs: openstax-v1-2.2-function-limits. See SOURCES.md for scope and adaptation boundaries; this note does not claim verbatim reproduction.

Standard | Fill in the blank | Methods | Classic-method variant

Q033 · Writing a Function as a Constant Plus an Infinitesimal

Problem

As $x \to 0$, write $f(x) = \frac{3+2x}{1+x}$ in the form $f(x) = A + \alpha(x)$, where $\alpha(x)$ is infinitesimal. Find A and $\alpha(x)$.

Answer

$A = 3$ **and** $\alpha(x) = -\frac{x}{1+x}$.

Knowledge points

- Equivalence between lim $f = A$ and $f = A + \alpha$
- Algebraic decomposition
- Verification of an infinitesimal

Reading and method choice

First obtain the candidate limit A by substitution, then construct the exact remainder $\alpha = f - A$ and verify that it tends to 0.

Detailed derivation

1. Since the denominator at $x = 0$ is $1 \neq 0, A = \lim_{x\to 0} \frac{3+2x}{1+x} = 3$.
2. Define the remainder by $\alpha(x) = f(x) - A$.
3. Compute $\alpha(x) = \frac{3+2x}{1+x} - 3 = \frac{3+2x-3-3x}{1+x} = -\frac{x}{1+x}$.
4. As $x \to 0$, the numerator $-x \to 0$ and the denominator $1 + x \to 1 \neq 0$, so $\alpha(x) \to 0$.
5. Hence the required decomposition is $f(x) = 3 - \frac{x}{1+x}$.

Common pitfalls

- Giving A without verifying that α is infinitesimal
- Expanding $3(1 + x)$ incorrectly
- Replacing an exact decomposition by an approximation

Verification

Combining $3 - \frac{x}{1+x}$ gives $\frac{3(1+x)-x}{1+x} = \frac{3+2x}{1+x}$, exactly the original function.

Method takeaway

When lim $f = A$, the natural exact remainder is $\alpha = f - A$, and it must be infinitesimal.

Extension

Conversely, $f = A + \alpha$ with $\alpha \to 0$ immediately implies lim $f = A$.

Source lineage: Classic-method variant · limit representation / infinitesimal decomposition · reference IDs: openstax-v1-2.2-function-limits, openstax-v1-2.5-precise-limit. See SOURCES.md for scope and adaptation boundaries; this note does not claim verbatim reproduction.

Standard Calculation Methods Classic-method variant

Q034 · Infinite Behavior of a Reciprocal Square

Problem

Find $\lim_{x\to 0} \frac{1}{x^2}$, and explain why the reciprocal relationship between $\frac{1}{x^2}$ and x^2 is valid as $x \to 0$.

Answer

$\lim_{x\to 0} \frac{1}{x^2} = +\infty$. **Since $x^2 \to 0$ and $x^2 > 0$ in a deleted neighborhood of 0, its reciprocal tends to $+\infty$.**

Knowledge points

- Definition of a positive infinite limit
- Reciprocal relation between infinitesimals and infinite quantities
- Deleted neighborhoods

Reading and method choice

Besides stating $+\infty$, one must check that the infinitesimal denominator is nonzero near the point. Squaring also guarantees the same sign from both sides.

Detailed derivation

1. As $x \to 0, x^2 \to 0$, and $x^2 > 0$ whenever $x \neq 0$.
2. Given any $M > 0$, choose $\delta = \frac{1}{\sqrt{M}}$.
3. If $0 < |x| < \delta$, then $x^2 < \delta^2 = \frac{1}{M}$, hence $\frac{1}{x^2} > M$.
4. This is exactly the definition of $\lim_{x\to 0} \frac{1}{x^2} = +\infty$.
5. Thus x^2 is an eventually positive nonzero infinitesimal, and its reciprocal $\frac{1}{x^2}$ is positive infinite.

Common pitfalls

- Assigning opposite signs from the two sides despite the square
- Taking a reciprocal without checking nonvanishing
- Treating $+\infty$ as an ordinary real number

Verification

For $M = 10^4$, choose $\delta = 0.01$. Every x with $0 < |x| < 0.01$ satisfies $\frac{1}{x^2} > 10^4$, exactly as required.

Method takeaway

An infinite limit should be verifiable against every threshold M; the square forces $+\infty$ from both sides.

Extension

If $\alpha(x) \to 0$ and $\alpha(x) \neq 0$ in a deleted neighborhood, then $|\frac{1}{\alpha(x)}| \to \infty$; its sign determines $+\infty, -\infty$, or only divergence in absolute value.

Source lineage: Classic-method variant · infinite limit / reciprocal relation · reference IDs: openstax-v1-2.2-function-limits, openstax-v1-2.5-precise-limit. See SOURCES.md for scope and adaptation boundaries; this note does not claim verbatim reproduction.

Standard Calculation Methods Open-text method adaptation

Q035 · Rationalizing with a Conjugate

Problem

Evaluate $\lim_{x\to 1} \frac{\sqrt{x+3}-2}{x-1}$ without derivatives.

Answer

$\frac{1}{4}$

Knowledge points

- Conjugate rationalization
- Cancellation in a deleted neighborhood
- Domains of radicals

Reading and method choice

Both numerator and denominator tend to 0. Multiply by the conjugate to turn the radical difference into $x-1$, then cancel.

Detailed derivation

1. Direct substitution gives $\frac{2-2}{1-1} = \frac{0}{0}$.
2. Multiply numerator and denominator by $\sqrt{x+3}+2$.
3. The numerator becomes $(x+3)-4 = x-1$, so for $x \neq 1$ the expression equals $\frac{1}{\sqrt{x+3}+2}$.
4. The simplified denominator tends to $2+2=4$ as $x \to 1$.
5. Therefore the limit is $\frac{1}{4}$.

Common pitfalls

- Expanding the conjugate product incorrectly
- Returning to the unsimplified $\frac{0}{0}$ expression
- Ignoring the local radical domain

Verification

At $x = 1.01$, the simplified expression is $\frac{1}{\sqrt{4.01}+2} \approx 0.249844$, close to 0.25.

Method takeaway

When a difference of radicals creates $\frac{0}{0}$, try conjugate rationalization first.

Extension

In general, the conjugate of $\sqrt{u} - \sqrt{v}$ is $\sqrt{u} + \sqrt{v}$.

Source lineage: Open-text method adaptation · rationalization / radical limit · reference IDs: openstax-v1-2.3-limit-laws. See SOURCES.md for scope and adaptation boundaries; this note does not claim verbatim reproduction.

Standard Calculation Methods Classic-method variant

Q036 · Algebraic Simplification of a Cube-Root Difference

Problem

Evaluate $\lim_{x \to 8} \frac{\sqrt[3]{x}-2}{x-8}$ without derivatives.

Answer

$\frac{1}{12}$

Knowledge points

- Factorization of $a^3 - b^3$
- Cube-root substitution
- Cancellation in a deleted neighborhood

Reading and method choice

Let $u = \sqrt[3]{x}$. Then $x - 8 = u^3 - 2^3$ factors by $u - 2$, exactly the numerator factor to cancel.

Detailed derivation

1. Direct substitution gives $\frac{0}{0}$. Set $u = \sqrt[3]{x}$, so $u \to 2$ as $x \to 8$.
2. Factor $x - 8 = u^3 - 8 = (u-2)(u^2 + 2u + 4)$.
3. For $x \neq 8$, equivalently $u \neq 2$, the expression equals $\frac{1}{u^2+2u+4}$.
4. As $u \to 2$, the denominator tends to $4 + 4 + 4 = 12$.
5. Therefore the limit is $\frac{1}{12}$.

Common pitfalls

- Using a square-root conjugate formula
- Omitting the middle term 2u
- Failing to state that $u \to 2$

Verification

At $u = 2.001$, the simplified value is about $\frac{1}{12.012001}$, close to $\frac{1}{12}$.

Method takeaway

A cube-root difference pairs with the difference-of-cubes identity, not a square conjugate.

Extension

For nth roots, use $a^n - b^n = (a-b)(a^{n-1} + \cdots + b^{n-1})$.

Source lineage: Classic-method variant · cube root / difference of cubes · reference IDs: openstax-v1-2.3-limit-laws. See SOURCES.md for scope and adaptation boundaries; this note does not claim verbatim reproduction.

Standard | Fill in the blank | Methods | Classic-method variant

Q037 · A Rational Expression from Known Limits

Problem

If $f(x) \to 2$ and $g(x) \to -1$ as $x \to a$, then $\lim_{x\to a} \frac{3f(x)-g(x)}{f(x)-g(x)} =$ ____.

Answer

$\frac{7}{3}$

Knowledge points

- Limits of linear combinations
- Quotient limit law
- Nonzero denominator check

Reading and method choice

Find the numerator and denominator limits separately, then verify that the denominator limit is nonzero before dividing.

Detailed derivation

1. The numerator limit is $3 \cdot 2 - (-1) = 7$.
2. The denominator limit is $2 - (-1) = 3$.
3. Because the denominator limit $3 \neq 0$, the quotient law applies.
4. Thus the requested limit is $\frac{7}{3}$.
5. The blank is $\frac{7}{3}$.

Common pitfalls

- Mishandling the sign of $-g$
- Not checking the denominator limit
- Confusing 3f with f^3

Verification

Temporarily replacing f by 2 and g by -1 gives $\frac{6+1}{2+1} = \frac{7}{3}$, matching the result.

Method takeaway

For nested limit operations, use the sequence: numerator, denominator, nonzero check.

Extension

The same procedure works for any rational combination whose denominator limit is nonzero.

Source lineage: Classic-method variant · combined limit laws / quotient law · reference IDs: openstax-v1-2.3-limit-laws. See SOURCES.md for scope and adaptation boundaries; this note does not claim verbatim reproduction.

Standard | True/false with justification | Methods | Classic-method variant

Q038 · A Quotient of Two Infinitesimals

Problem

True or false, with justification: If $\alpha(x) \to 0$ and $\beta(x) \to 0$, then $\frac{\alpha(x)}{\beta(x)} \to 1$.

Answer

False. Take $\alpha(x) = x$ and $\beta(x) = x^2$ as $x \to 0$. Both tend to 0, but $\frac{\alpha}{\beta} = \frac{1}{x}$ has no finite two-sided limit.

Knowledge points

- The indeterminate form $\frac{0}{0}$
- Counterexample method
- Two-sided limits

Reading and method choice

The facts that both quantities tend to 0 describe their individual sizes but do not determine the ratio of their rates of decay.

Detailed derivation

1. Let $\alpha(x) = x$ and $\beta(x) = x^2$, with $x \to 0$.
2. Clearly $\lim \alpha = 0$ and $\lim \beta = 0$, so every hypothesis is satisfied.
3. For $x \neq 0$, $\frac{\alpha}{\beta} = \frac{x}{x^2} = \frac{1}{x}$.
4. As $x \to 0^+$, $\frac{1}{x} \to +\infty$, while as $x \to 0^-$, $\frac{1}{x} \to -\infty$.
5. The two-sided limit therefore does not exist and is certainly not 1.

Common pitfalls

- Canceling $\frac{0}{0}$ to get 1
- Checking only the premises without the quotient
- Ignoring opposite one-sided behavior

Verification

At $x = 0.01$ the quotient is 100, while at $x = -0.01$ it is -100, so it is not approaching 1.

Method takeaway

The form $\frac{0}{0}$ is indeterminate; compare orders or simplify algebraically.

Extension

With the additional condition $\alpha(x) \sim \beta(x)$, the definition is precisely $\frac{\alpha(x)}{\beta(x)} \to 1$.

Source lineage: Classic-method variant · $\frac{0}{0}$ form / quotient law · reference IDs: openstax-v1-2.3-limit-laws. See SOURCES.md for scope and adaptation boundaries; this note does not claim verbatim reproduction.

Standard | Multiple choice | Methods | Classic-method variant

Q039 · A Rational Limit at Negative Infinity

Problem

For $L = \lim_{x \to -\infty} \frac{3x^2 - x + 1}{2x^2 + 5}$, which statements are correct? Select all that apply.

A. $L = \frac{3}{2}$

B. Replacing $x \to -\infty$ by $x \to +\infty$ still gives $\frac{3}{2}$

C. After division by x^2, both $\frac{1}{x}$ and $\frac{1}{x^2}$ tend to 0

D. Since $x \to -\infty, x^2 \to -\infty$

Answer

A, B, and C

Knowledge points

- Limits at infinity
- Division by the highest power
- Algebraic limit laws

Reading and method choice

Both numerator and denominator are quadratic. Dividing by x^2 turns every lower-order term into a quantity tending to 0.

Detailed derivation

1. As $x \to -\infty, x \neq 0$, so divide numerator and denominator by x^2.
2. The expression becomes $\frac{3 - \frac{1}{x} + \frac{1}{x^2}}{2 + \frac{5}{x^2}}$.
3. Both $\frac{1}{x} \to 0$ and $\frac{1}{x^2} \to 0$.
4. The numerator limit is 3 and the denominator limit is $2 \neq 0$.
5. The quotient law gives $L = \frac{3}{2}$, so A is true. The same analysis holds for $x \to +\infty$, so B is true; C is the valid simplification, while D is false because a square is nonnegative.

Common pitfalls

- Treating x^2 as negative because $x \to -\infty$
- Dividing only some terms
- Forgetting the nonzero denominator-limit check

Verification

At $x = -100$ the value is $\frac{30101}{20005} \approx 1.50467$, close to 1.5.

Method takeaway

For equal-degree rational functions at infinity, the limit is the ratio of leading coefficients.

Extension

If the numerator degree is lower, the limit is 0; if higher, analyze the sign of the dominant power.

Source lineage: Classic-method variant · limit at infinity / leading terms · reference IDs: openstax-v1-2.3-limit-laws. See SOURCES.md for scope and adaptation boundaries; this note does not claim verbatim reproduction.

Standard Calculation Methods Open-text method adaptation

Q040 · Standardizing Two Sine Factors

Problem

Evaluate $\lim_{x\to 0} \frac{\sin(2x)\sin(3x)}{x^2}$.

Answer

6

Knowledge points

- First fundamental limit
- Product limit theorem
- Scale factors

Reading and method choice

Pair each sine with its own argument and distribute x^2 into (2x)(3x), retaining the factor 6.

Detailed derivation

1. Rewrite the expression as $6 \cdot [\frac{\sin(2x)}{2x}] \cdot [\frac{\sin(3x)}{3x}]$.
2. As $x \to 0$, both $2x \to 0$ and $3x \to 0$.
3. The first fundamental limit makes each bracket tend to 1.
4. By the product law, the total limit is $6 \cdot 1 \cdot 1$.
5. Therefore the answer is 6.

Common pitfalls

- Pairing x^2 with only one sine
- Dropping $2 \cdot 3 = 6$
- Using the false identity $\sin(2x)\sin(3x) = \sin(6x)$

Verification

At $x = 0.001$ the expression is about 5.999987, close to 6.

Method takeaway

For products of small-angle sines, create $\frac{\sin u}{u}$ factor by factor and preserve every scale constant.

Extension

$\lim_{x\to 0} \frac{\prod_{k=1}^{m} \sin(a_k x)}{x^m} = \prod_{k=1}^{m} a_k.$

Source lineage: Open-text method adaptation · first fundamental limit / product limit · reference IDs: openstax-v1-2.3-limit-laws, mit-18.01sc-session-8-trig-limits. See SOURCES.md for scope and adaptation boundaries; this note does not claim verbatim reproduction.

Standard Calculation Methods Classic-method variant

Q041 · Deriving a Cosine Limit from the First Fundamental Limit

Problem

Evaluate $\lim_{x\to 0} \frac{1-\cos(2x)}{x^2}$ without Taylor expansions.

Answer

2

Knowledge points

- Identity $1 - \cos 2x = 2\sin^2 x$
- First fundamental limit
- Limits of squares

Reading and method choice

Convert the cosine difference into a sine square so that $\frac{\sin x}{x} \to 1$ applies directly.

Detailed derivation

1. Use $1 - \cos(2x) = 2\sin^2 x$.
2. The expression becomes $\frac{2\sin^2 x}{x^2} = 2[\frac{\sin x}{x}]^2$.
3. As $x \to 0$, $\frac{\sin x}{x} \to 1$.
4. Therefore $[\frac{\sin x}{x}]^2 \to 1$.
5. The requested limit is $2 \cdot 1 = 2$.

Common pitfalls

- Writing $1 - \cos 2x = \sin^2 x$
- Losing the square in the denominator
- Using Taylor expansions in a Chapter 1 solution

Verification

At $x = 0.01$, $\frac{1-\cos 0.02}{0.0001} \approx 1.99993$, close to 2.

Method takeaway

For a cosine difference, first use a half-angle identity to reach the sine fundamental limit.

Extension

Similarly, $\lim_{x\to 0} \frac{1-\cos(ax)}{x^2} = \frac{a^2}{2}$.

Source lineage: Classic-method variant · trigonometric identity / first fundamental limit · reference IDs: openstax-v1-2.3-limit-laws, mit-18.01sc-session-8-trig-limits. See SOURCES.md for scope and adaptation boundaries; this note does not claim verbatim reproduction.

Standard Calculation Methods Classic-method variant

Q042 · Creating the Second Fundamental Limit

Problem

Evaluate $\lim_{x\to 0}(1+2x)^{\frac{3}{x}}$.

Answer

e^6

Knowledge points

- $\lim_{u\to 0}(1+u)^{\frac{1}{u}}=e$
- Exponent decomposition
- Composite limits

Reading and method choice

Set $u=2x$ and rewrite $\frac{3}{x}$ as $\frac{6}{u}$, producing the sixth power of the standard fundamental-limit expression.

Detailed derivation

1. Let $u=2x$; then $u\to 0$ as $x\to 0$.
2. The exponent is $\frac{3}{x}=\frac{3}{\frac{u}{2}}=\frac{6}{u}$.
3. Thus $(1+2x)^{\frac{3}{x}}=[(1+u)^{\frac{1}{u}}]^6$.
4. By the second fundamental limit, $(1+u)^{\frac{1}{u}}\to e$.
5. Hence the original limit is e^6.

Common pitfalls

- Computing the exponent scale as $\frac{3}{2}$
- Concluding 1 merely because the base tends to 1
- Ignoring positivity of the base near 0

Verification

At $x=0.001, 1.002^{3000}\approx e^{5.994}$, close to e^6.

Method takeaway

For a 1^{∞} form, standardize both the base increment and exponent; their product controls the final exponent.

Extension

In general, $\lim_{x \to 0}(1+ax)^{\frac{b}{x}} = e^{ab}$.

Source lineage: Classic-method variant · second fundamental limit / exponent scaling · reference IDs: openstax-v1-2.3-limit-laws, mit-18.01sc-session-19-limit-involving-e. See SOURCES.md for scope and adaptation boundaries; this note does not claim verbatim reproduction.

Standard Calculation Methods Classic-method variant

Q043 · An Exponential Sequence with a Negative Increment

Problem

Evaluate $\lim_{n\to\infty}(1-\frac{3}{n})^n$.

Answer

e^{-3}

Knowledge points

- Sequence form of the second fundamental limit
- Negative approach to zero
- Exponent scaling

Reading and method choice

Let $u_n = -\frac{3}{n}$, so $u_n \to 0$ and $n = -\frac{3}{u_n}$. The base is positive for all sufficiently large n.

Detailed derivation

1. For $n > 3, 1 - \frac{3}{n} > 0$, so the real power is well defined.
2. Set $u_n = -\frac{3}{n}$; then $u_n \to 0$.
3. Since $n = -\frac{3}{u_n}, (1-\frac{3}{n})^n = [(1+u_n)^{\frac{1}{u_n}}]^{-3}$.
4. The second fundamental limit gives $(1+u_n)^{\frac{1}{u_n}} \to e$.
5. Therefore the sequence limit is e^{-3}.

Common pitfalls

- Dropping the minus sign and obtaining e^3
- Letting finitely many early terms affect the tail limit
- Treating 1^n as an ordinary power

Verification

At $n = 1000, (0.997)^{1000} \approx 0.04956$, while $e^{-3} \approx 0.04979$.

Method takeaway

The second fundamental limit also permits a negative increment; the increment-exponent product remains decisive.

Extension

In general, $\lim_{n\to\infty}(1+\frac{c}{n})^n = e^c$.

Source lineage: Classic-method variant · second fundamental limit / sequence limit · reference IDs: openstax-v1-2.3-limit-laws, mit-18.01sc-session-19-limit-involving-e, openstax-v2-5.1-sequences. See SOURCES.md for scope and adaptation boundaries; this note does not claim verbatim reproduction.

Standard | Multiple choice | Methods | Classic-method variant

Q044 · Recognizing Common Indeterminate Forms

Problem

Which limit structures are indeterminate forms? Select all that apply.

A. $\frac{0}{0}$

B. $\frac{\infty}{\infty}$

C. 1^{∞}

D. 0^{1}

Answer

A, B, and C

Knowledge points

- Indeterminate-form types
- Limit structure
- Second fundamental limit

Reading and method choice

An indeterminate form means the component limits alone do not determine the whole. The forms $\frac{0}{0}$, $\frac{\infty}{\infty}$, and 1^{∞} can lead to many outcomes; 0^{1} normally tends to 0 under the usual real-domain conditions.

Detailed derivation

1. A can yield 0, a finite nonzero value, infinity, or no limit, so $\frac{0}{0}$ is indeterminate.
2. B depends on relative growth rates, so $\frac{\infty}{\infty}$ is indeterminate.
3. C depends on the product of the base's deviation from 1 and the exponent; it is indeterminate.
4. D has a base tending to 0 and an exponent tending to 1, with no 0^{0}-type competition, so it normally tends to 0.
5. Thus select A, B, and C.

Common pitfalls

- Treating symbols involving ∞ as arithmetic numbers
- Saying every power with base near 1 tends to 1

- Confusing 0^1 with 0^0

Verification

As $x \to 0$, compare $\frac{x}{x}$ with $\frac{x^2}{x}$ for $\frac{0}{0}$; as $x \to +\infty$, compare $\frac{x}{x}$ with $\frac{x^2}{x}$ for $\frac{\infty}{\infty}$; and compare $(1+\frac{1}{n})^n$ with $(1+\frac{1}{n})^{2n}$ for 1^∞. Each form can produce different limits.

Method takeaway

An indeterminate form is a structural warning, not an answer; further transformation is required.

Extension

Other common forms in this chapter include $0 \cdot \infty, \infty - \infty$, and 0^0.

Source lineage: Classic-method variant · indeterminate forms / multiple choice · reference IDs: openstax-v1-2.3-limit-laws. See SOURCES.md for scope and adaptation boundaries; this note does not claim verbatim reproduction.

Standard **Fill in the blank** **Methods** **Classic-method variant**

Q045 · Comparing Orders of Power Infinitesimals

Problem

As $x \to 0, |x|^3$ is a ____-order infinitesimal relative to x^2 because $\frac{|x|^3}{x^2} \to$ ____.

Answer

higher; 0.

Knowledge points

- Definition of higher-order infinitesimals
- Absolute-value powers
- Order comparison by a ratio

Reading and method choice

If $\frac{\alpha}{\beta} \to 0$, then α is of higher order than β. Here the ratio simplifies directly to $|x|$.

Detailed derivation

1. Let $\alpha(x) = |x|^3$ and $\beta(x) = x^2 = |x|^2$.
2. For $x \neq 0$, $\frac{\alpha}{\beta} = \frac{|x|^3}{|x|^2} = |x|$.
3. As $x \to 0, |x| \to 0$.
4. Thus $\frac{\alpha}{\beta} \to 0$, so α is of higher order than β.
5. The blanks are higher and 0.

Common pitfalls

- Calling a larger exponent lower order
- Ignoring $x^2 = |x|^2$
- Thinking higher order means larger numerical size

Verification

At $x = 0.1$ the ratio is 0.1, and at $x = 0.01$ it is 0.01, clearly approaching 0.

Method takeaway

Near 0, a larger positive power usually vanishes faster and is therefore of higher order.

Extension

If $p > q > 0$, then $|x|^p$ is higher order than $|x|^q$.

Source lineage: Classic-method variant · order of infinitesimals / higher-order infinitesimal · reference IDs: openstax-v1-2.3-limit-laws. See SOURCES.md for scope and adaptation boundaries; this note does not claim verbatim reproduction.

Standard Calculation Methods Classic-method variant

Q046 · Equivalent Substitution in a Product-Quotient Structure

Problem

Evaluate $\lim_{x\to 0} \frac{\sin(3x)\cdot\ln(1+2x)}{x(e^x-1)}$.

Answer

6

Knowledge points

- $\sin u \sim u$
- $\ln(1+u) \sim u$
- $e^u - 1 \sim u$
- Equivalent substitution in products and quotients

Reading and method choice

Every substituted infinitesimal appears as a multiplicative or divisive factor, so the replacements are safe.

Detailed derivation

1. As $x \to 0, \sin(3x) \sim 3x$.
2. Also $\ln(1+2x) \sim 2x$ and $e^x - 1 \sim x$.
3. After factorwise substitution, the expression is equivalent to $\frac{(3x)(2x)}{x\cdot x}$.
4. This simplifies to the constant 6.
5. Therefore the limit is 6.

Common pitfalls

- Replacing $\ln(1+2x)$ by x instead of 2x
- Dropping the existing x in the denominator
- Using equivalent substitution without checking the algebraic position

Verification

Alternatively factor it as $[\frac{\sin 3x}{3x}] \cdot [\frac{\ln(1+2x)}{2x}] \cdot [\frac{x}{e^x-1}] \cdot 6$; the first three factors tend to 1.

Method takeaway

In products and quotients, apply equivalences factor by factor and preserve every scale constant.

Extension

A factor not tending to 0 can simply be handled by its ordinary limit rather than forced into an infinitesimal equivalence.

Source lineage: Classic-method variant · equivalent-infinitesimal substitution / product and quotient structure · reference IDs: openstax-v1-2.3-limit-laws. See SOURCES.md for scope and adaptation boundaries; this note does not claim verbatim reproduction.

Standard **Calculation** **Methods** **Classic-method variant**

Q047 · Balancing Second- and First-Order Factors

Problem

Evaluate $\lim_{x\to 0} \frac{1-\cos(2x)}{x\sin(3x)}$.

Answer

$\frac{2}{3}$

Knowledge points

- $1-\cos u \sim \frac{u^2}{2}$
- $\sin u \sim u$
- Equivalent infinitesimals

Reading and method choice

The numerator is second order, and the denominator is a product of two first-order factors, also second order. The remaining constants determine the limit.

Detailed derivation

1. As $x \to 0, 1-\cos(2x) \sim \frac{(2x)^2}{2} = 2x^2$.
2. Also $\sin(3x) \sim 3x$.
3. Hence $x\sin(3x) \sim x \cdot 3x = 3x^2$.
4. The quotient is equivalent to $\frac{2x^2}{3x^2} = \frac{2}{3}$.
5. Therefore the limit is $\frac{2}{3}$.

Common pitfalls

- Replacing $1-\cos u$ by u
- Dropping the factor $\frac{1}{2}$
- Forgetting the extra x in the denominator

Verification

Using $1-\cos 2x = 2\sin^2 x$ gives $2[\frac{\sin x}{x}]^2 \cdot [\frac{x}{\sin 3x}]$, with the same limit $\frac{2}{3}$.

Method takeaway

Match the total orders first, then collect the constants left by the equivalences.

Extension

$\lim_{x\to 0} \frac{1-\cos(ax)}{x\sin(bx)} = \frac{a^2}{2b}$ for $b \neq 0$.

Source lineage: Classic-method variant · cosine equivalence / sine equivalence · reference IDs: openstax-v1-2.3-limit-laws, mit-18.01sc-session-8-trig-limits. See SOURCES.md for scope and adaptation boundaries; this note does not claim verbatim reproduction.

Standard | True/false with justification | Methods | Open-text method adaptation

Q048 · Can a Removable Discontinuity Always Be Repaired?

Problem

True or false: If $x = a$ is a removable discontinuity of f, then redefining only $f(a)$ can always make f continuous at a.

Answer

True

Knowledge points

- definition of a removable discontinuity
- finite limit
- uniqueness of a continuous extension

Reading and method choice

At a removable discontinuity, the finite two-sided limit already exists. The only defect is that the function is undefined at the point or has a value different from that limit.

Detailed derivation

1. By definition of a removable discontinuity, there is a finite number L such that $\lim_{x\to a} f(x) = L$.
2. The original $f(a)$ is either undefined or is defined with $f(a) \neq L$.
3. Define a new function F by $F(x) = f(x)$ for $x \neq a$ and $F(a) = L$.
4. Changing one point does not change a punctured limit, so $\lim_{x\to a} F(x) = L$.
5. Since $F(a) = L$, we have $\lim_{x\to a} F(x) = F(a)$, and therefore F is continuous at a.

Common pitfalls

- trying to repair a jump discontinuity by changing one point
- ignoring the requirement that the one-sided limits agree
- thinking that any newly assigned value will work

Verification

The repaired function satisfies all three continuity checks: $F(a)$ exists, the two-sided limit exists, and both equal L.

Method takeaway

The word removable means that the point can be repaired by assigning the unique limiting value.

Extension

If the two one-sided limits are finite but unequal, the point is a jump discontinuity and no single assigned value can repair it.

Source lineage: Open-text method adaptation · removable discontinuity / redefinition · reference IDs: openstax-v1-2.4-continuity, mit-18.01sc-session-5-discontinuity. See SOURCES.md for scope and adaptation boundaries; this note does not claim verbatim reproduction.

Standard Calculation Methods Classic-method variant

Q049 · Parameters for Continuity at Two Junctions

Problem

Let $f(x) = ax + b$ for $x < 1$, $f(x) = x^2 + 1$ for $1 \le x \le 2$, and $f(x) = 3x - a$ for $x > 2$. Find a, b so that f is continuous on $\mathbb{R}$.

Answer

$a = 1,\ b = 1$

Knowledge points

- continuity of a piecewise function
- one-sided limits at junctions
- linear system

Reading and method choice

Each formula is a polynomial and is therefore continuous inside its own interval. Only the junctions $x = 1$ and $x = 2$ can cause a loss of continuity, so we impose a matching condition at each one.

Detailed derivation

1. On $(-\infty, 1)$, $(1, 2)$, and $(2, +\infty)$, each piece is a polynomial and needs no further condition.
2. At $x = 1$, the left-hand limit is $a + b$, while the function value and right-hand limit are $1^2 + 1 = 2$; hence $a + b = 2$.
3. At $x = 2$, the left-hand limit and the function value are $2^2 + 1 = 5$.
4. The right-hand limit at $x = 2$ is $3 \cdot 2 - a = 6 - a$, so continuity requires $6 - a = 5$.
5. The second equation gives $a = 1$; substituting into $a + b = 2$ gives $b = 1$.
6. Thus the unique parameters making both junctions continuous are $a = 1, b = 1$.

Common pitfalls

- checking only one junction
- using the left piece to compute the value at $x = 1$ even though it excludes that point
- matching one-sided limits without checking the actual function value

Verification

With these values, the left-hand limit at 1 is $1 + 1 = 2 = f(1)$, and the right-hand limit at 2 is $6 - 1 = 5 = f(2)$. Every other point lies inside a continuous polynomial piece.

Method takeaway

For a piecewise function, first reduce the work to the points where the defining formula changes.

Extension

If continuity were required only at $x = 1$, the parameters would satisfy the line $a + b = 2$; the second junction makes the pair unique.

Source lineage: Classic-method variant · piecewise function / parameters for continuity · reference IDs: openstax-v1-2.4-continuity, mit-18.01sc-session-5-discontinuity. See SOURCES.md for scope and adaptation boundaries; this note does not claim verbatim reproduction.

Standard Multiple choice Methods Classic-method variant

Q050 · Distinguishing Types of Discontinuity

Problem

Which statements about a discontinuity at $x = a$ are correct?

A. If both one-sided limits exist, are equal and finite, but their common value differs from $f(a)$ (or $f(a)$ is undefined), then a is a removable discontinuity

B. If both one-sided limits are finite but unequal, then a is a jump discontinuity

C. If at least one one-sided limit is infinite, then a is an infinite discontinuity and is of the second kind

D. If the limit fails to exist because of oscillation, then a is a discontinuity of the first kind

Answer

A, B, C

Knowledge points

- removable discontinuity
- jump discontinuity
- infinite discontinuity
- oscillatory discontinuity

Reading and method choice

The main dividing line is whether both one-sided limits exist as finite numbers. If they do, the discontinuity is of the first kind; if at least one finite one-sided limit fails to exist, it is of the second kind.

Detailed derivation

1. In A, the equal finite one-sided limits give a finite two-sided limit, and only the point value is defective; thus A correctly describes a removable discontinuity.
2. In B, the two finite one-sided limits are unequal, producing a finite jump; thus B is correct.
3. In C, an infinite one-sided limit is not a finite limit, so the point is an infinite discontinuity of the second kind; thus C is correct.
4. In D, persistent oscillation prevents at least one one-sided limit from existing, so the point is of the second kind, not the first; thus D is false.
5. The correct selections are therefore A, B, and C.

Common pitfalls

- interpreting first kind as the more severe kind
- classifying two infinite sides as first kind
- forgetting that both infinite and oscillatory discontinuities are of the second kind

Verification

Apply one uniform test: A and B have two finite one-sided limits, whereas C and D lose at least one finite one-sided limit. The classifications are therefore consistent.

Method takeaway

First determine whether both finite one-sided limits exist; only then distinguish removable, jump, infinite, and oscillatory cases.

Extension

A second-kind discontinuity can be bounded: $\sin(\frac{1}{x})$ near $x = 0$ is the standard oscillatory example.

Source lineage: Classic-method variant · classification of discontinuities / first-kind discontinuity · reference IDs: openstax-v1-2.4-continuity, mit-18.01sc-session-5-discontinuity. See SOURCES.md for scope and adaptation boundaries; this note does not claim verbatim reproduction.

Standard Proof Methods Classic-method variant

Q051 · Proving Continuity of a Quadratic from the Definition

Problem

Without citing the theorem that polynomials are continuous, use the ε-δ definition to prove that $f(x) = x^2 + 1$ is continuous at every $a \in \mathbb{R}$.

Answer

For any $\varepsilon > 0$, take $\delta = \min\{1, \frac{\varepsilon}{2|a|+1}\}$; then $|(x^2+1) - (a^2+1)| < \varepsilon$ follows.

Knowledge points

- epsilon-delta definition of continuity
- difference of squares
- auxiliary restriction

Reading and method choice

The target difference factors as $|x-a||x+a|$. The first factor is controlled by $|x-a| < \delta$, while the second still contains the variable x. We first impose $|x-a| < 1$ to bound it by a constant depending only on the fixed point a.

Detailed derivation

1. Let $\varepsilon > 0$ be arbitrary. We seek $\delta > 0$ such that $|x-a| < \delta$ implies $|f(x) - f(a)| < \varepsilon$.
2. Factor the target difference: $|f(x) - f(a)| = |x^2 - a^2| = |x-a||x+a|$.
3. If $|x-a| < 1$, then $|x| \le |x-a| + |a| < 1 + |a|$, so $|x+a| \le |x| + |a| < 2|a| + 1$.
4. Choose $\delta = \min\{1, \frac{\varepsilon}{2|a|+1}\}$. This is positive and enforces both the auxiliary bound and the desired accuracy.
5. Whenever $|x-a| < \delta$, we have $|f(x) - f(a)| < |x-a|(2|a|+1) < \delta(2|a|+1) \le \varepsilon$.
6. Thus the definition is satisfied at a. Since a was arbitrary, $f(x) = x^2 + 1$ is continuous on $\mathbb{R}$.

Common pitfalls

- setting $\delta = \frac{\varepsilon}{|x+a|}$, which still depends on the variable x
- forgetting that a is fixed while x varies
- failing to control $|x-a|$ before bounding the second factor

Verification

The chosen δ depends only on the given ε and the fixed point a, not on x, and the final estimate is strictly below ε in the required order of quantifiers.

Method takeaway

When a product contains an uncontrolled variable factor, first use a simple neighborhood restriction to bound that factor by a constant.

Extension

The same factorization-and-bounding strategy can be iterated to prove continuity of every power x^n and hence every polynomial.

Source lineage: Classic-method variant · epsilon-delta definition / proof of continuity · reference IDs: openstax-v1-2.4-continuity, mit-18.01sc-session-5-discontinuity, openstax-v1-2.5-precise-limit. See SOURCES.md for scope and adaptation boundaries; this note does not claim verbatim reproduction.

Standard | Calculation | Methods | Open-text method adaptation

Q052 · Evaluating a Limit by Continuity of Elementary Functions

Problem

Use continuity to evaluate $\lim_{x \to \frac{\pi}{6}} \frac{\ln(2+\cos x)}{\sqrt{1+\sin x}}$, and explain why direct substitution is valid.

Answer

$$\frac{\ln(2+\frac{\sqrt{3}}{2})}{\sqrt{\frac{3}{2}}}$$

Knowledge points

- continuity of elementary functions
- composition
- continuity of a quotient
- direct substitution

Reading and method choice

The expression is composed of sine, cosine, logarithm, square root, and division. Once the logarithm's argument is positive and the denominator is nonzero near the target point, continuity turns the limit into an ordinary function value.

Detailed derivation

1. At $x = \frac{\pi}{6}$, $2 + \cos x = 2 + \frac{\sqrt{3}}{2} > 0$, so the logarithm is defined and continuous at the corresponding value.
2. Also, $1 + \sin(\frac{\pi}{6}) = 1 + \frac{1}{2} = \frac{3}{2} > 0$, so the square root is defined and the denominator $\sqrt{\frac{3}{2}}$ is nonzero.
3. Sine and cosine are continuous on $\mathbb{R}$; adding constants and composing with logarithm and square root preserve continuity here.
4. Because the denominator is nonzero at the target point, the entire quotient is continuous at $x = \frac{\pi}{6}$.
5. Therefore the limit equals $\frac{\ln(2+\cos(\frac{\pi}{6}))}{\sqrt{1+\sin(\frac{\pi}{6})}}$ by direct substitution.
6. Using $\cos(\frac{\pi}{6}) = \frac{\sqrt{3}}{2}$ and $\sin(\frac{\pi}{6}) = \frac{1}{2}$ gives $\frac{\ln(2+\frac{\sqrt{3}}{2})}{\sqrt{\frac{3}{2}}}$.

Common pitfalls

- citing elementary continuity without checking the domain and nonzero denominator
- interchanging the values of sine and cosine at $\frac{\pi}{6}$
- making an unjustified algebraic change to the square-root denominator

Verification

At the target point the logarithm's argument is about $2.866 > 0$ and the denominator about $1.225 > 0$, so there is no domain singularity and the substituted value is finite.

Method takeaway

Direct substitution is justified by continuity; it is not merely a mechanical rule.

Extension

If the logarithm's argument approached 0^+ or the denominator approached 0, continuity would no longer justify direct substitution and a separate limit analysis would be required.

Source lineage: Open-text method adaptation · continuity of compositions / direct substitution · reference IDs: openstax-v1-2.4-continuity, mit-18.01sc-session-4-limits-continuity. See SOURCES.md for scope and adaptation boundaries; this note does not claim verbatim reproduction.

Standard Proof Methods Classic-method variant

Q053 · The Square Root of a Positive Continuous Function

Problem

Suppose f is continuous at a and $f(a) > 0$. Prove that in some neighborhood of a, $f(x) > \frac{f(a)}{2} > 0$, and then prove that $g(x) = \sqrt{f(x)}$ is continuous at a.

Answer

Use continuity with $|f(x) - f(a)| < \frac{f(a)}{2}$ to obtain $f(x) > \frac{f(a)}{2}$. Then continuity of the square-root function, or a rationalization estimate, gives $\sqrt{f(x)} \to \sqrt{f(a)}$.

Knowledge points

- local sign preservation of a continuous function
- epsilon-delta definition
- continuity of the square-root function
- rationalization

Reading and method choice

Choose the positive error threshold $\frac{f(a)}{2}$ in the continuity definition to keep $f(x)$ positive nearby. The square root is then defined throughout that neighborhood. Continuity follows either by composition or directly by rationalizing the difference.

Detailed derivation

1. Because $f(a) > 0$, the number $\frac{f(a)}{2}$ is a valid positive error threshold.
2. Continuity of f at a gives $\delta_1 > 0$ such that $|x - a| < \delta_1$ implies $|f(x) - f(a)| < \frac{f(a)}{2}$.
3. Therefore $f(x) > f(a) - \frac{f(a)}{2} = \frac{f(a)}{2} > 0$. This proves local positivity and ensures that $\sqrt{f(x)}$ is defined.
4. Rationalize: $|g(x) - g(a)| = |\sqrt{f(x)} - \sqrt{f(a)}| = \frac{|f(x)-f(a)|}{\sqrt{f(x)}+\sqrt{f(a)}}$.
5. The denominator is at least $\sqrt{f(a)}$, so $|g(x) - g(a)| \le \frac{|f(x)-f(a)|}{\sqrt{f(a)}}$.
6. Given $\varepsilon > 0$, continuity of f supplies $\delta_2 > 0$ such that $|x - a| < \delta_2$ implies $|f(x) - f(a)| < \varepsilon\sqrt{f(a)}$.
7. Taking $\delta = \min\{\delta_1, \delta_2\}$ simultaneously guarantees a valid square root and $|g(x) - g(a)| < \varepsilon$. Hence g is continuous at a.

Common pitfalls

- writing square roots nearby before proving $f(x) > 0$
- forgetting the denominator in the rationalization step
- using $\frac{f(a)}{2}$ without noting that it is positive

Verification

The same final δ handles both the domain issue and the accuracy requirement, and the last estimate depends only on the controlled quantity $|f(x) - f(a)|$.

Method takeaway

For a composition with domain restrictions, first use continuity to preserve the needed sign locally, then apply composition or a direct estimate.

Extension

Similarly, if $f(a) \neq 0$, then $f(x)$ stays nonzero in a sufficiently small neighborhood, which proves continuity of $\frac{1}{f(x)}$ at a.

Source lineage: Classic-method variant · sign preservation / composition · reference IDs: openstax-v1-2.4-continuity, mit-18.01sc-session-4-limits-continuity. See SOURCES.md for scope and adaptation boundaries; this note does not claim verbatim reproduction.

Standard **Fill in the blank** **Methods** **Open-text method adaptation**

Q054 · A Value Between the Endpoint Values

Problem

Suppose f is continuous on $[0, 2]$, with $f(0) = -3$ and $f(2) = 5$. By the intermediate value theorem, there is at least one $\xi \in$ ____ such that $f(\xi) = 1$.

Answer

$(0, 2)$

Knowledge points

- intermediate value theorem
- endpoint values
- attainment in the open interval

Reading and method choice

The target value 1 lies strictly between the endpoint values -3 and 5. A continuous function cannot move between those values while skipping 1, and because 1 differs from both endpoint values, the corresponding point is interior.

Detailed derivation

1. The function is continuous on the closed interval $[0, 2]$, satisfying the continuity hypothesis of the intermediate value theorem.
2. Numerically, $-3 < 1 < 5$, so $f(0) < 1 < f(2)$.
3. The intermediate value theorem gives some $\xi \in [0, 2]$ with $f(\xi) = 1$.
4. Since $f(0) = -3 \neq 1$ and $f(2) = 5 \neq 1$, this point cannot be either endpoint.
5. Therefore at least one such point satisfies $\xi \in (0, 2)$.

Common pitfalls

- writing only $[0, 2]$ without using the fact that the target differs from both endpoint values
- mistaking existence for uniqueness
- failing to verify that the target lies between the endpoint values

Verification

Both endpoints are excluded directly: their values are -3 and 5, not 1. Hence the point supplied by the theorem must lie in the interior.

Method takeaway

The intermediate value theorem guarantees existence, not uniqueness; a target strictly between endpoint values is attained at an interior point.

Extension

For every $\mu \in (-3, 5)$, there is likewise at least one $\xi \in (0, 2)$ such that $f(\xi) = \mu$.

Source lineage: Open-text method adaptation · intermediate value theorem / continuity on a closed interval · reference IDs: openstax-v1-2.4-continuity. See SOURCES.md for scope and adaptation boundaries; this note does not claim verbatim reproduction.

Standard Calculation Methods Open-text method adaptation

Q055 · Locating and Proving Uniqueness of a Cubic Root

Problem

Without using derivatives, prove that $x^3+x-1=0$ has exactly one real root in $(\frac{1}{2},\frac{3}{4})$.

Answer

Let $F(x)=x^3+x-1$. Since $F(\frac{1}{2})=-\frac{3}{8}<0$ and $F(\frac{3}{4})=\frac{11}{64}>0$, a root exists in the interval. Since F is strictly increasing on $\mathbb{R}$, that root is unique.

Knowledge points

- zero theorem
- continuity of polynomials
- algebraic proof of strict monotonicity
- separating existence from uniqueness

Reading and method choice

The phrase exactly one contains two tasks. Continuity and opposite endpoint signs prove existence. The zero theorem alone does not prove uniqueness, so strict monotonicity must be established separately, here by factoring a difference rather than using derivatives.

Detailed derivation

1. Let $F(x)=x^3+x-1$. Being a polynomial, it is continuous on the closed interval $[\frac{1}{2},\frac{3}{4}]$.
2. Compute $F(\frac{1}{2})=\frac{1}{8}+\frac{1}{2}-1=-\frac{3}{8}<0$.
3. Compute $F(\frac{3}{4})=\frac{27}{64}+\frac{3}{4}-1=\frac{27}{64}+\frac{48}{64}-\frac{64}{64}=\frac{11}{64}>0$.
4. The endpoint values have opposite signs, so the zero theorem gives at least one $\xi\in(\frac{1}{2},\frac{3}{4})$ with $F(\xi)=0$.
5. For uniqueness, let $x>y$. Then $F(x)-F(y)=x^3-y^3+x-y=(x-y)(x^2+xy+y^2+1)$.
6. Here $x-y>0$, and $x^2+xy+y^2=(x+\frac{y}{2})^2+\frac{3y^2}{4}\geq 0$, so the second factor is strictly positive.
7. Thus $F(x)>F(y)$, proving that F is strictly increasing on $\mathbb{R}$. A strictly increasing function has at most one zero.
8. Combining at least one with at most one proves that there is exactly one root in $(\frac{1}{2},\frac{3}{4})$.

Common pitfalls

- turning the zero theorem's at least one into exactly one
- making an arithmetic error at the fractional endpoints
- using derivatives when the requested method is pre-derivative

Verification

Continuity and opposite signs are explicitly checked for existence, while the global inequality $F(x) - F(y) > 0$ for arbitrary $x > y$ independently guarantees uniqueness.

Method takeaway

A standard unique-root proof has two parts: continuity plus a sign change for existence, then strict monotonicity for at most one root.

Extension

The same argument applies to $x^{2m+1} + cx - d = 0$ with $c > 0$: the odd power and positive linear term together give strict increase.

Source lineage: Open-text method adaptation · zero theorem / existence of a root · reference IDs: openstax-v1-2.4-continuity. See SOURCES.md for scope and adaptation boundaries; this note does not claim verbatim reproduction.

Advanced Proof Synthesis Original synthesis / diagnosis

Q056 · Parity of a Composite Function

Problem

Let D be symmetric about the origin. Suppose $g : D \to E$ is even, $f : E \to R$ is arbitrary, and $h = f \circ g$ is defined on D. Prove that h is even on D.

Answer

For every $x \in D, h(-x) = f(g(-x)) = f(g(x)) = h(x)$; hence h is even.

Knowledge points

- definition of an even function
- composite functions
- symmetric-domain requirement

Reading and method choice

The outer function f needs no parity property. Because the even inner function g assigns the same inner value to x and $-x$, f receives the same input in both cases.

Detailed derivation

1. Take any $x \in D$. Since D is symmetric about the origin, $-x \in D$ and $h(-x)$ is defined.
2. By the definition of composition, $h(-x) = f(g(-x))$.
3. Since g is even, $g(-x) = g(x)$, so $h(-x) = f(g(x))$.
4. Because $h(x) = f(g(x))$, we obtain $h(-x) = h(x)$ for every $x \in D$; therefore h is even.

Common pitfalls

- omitting the symmetric-domain condition required for parity
- incorrectly assuming that the outer function f must also be even

Verification

For example, with $g(x) = x^2$ and $f(u) = u + u^3$, one gets $h(x) = x^2 + x^6$, which indeed satisfies $h(-x) = h(x)$.

Method takeaway

An even inner function maps x and $-x$ to the same point, so any admissible outer function preserves that symmetry.

Extension

If g is odd, the parity of $f \circ g$ depends on f : an even f gives an even composition, while an odd f gives an odd composition.

Source lineage: Original synthesis / diagnosis · parity / composite function · reference IDs: openstax-v1-1.1-functions. See SOURCES.md for scope and adaptation boundaries; this note does not claim verbatim reproduction.

Advanced | **Parameter / synthesis / application** | **Synthesis** | **Original synthesis / diagnosis**

Q057 · A Piecewise Parking-Fee Model

Problem

A parking lot charges by continuous parking time t in hours. For $0 < t \leq 2$, the fee is 6 yuan. For $2 < t \leq 5$, it charges 6 yuan plus 3 yuan per hour for the portion beyond 2 hours. Write the piecewise formula $C(t)$, find its domain and range, and describe its monotonicity and boundedness.

Answer

$C(t) = \begin{cases} 6, & 0 < t \leq 2, \\ 3t, & 2 < t \leq 5. \end{cases}$ **The domain is** $(0, 5]$**, the range is** $[6, 15]$**,** C **is nondecreasing, and** $6 \leq C(t) \leq 15$.

Knowledge points

- constructing a piecewise model
- range of a piecewise function
- nondecreasing functions
- boundedness

Reading and method choice

The pricing rule changes at $t = 2$. The first piece is constant, and the second charges only for time beyond 2 hours. The two formulas should also be checked at the junction.

Detailed derivation

1. The permitted parking time is $0 < t \leq 5$, so the domain of C is $(0, 5]$.
2. For $0 < t \leq 2$, the fixed charge gives $C(t) = 6$.
3. For $2 < t \leq 5$, the extra time is $t - 2$, so $C(t) = 6 + 3(t - 2) = 3t$.
4. The first piece contributes the value 6; the second contributes $(6, 15]$, so the total range is $[6, 15]$.
5. C is constant on the first piece and increases on the second, with no decrease across $t = 2$; hence it is nondecreasing and satisfies $6 \leq C(t) \leq 15$.

Common pitfalls

- writing $6 + 3t$ and charging the first two hours twice
- calling the function strictly increasing even though the fee is constant for the first two hours

Verification

At $t = 2$ the charge is 6. The right-hand formula tends to $3 \cdot 2 = 6$ near the junction. At $t = 5$ it gives 15, matching $6 + 3 \times 3 = 15$.

Method takeaway

In modeling, identify the variable range and breakpoints, translate each rule, and then check junctions and global properties.

Extension

If every started half-hour were charged as a full half-hour, the model would become a step function rather than a linear continuous-time model.

Source lineage: Original synthesis / diagnosis · piecewise function / modeling · reference IDs: openstax-v1-1.1-functions. See SOURCES.md for scope and adaptation boundaries; this note does not claim verbatim reproduction.

Advanced Calculation Synthesis Original synthesis / diagnosis

Q058 · Rationalize a Difference of Radicals

Problem

Evaluate $\lim_{n\to\infty}[\sqrt{n^2+3n}-n]$.

Answer

$\frac{3}{2}$.

Knowledge points

- conjugate rationalization
- factoring n^2 from a radical
- limit laws

Reading and method choice

The original expression is a difference of two quantities that both grow without bound, so their limits cannot be subtracted. Multiplying by the conjugate removes the cancellation and permits normalization.

Detailed derivation

1. Multiply by the conjugate: $\sqrt{n^2+3n}-n=\frac{(n^2+3n)-n^2}{\sqrt{n^2+3n}+n}$.
2. The numerator simplifies to 3n, so the expression is $\frac{3n}{\sqrt{n^2+3n}+n}$.
3. Since $n>0$, factor n from the radical: $\sqrt{n^2+3n}=n\sqrt{1+\frac{3}{n}}$.
4. Cancel n to obtain $\frac{3}{\sqrt{1+\frac{3}{n}}+1}$.
5. As $n\to\infty$, $\frac{3}{n}\to 0$, so the denominator tends to 2 and the limit is $\frac{3}{2}$.

Common pitfalls

- subtracting the two infinite limits directly
- factoring n from $\sqrt{n^2+3n}$ without noting that n is positive

Verification

At $n=100$, the original expression is $\sqrt{10300}-100\approx 1.4889$, close to 1.5; the rationalized form gives the same value.

Method takeaway

When a difference of radicals contains strong cancellation, use the conjugate to turn it into a manageable quotient.

Extension

Replacing 3n by cn gives, by the same argument, $\lim[\sqrt{n^2+cn}-n]=\frac{c}{2}$.

Source lineage: Original synthesis / diagnosis · radical sequence limit / rationalization · reference IDs: openstax-v2-5.1-sequences. See SOURCES.md for scope and adaptation boundaries; this note does not claim verbatim reproduction.

Advanced | Proof | Synthesis | Original synthesis / diagnosis

Q059 · Eventual Positivity from a Positive Limit

Problem

Suppose $a_n \to a$ and $a > 0$. Prove that there exists a positive integer N such that $a_n > \frac{a}{2}$ whenever $n > N$.

Answer

Choose $\varepsilon = \frac{a}{2}$. Since $a_n \to a$, some N satisfies $|a_n - a| < \frac{a}{2}$ for $n > N$. Hence $a_n > a - \frac{a}{2} = \frac{a}{2}$.

Knowledge points

- definition of convergence
- eventual sign preservation
- extracting a one-sided bound from an absolute-value inequality

Reading and method choice

We want the tail to stay positive by a definite margin. Choosing the allowed error to be half of the positive limit places a_n inside $(\frac{a}{2}, \frac{3a}{2})$.

Detailed derivation

1. Because $a > 0$, the choice $\varepsilon = \frac{a}{2}$ is valid and positive.
2. From $a_n \to a$, there exists a positive integer N such that $|a_n - a| < \frac{a}{2}$ whenever $n > N$.
3. This absolute-value inequality is equivalent to $-\frac{a}{2} < a_n - a < \frac{a}{2}$.
4. Using its left half gives $a_n > a - \frac{a}{2} = \frac{a}{2}$ for every $n > N$.

Common pitfalls

- claiming eventual positivity without specifying a useful ε
- incorrectly concluding $a_n > a$ from $|a_n - a| < \frac{a}{2}$

Verification

The resulting neighborhood is $(a - \frac{a}{2}, a + \frac{a}{2}) = (\frac{a}{2}, \frac{3a}{2})$, whose left endpoint is strictly positive; the conclusion is stronger than mere eventual positivity.

Method takeaway

To preserve a nonzero sign, choose ε as a fixed fraction of the absolute value of the limit.

Extension

If $a < 0$, choosing $\varepsilon = \frac{|a|}{2}$ similarly proves that sufficiently large terms satisfy $a_n < \frac{a}{2} < 0$.

Source lineage: Original synthesis / diagnosis · properties of convergent sequences / eventual sign · reference IDs: openstax-v2-5.1-sequences. See SOURCES.md for scope and adaptation boundaries; this note does not claim verbatim reproduction.

Advanced Multiple choice Synthesis Original synthesis / diagnosis

Q060 · Necessary Consequences of a Known Limit

Problem

Suppose $a_n \to 2$. Which conclusions must be true?

A. $\{a_n\}$ is bounded

B. There exists N such that $a_n > 0$ whenever $n > N$

C. There exists N such that the tail a_n is strictly increasing

D. Every subsequence of $\{a_n\}$ converges to 2

Answer

A, B, D

Knowledge points

- boundedness of convergent sequences
- eventual sign from a positive limit
- limits of subsequences
- convergence does not imply monotonicity

Reading and method choice

Convergence keeps the tail close to 2, which ensures boundedness, eventual positivity, and the same limit for every subsequence. However, terms may still oscillate slightly around 2 and need not be monotone.

Detailed derivation

1. Every convergent sequence is bounded, so A must be true.
2. Choose $\varepsilon = 1$. For all sufficiently large n, $|a_n - 2| < 1$, hence $a_n > 1 > 0$; therefore B is true.
3. The sequence $a_n = 2 + \frac{(-1)^n}{n}$ converges to 2 but alternates above and below 2, so it is not eventually strictly increasing; C need not hold.
4. For any subsequence, its indices still tend to infinity and therefore satisfy the same tail estimates; every subsequence converges to 2, so D is true.

Common pitfalls

- interpreting 'closer and closer' as numerical monotonicity
- remembering boundedness but overlooking that every subsequence inherits the original limit

Verification

In the counterexample $2+\frac{(-1)^n}{n}$, even terms lie above 2, odd terms below 2, and the error $\frac{1}{n}$ tends to 0, directly disproving the necessity of C.

Method takeaway

Convergence controls distance to the limit, not the direction or monotone pattern of approach.

Extension

If any subsequence has a limit different from 2, the original sequence cannot converge to 2.

Source lineage: Original synthesis / diagnosis · convergent sequence properties / subsequences · reference IDs: openstax-v2-5.1-sequences. See SOURCES.md for scope and adaptation boundaries; this note does not claim verbatim reproduction.

Advanced | Proof | Synthesis | Original synthesis / diagnosis

Q061 · An Epsilon-Delta Proof for a Quadratic Limit

Problem

Use the epsilon-delta definition to prove $\lim_{x\to 2}(x^2+1)=5$.

Answer

Given $\varepsilon>0$, choose $\delta=\min\{1,\frac{\varepsilon}{5}\}$. If $0<|x-2|<\delta$, then $|x+2|<5$, and $|(x^2+1)-5|=|x-2||x+2|<5\delta\le\varepsilon$.

Knowledge points

- epsilon-delta definition
- use a fixed neighborhood to bound an extra factor
- product error estimate

Reading and method choice

The error factors as $|x-2||x+2|$. The first factor is directly controlled by δ, while the second varies with x and must first be bounded uniformly in a fixed neighborhood of 2.

Detailed derivation

1. Let $\varepsilon>0$ be arbitrary; the goal is to make $|(x^2+1)-5|<\varepsilon$.
2. Factor the error: $|(x^2+1)-5|=|x^2-4|=|x-2||x+2|$.
3. First require $|x-2|<1$. Then $1<x<3$, so $3<x+2<5$ and hence $|x+2|<5$.
4. To satisfy both the fixed-neighborhood bound and the accuracy requirement, choose $\delta=\min\{1,\frac{\varepsilon}{5}\}$.
5. If $0<|x-2|<\delta$, then $|(x^2+1)-5|<5|x-2|<5\delta\le\varepsilon$.
6. Thus an appropriate δ exists for every $\varepsilon>0$, and the definition yields $\lim_{x\to 2}(x^2+1)=5$.

Common pitfalls

- setting $\delta=\frac{\varepsilon}{|x+2|}$ and thereby allowing δ to depend on the varying x
- working backward only and failing to verify the chosen δ in the forward direction

Verification

For $\varepsilon=0.1$, choose $\delta=\min\{1,0.02\}=0.02$. If $|x-2|<0.02$, then $|x+2|<4.02<5$ and the error is below $5\times 0.02=0.1$.

Method takeaway

In polynomial epsilon-delta proofs, first use a fixed small neighborhood to bound extra factors, then use δ to meet the requested accuracy.

Extension

A sharper estimate can use $|x+2|<4+\delta$, but the minimum construction is clearer and fully sufficient.

Source lineage: Original synthesis / diagnosis · epsilon-delta proof / definition of a function limit · reference IDs: openstax-v1-2.2-function-limits, openstax-v1-2.5-precise-limit. See SOURCES.md for scope and adaptation boundaries; this note does not claim verbatim reproduction.

Advanced | Fill in the blank | Synthesis | Original synthesis / diagnosis

Q062 · Opposite Signs Near a Vertical Blow-Up

Problem

Fill in the blanks: $\lim_{x\to 2^-}\frac{1}{x-2}=$ ____, $\lim_{x\to 2^+}\frac{1}{x-2}=$ ____, and therefore $\lim_{x\to 2}\frac{1}{x-2}$ ____.

Answer

$-\infty$; $+\infty$; does not exist.

Knowledge points

- infinite limits
- sign of the denominator from each side
- criterion for a two-sided limit

Reading and method choice

The denominator tends to zero in magnitude on both sides, but its sign differs: it is a small negative quantity on the left and a small positive quantity on the right. Their reciprocals diverge in opposite directions.

Detailed derivation

1. As $x\to 2^-, x-2<0$ and $|x-2|\to 0$, so $\frac{1}{x-2}$ is negative with increasing magnitude.
2. Thus $\lim_{x\to 2^-}\frac{1}{x-2}=-\infty$.
3. As $x\to 2^+, x-2>0$ and tends to zero, so its reciprocal becomes arbitrarily large and positive; the right-hand limit is $+\infty$.
4. The one-sided infinite limits have opposite signs and cannot form one two-sided limit, so the two-sided limit does not exist.

Common pitfalls

- writing simply ∞ whenever a denominator tends to zero
- treating $-\infty$ and $+\infty$ as equal because both are unbounded

Verification

At $x=1.99$ the value is -100, while at $x=2.01$ it is 100. Closer inputs increase the magnitudes but retain opposite signs.

Method takeaway

For reciprocal blow-ups, inspect both the direction toward zero and the sign of the denominator.

Extension

For $\frac{1}{(x-2)^2}$ the denominator is positive on both sides, so both one-sided limits are $+\infty$ and the two-sided infinite limit is $+\infty$.

Source lineage: Original synthesis / diagnosis · infinite limit / one-sided limits · reference IDs: openstax-v1-2.2-function-limits. See SOURCES.md for scope and adaptation boundaries; this note does not claim verbatim reproduction.

Advanced Calculation Synthesis Original synthesis / diagnosis

Q063 · Limit of a Difference of Two Radicals

Problem

Evaluate $\lim_{x\to 0} \frac{\sqrt{1+x}-\sqrt{1-x}}{x}$.

Answer

1.

Knowledge points

- conjugate rationalization
- cancellation in a deleted neighborhood
- limit laws

Reading and method choice

Direct substitution gives $\frac{0}{0}$. Because the numerator is a difference of radicals, multiplying by the conjugate sum changes the numerator to 2x, which can then be canceled.

Detailed derivation

1. Multiply by $\frac{\sqrt{1+x}+\sqrt{1-x}}{\sqrt{1+x}+\sqrt{1-x}}$.
2. The numerator becomes $(1+x)-(1-x)=2x$.
3. For $x \neq 0$ in the deleted neighborhood, cancel x to obtain $\frac{2}{\sqrt{1+x}+\sqrt{1-x}}$.
4. As $x \to 0$, both radicals tend to 1, so the denominator tends to 2 and the limit is $\frac{2}{2}=1$.

Common pitfalls

- splitting the difference into separate limits and remaining stuck at $\frac{0}{0}$
- canceling x without noting that a limit uses $x \neq 0$ in a deleted neighborhood

Verification

At $x=0.01$, the original quotient is approximately $\frac{\sqrt{1.01}-\sqrt{0.99}}{0.01} \approx 1.0000$, consistent with 1.

Method takeaway

When a difference of radicals gives $\frac{0}{0}$, conjugate rationalization often exposes the factor that can be canceled.

Extension

Replacing $1 \pm x$ by $1 \pm ax$ gives, by the same method, the limit a.

Source lineage: Original synthesis / diagnosis · radical limit / rationalization · reference IDs: openstax-v1-2.2-function-limits. See SOURCES.md for scope and adaptation boundaries; this note does not claim verbatim reproduction.

Advanced Calculation Synthesis Original synthesis / diagnosis

Q064 · Parameter Classification with an Oscillatory Factor

Problem

Let $f_p(x) = |x|^p \sin(\frac{1}{x})$ for $x \neq 0$. For $p > 0, p = 0$, and $p < 0$, decide whether f_p is infinitesimal, bounded with no limit, or unbounded without tending to $+\infty$ or $-\infty$ as $x \to 0$.

Answer

For $p > 0, f_p(x) \to 0$ and is infinitesimal. For $p = 0$, it is bounded but has no limit. For $p < 0$, it is unbounded and tends to neither $+\infty$ nor $-\infty$.

Knowledge points

- Bounded oscillatory factors
- Squeeze theorem
- Construction of approaching sequences
- Unboundedness versus an infinite limit

Reading and method choice

The key bound is $|\sin(\frac{1}{x})| \leq 1$. For $p > 0$ the power suppresses oscillation; for $p \leq 0$, choose sequences on which the sine equals $1, -1$, or 0.

Detailed derivation

1. If $p > 0$, then $|f_p(x)| \leq |x|^p$ and $|x|^p \to 0$; the squeeze theorem gives $f_p(x) \to 0$.
2. If $p = 0$, then $f_0(x) = \sin(\frac{1}{x})$, which always lies in $[-1, 1]$, so it is bounded.
3. Let $x_n = \frac{1}{\frac{\pi}{2}+2\pi n}$ and $y_n = \frac{1}{\frac{3\pi}{2}+2\pi n}$. Both tend to 0, but $f_0(x_n) = 1$ and $f_0(y_n) = -1$, so no limit exists.
4. If $p < 0$, then $f_p(x_n) = |x_n|^p \to +\infty$ while $f_p(y_n) = -|y_n|^p \to -\infty$, so the function is unbounded.
5. Also let $z_n = \frac{1}{\pi n}$; then $f_p(z_n) = 0$. Hence the whole function tends to neither $+\infty$ nor $-\infty$.

Common pitfalls

- Ignoring the zeros and sign changes of $\sin(\frac{1}{x})$
- Equating unboundedness with an infinite limit
- Using one sequence as if it established a full function limit

Verification

All three sequences approach 0 while forcing the sine to be $1, -1$, and 0. This confirms oscillation for $p \leq 0$, whereas the absolute bound tends directly to 0 for $p > 0$.

Method takeaway

A bounded factor times an infinitesimal tends to 0; an unbounded factor times an oscillatory term requires sequence-by-sequence analysis.

Extension

Replacing $|x|^p$ by a positive $a(x)$: if $a(x) \to 0$, the product tends to 0; if $a(x) \to \infty$, sequences usually decide whether the result is merely unbounded rather than infinite.

Source lineage: Original synthesis / diagnosis · parameter classification / oscillation · reference IDs: openstax-v1-2.2-function-limits, openstax-v1-2.5-precise-limit. See SOURCES.md for scope and adaptation boundaries; this note does not claim verbatim reproduction.

Advanced | True/false with justification | Synthesis | Original synthesis / diagnosis

Q065 · Does Local Unboundedness Imply an Infinite Limit?

Problem

True or false, with justification: If f is unbounded in every deleted neighborhood of $x = 0$, then $\lim_{x\to 0} |f(x)| = +\infty$.

Answer

False. Let $f(x) = \frac{1}{|x|}$ for rational x and $f(x) = 0$ for irrational x. It is unbounded in every deleted neighborhood, but remains 0 along irrational points approaching 0.

Knowledge points

- Local unboundedness
- Definition of divergence to infinity
- Dense-set counterexamples
- Necessary versus sufficient conditions

Reading and method choice

Local unboundedness only asks for some arbitrarily large values in every neighborhood. A limit of $+\infty$ requires every sufficiently nearby value to exceed each threshold, which is much stronger.

Detailed derivation

1. Define $f(x) = \frac{1}{|x|}$ for rational x and $f(x) = 0$ for irrational x, with $x \neq 0$.
2. Given $\delta > 0$ and $M > 0$, choose a rational x with $0 < |x| < \min(\delta, \frac{1}{M})$; then $f(x) = \frac{1}{|x|} > M$, so f is unbounded in every deleted neighborhood.
3. If $|f(x)| \to +\infty$, then for $M = 1$ there must be $\delta > 0$ such that every x with $0 < |x| < \delta$ satisfies $|f(x)| > 1$.
4. Every such neighborhood contains an irrational y, but $f(y) = 0 \leq 1$, so the universal condition fails.
5. Hence local unboundedness does not imply an infinite absolute-value limit, and the statement is false.

Common pitfalls

- Confusing the existence of a large value with all values being large
- Looking only along rational points
- Giving a counterexample without proving unboundedness in every neighborhood

Verification

Along rational $x_n = \frac{1}{n}, f(x_n) = n \to \infty$; along any irrational $y_n \to 0, f(y_n) = 0$. These two behaviors rule out $|f(x)| \to \infty$.

Method takeaway

Unboundedness is existential, while an infinite limit is an eventual universal statement.

Extension

If $|f(x)| \to \infty$, then f is locally unbounded; the converse is generally false.

Source lineage: Original synthesis / diagnosis · unbounded function / infinite limit · reference IDs: openstax-v1-2.2-function-limits, openstax-v1-2.5-precise-limit. See SOURCES.md for scope and adaptation boundaries; this note does not claim verbatim reproduction.

Advanced Calculation Synthesis Original synthesis / diagnosis

Q066 · Double Rationalization of Nested Radicals

Problem

Evaluate $\lim_{x\to 0}\frac{\sqrt{1+\sqrt{1+x}}-\sqrt{2}}{x}$ without derivatives or Taylor expansions.

Answer

$\frac{1}{4\sqrt{2}}$

Knowledge points

- Two successive conjugate rationalizations
- $\frac{0}{0}$ limits
- Product and quotient limit laws

Reading and method choice

Rationalizing the outer radical still leaves $\sqrt{1+x}-1$, so a second rationalization is needed to expose the factor x.

Detailed derivation

1. Call the limit L and multiply first by the outer conjugate $\sqrt{1+\sqrt{1+x}}+\sqrt{2}$.
2. The numerator becomes $\sqrt{1+x}-1$, so L equals $\frac{\sqrt{1+x}-1}{x[\sqrt{1+\sqrt{1+x}}+\sqrt{2}]}$.
3. Rationalize $\sqrt{1+x}-1$ with $\sqrt{1+x}+1$, obtaining $\frac{x}{\sqrt{1+x}+1}$.
4. Cancel x to get $L=\frac{1}{[\sqrt{1+x}+1][\sqrt{1+\sqrt{1+x}}+\sqrt{2}]}$.
5. As $x\to 0$, the brackets tend to 2 and $2\sqrt{2}$, so $L=\frac{1}{4\sqrt{2}}$.

Common pitfalls

- Stopping after only one rationalization
- Losing the term $\sqrt{1+x}-1$
- Multiplying $2\cdot 2\sqrt{2}$ incorrectly

Verification

At $x = 0.001$ the original expression is about 0.17672, while $\frac{1}{4\sqrt{2}} \approx 0.17678.$

Method takeaway

Nested radicals often require layer-by-layer rationalization aimed at exposing a linear factor.

Extension

The same strategy can be repeated for deeper nests, while tracking the limit of each conjugate factor.

Source lineage: Original synthesis / diagnosis · double rationalization / nested radicals · reference IDs: openstax-v1-2.3-limit-laws. See SOURCES.md for scope and adaptation boundaries; this note does not claim verbatim reproduction.

Advanced Calculation Synthesis Original synthesis / diagnosis

Q067 · Determining Parameters from a Finite Limit

Problem

Find real numbers a,b such that $\lim_{x\to 1} \frac{x^2+ax+b}{x-1} = 3$.

Answer

$a = 1$ and $b = -2$.

Knowledge points

- Necessary numerator cancellation for a finite limit
- Polynomial factorization
- Parameter equations

Reading and method choice

The denominator tends to 0. A finite quotient limit forces the numerator to tend to 0 first; then match the limit after cancellation.

Detailed derivation

1. A finite limit requires the numerator at $x = 1$ to vanish, so $1 + a + b = 0$.
2. Using $b = -1 - a$, rewrite the numerator as $x^2 + ax - 1 - a$.
3. Factor $x^2 + ax - 1 - a = (x - 1)(x + a + 1)$.
4. For $x \neq 1$ the quotient is $x + a + 1$, whose limit at 1 is $a + 2$.
5. Set $a + 2 = 3$ to get $a = 1$, then $1 + a + b = 0$ gives $b = -2$.

Common pitfalls

- Stopping after $1 + a + b = 0$
- Treating the necessary cancellation as sufficient by itself
- Making a sign error in factorization

Verification

With $a = 1, b = -2$, the numerator is $x^2 + x - 2 = (x - 1)(x + 2)$; the quotient tends to 3.

Method takeaway

Parameter-dependent $\frac{0}{0}$ limits usually require two stages: cancel the constant mismatch, then match the reduced limit.

Extension

If the target is L, the same work gives $a = L - 2$ and $b = 1 - L$.

Source lineage: Original synthesis / diagnosis · parameter limit / removable zero factor · reference IDs: openstax-v1-2.3-limit-laws. See SOURCES.md for scope and adaptation boundaries; this note does not claim verbatim reproduction.

Advanced | Calculation | Synthesis | Original synthesis / diagnosis

Q068 · Using a Known Ratio of Deviations

Problem

As $x \to a$, suppose $f(x) \to 1, g(x) \to 2$, and $\frac{f(x)-1}{g(x)-2} \to 3$. Find $\lim_{x\to a} \frac{f(x)^2-1}{g(x)^2-4}$.

Answer

$\frac{3}{2}$

Knowledge points

- Difference-of-squares factorization
- Product limit law
- Preserving a known ratio

Reading and method choice

Do not take the two squared differences separately, because that remains $\frac{0}{0}$. Factor them and preserve the given ratio as one factor.

Detailed derivation

1. Factor $f^2 - 1 = (f-1)(f+1)$ and $g^2 - 4 = (g-2)(g+2)$.
2. In the relevant deleted neighborhood, write the quotient as $[\frac{f-1}{g-2}] \cdot [\frac{f+1}{g+2}]$.
3. The first factor tends to 3 by hypothesis.
4. The second tends to $\frac{1+1}{2+2} = \frac{2}{4} = \frac{1}{2}$, with denominator limit $4 \neq 0$.
5. The product law yields $\frac{3\cdot1}{2} = \frac{3}{2}$.

Common pitfalls

- Calling the $\frac{0}{0}$ squared-difference quotient zero
- Dropping $f+1$ or $g+2$
- Inverting the given ratio

Verification

Set $g-2=t$ and $f-1=3t$. Then the quotient is $\frac{6t+9t^2}{4t+t^2} \to \frac{6}{4} = \frac{3}{2}$.

Method takeaway

When a ratio of deviations is given, factor expressions so that the known ratio remains intact.

Extension

The same idea handles $\frac{f^n-1}{g^m-2^m}$ using difference-of-powers identities.

Source lineage: Original synthesis / diagnosis · known ratio limit / factorization · reference IDs: openstax-v1-2.3-limit-laws. See SOURCES.md for scope and adaptation boundaries; this note does not claim verbatim reproduction.

Advanced Proof Synthesis Original synthesis / diagnosis

Q069 · Proving the Product Limit Theorem from Definitions

Problem

Suppose $f(x) \to A$ and $g(x) \to B$ as $x \to a$. Using only the limit definition and local boundedness of convergent functions, prove that $f(x)g(x) \to AB$.

Answer

Use $f(x)g(x) - AB = f(x)[g(x) - B] + B[f(x) - A]$ and control the two terms separately.

Knowledge points

- Product limit theorem
- Local boundedness of convergent functions
- Error decomposition
- Epsilon-delta proof

Reading and method choice

The decomposition $f(g - B) + B(f - A)$ needs only a uniform bound on f, followed by a split of the epsilon budget.

Detailed derivation

1. Since $f(x) \to A$, choose a deleted neighborhood where $|f(x) - A| < 1$; then $|f(x)| \le |A| + 1$. Set $M = |A| + 1 > 0$.
2. Given $\varepsilon > 0, g(x) \to B$ gives a neighborhood where $|g(x) - B| < \frac{\varepsilon}{2M}$.
3. Also $f(x) \to A$ gives a neighborhood where $|f(x) - A| < \frac{\varepsilon}{2(|B|+1)}$.
4. Take the minimum of the three neighborhood radii so that all estimates hold simultaneously.
5. Using $fg - AB = f(g - B) + B(f - A)$, obtain $|fg - AB| \le \frac{M \cdot \varepsilon}{2M} + \frac{|B| \cdot \varepsilon}{2(|B|+1)} < \varepsilon$.
6. Therefore $\lim_{x \to a} f(x)g(x) = AB$ by definition.

Common pitfalls

- Assuming boundedness without deriving it
- Dividing by $|B|$ when $B = 0$
- Failing to take the minimum of several deltas
- Dropping a term in the error decomposition

Verification

Each error contribution is below $\frac{\varepsilon}{2}$, and the denominators M and $|B| + 1$ are always positive.

Method takeaway

Definition proofs of limit laws often first establish a local bound and then split the total error into controllable pieces.

Extension

Repeated application proves that the limit of any finite product equals the product of the limits.

Source lineage: Original synthesis / diagnosis · product limit theorem / local boundedness · reference IDs: openstax-v1-2.3-limit-laws. See SOURCES.md for scope and adaptation boundaries; this note does not claim verbatim reproduction.

Advanced Calculation Synthesis Original synthesis / diagnosis

Q070 · A Ratio of Sines with Different Scales

Problem

Evaluate $\lim_{x\to 0} \frac{\sin(3x)}{\sin(5x)}$.

Answer

$\frac{3}{5}$

Knowledge points

- First fundamental limit
- Multiplicative decomposition
- Reciprocal of a nonzero limit

Reading and method choice

Create $\frac{\sin u}{u}$ separately in numerator and denominator, retaining the scale ratio $\frac{3}{5}$.

Detailed derivation

1. Write $\frac{\sin(3x)}{\sin(5x)} = [\frac{\sin(3x)}{3x}] \cdot (\frac{3}{5}) \cdot [\frac{5x}{\sin(5x)}]$.
2. As $x \to 0$, both 3x and 5x tend to 0.
3. The first bracket tends to 1.
4. Since $\frac{\sin(5x)}{5x} \to 1$, its reciprocal $\frac{5x}{\sin(5x)}$ also tends to 1.
5. Therefore the limit is $1 \cdot (\frac{3}{5}) \cdot 1 = \frac{3}{5}$.

Common pitfalls

- Replacing $\frac{\sin(3x)}{\sin(5x)}$ by $\sin(\frac{3}{5})$
- Dropping the scale ratio
- Taking a reciprocal without a nonzero limit

Verification

At $x = 0.001$, $\frac{\sin 0.003}{\sin 0.005} \approx 0.6000016$, close to 0.6.

Method takeaway

The limiting ratio of small-angle sines equals the ratio of their linear scales.

Extension

For $a, b \neq 0$, $\lim_{x\to 0} \frac{\sin(ax)}{\sin(bx)} = \frac{a}{b}$.

Source lineage: Original synthesis / diagnosis · first fundamental limit / ratio of sines · reference IDs: openstax-v1-2.3-limit-laws, mit-18.01sc-session-8-trig-limits. See SOURCES.md for scope and adaptation boundaries; this note does not claim verbatim reproduction.

Advanced Calculation Synthesis Original synthesis / diagnosis

Q071 · The Second Fundamental Limit with a Fractional Base

Problem

Evaluate $\lim_{x\to0}[\frac{1+x}{1-x}]^{\frac{1}{x}}$.

Answer

e^2

Knowledge points

- The 1^∞ form
- Second fundamental limit
- Rewriting the increment

Reading and method choice

Write the fractional base as 1 plus $t = \frac{2x}{1-x}$, then compute the limit of t times the original exponent $\frac{1}{x}$.

Detailed derivation

1. $\frac{1+x}{1-x} = 1 + \frac{2x}{1-x}$. Set $t = \frac{2x}{1-x}$, so $t \to 0$.
2. Rewrite the expression as $(1+t)^{\frac{1}{x}} = [(1+t)^{\frac{1}{t}}]^{\frac{t}{x}}$.
3. The second fundamental limit gives $(1+t)^{\frac{1}{t}} \to e$.
4. Also $\frac{t}{x} = \frac{2}{1-x} \to 2$.
5. Therefore the full limit is e^2.

Common pitfalls

- Replacing the increment by 2x exactly
- Concluding 1 because the base tends to 1
- Dropping $1-x$ when computing $\frac{t}{x}$

Verification

At $x = 0.001$, $[\frac{1.001}{0.999}]^{1000} \approx 7.38906$, close to e^2.

Method takeaway

For a complicated 1^{∞} form, write the base as $1 + t$ and find the limit of t times the original exponent.

Extension

The same method gives $\lim_{x\to 0}[\frac{1+ax}{1+bx}]^{\frac{1}{x}} = e^{a-b}$.

Source lineage: Original synthesis / diagnosis · second fundamental limit / fractional base · reference IDs: openstax-v1-2.3-limit-laws, mit-18.01sc-session-19-limit-involving-e. See SOURCES.md for scope and adaptation boundaries; this note does not claim verbatim reproduction.

Advanced Calculation Synthesis Original synthesis / diagnosis

Q072 · Determining Parameters from an Exponential Limit

Problem

Let a,b be positive real numbers. Given $\lim_{x\to 0}(1+ax)^{\frac{b}{x}}=e^6$ and $a+b=5$, find all ordered pairs (a,b).

Answer

$(a,b)=(2,3)$ or $(3,2)$.

Knowledge points

- Parameterized second fundamental limit
- Injectivity of the exponential
- Quadratic equation

Reading and method choice

Standardization gives the limit e^{ab}. Comparing with e^6 yields $ab=6$, which combines with $a+b=5$.

Detailed derivation

1. Set $u=ax$; then $(1+ax)^{\frac{b}{x}}=[(1+u)^{\frac{1}{u}}]^{ab}$.
2. As $x\to 0, u\to 0$, so the limit is e^{ab}.
3. Since $e^{ab}=e^6$ and the exponential is one-to-one, $ab=6$.
4. Together with $a+b=5, a$ and b are the two positive roots of $t^2-5t+6=0$.
5. Factoring gives $(t-2)(t-3)=0$, hence $(a,b)=(2,3)$ or $(3,2)$.

Common pitfalls

- Using $\frac{b}{a}$ instead of ab
- Giving only one ordered pair
- Ignoring positivity and local positivity of the base

Verification

Both pairs have product 6 and sum 5, and substitution into the general limit gives e^6.

Method takeaway

For a parameterized 1^∞ limit, extract the increment-exponent product and convert the result into algebraic conditions.

Extension

Without the sum condition, every positive pair satisfying $ab = 6$ would work.

Source lineage: Original synthesis / diagnosis · parameter limit / second fundamental limit · reference IDs: openstax-v1-2.3-limit-laws, mit-18.01sc-session-19-limit-involving-e. See SOURCES.md for scope and adaptation boundaries; this note does not claim verbatim reproduction.

Advanced Proof Synthesis Original synthesis / diagnosis

Q073 · Squeezing the Floor-Function Error

Problem

Use the squeeze theorem to prove $\lim_{x\to+\infty}\frac{\lfloor x\rfloor}{x}=1$.

Answer

From $x-1<\lfloor x\rfloor\le x$ and $x>0$, obtain $1-\frac{1}{x}<\frac{\lfloor x\rfloor}{x}\le 1$; both bounds tend to 1.

Knowledge points

- Basic floor-function inequalities
- Squeeze at infinity
- Division by a positive quantity

Reading and method choice

The error between $\lfloor x\rfloor$ and x is less than 1. Dividing this bounded absolute error by the growing scale x makes the relative error vanish.

Detailed derivation

1. By definition, $\lfloor x\rfloor\le x<\lfloor x\rfloor+1$ for every real x.
2. The right inequality gives $x-1<\lfloor x\rfloor$.
3. For $x\to+\infty$ we eventually have $x>0$; divide $x-1<\lfloor x\rfloor\le x$ by x to get $1-\frac{1}{x}<\frac{\lfloor x\rfloor}{x}\le 1$.
4. The lower and upper bounds both tend to 1.
5. The squeeze theorem yields $\frac{\lfloor x\rfloor}{x}\to 1$.

Common pitfalls

- Writing $\lfloor x\rfloor=x-1$ identically
- Dividing before confirming $x>0$
- Mentioning a bounded error without converting it to a relative error

Verification

The proof gives $0\le 1-\frac{\lfloor x\rfloor}{x}<\frac{1}{x}$, whose right side explicitly tends to 0.

Method takeaway

A uniformly bounded absolute error becomes a vanishing relative error after division by a scale tending to infinity.

Extension

Similarly, $\lim_{n\to\infty} \frac{\lfloor \alpha n \rfloor}{n} = \alpha$ for every $\alpha > 0$.

Source lineage: Original synthesis / diagnosis · squeeze theorem / floor function · reference IDs: openstax-v1-2.3-limit-laws. See SOURCES.md for scope and adaptation boundaries; this note does not claim verbatim reproduction.

Advanced Proof Synthesis Original synthesis / diagnosis

Q074 · A Monotone-Bounded Proof for a Linear Recurrence

Problem

Let $a_1 > 2$ and $a_{n+1} = \frac{a_n+2}{3}$. Prove that $\{a_n\}$ converges and find its limit, without first deriving a closed form.

Answer

$\{a_n\}$ is decreasing and bounded below by 1, hence convergent; its limit is 1.

Knowledge points

- Mathematical induction
- Monotone convergence theorem
- Limit equation for a recurrence

Reading and method choice

First prove $a_n > 1$ to obtain a lower bound and determine the sign of $a_{n+1} - a_n$. Then apply monotone convergence and only afterward pass to the limit.

Detailed derivation

1. Since $a_1 > 2 > 1$, and $a_n > 1$ implies $a_{n+1} = \frac{a_n+2}{3} > \frac{1+2}{3} = 1$, induction gives $a_n > 1$ for all n.
2. Compute $a_{n+1} - a_n = \frac{a_n+2-3a_n}{3} = \frac{2(1-a_n)}{3} < 0.$
3. Thus $\{a_n\}$ is decreasing and bounded below by 1.
4. The monotone convergence theorem guarantees convergence. Let the limit be L.
5. Passing to the limit in the recurrence gives $L = \frac{L+2}{3}$, so $3L = L + 2$ and $L = 1$.
6. Therefore the sequence converges to 1.

Common pitfalls

- Declaring the difference negative before proving $a_n > 1$
- Solving a limit equation before proving existence
- Calling the lower bound 1 an attained minimum

Verification

For $a_1 = 4$, the first terms are $4, 2, \frac{4}{3}, \frac{10}{9}$; they remain above 1 and decrease toward 1.

Method takeaway

For a recursive sequence: preserve a bound, determine monotonicity, prove convergence, then solve the limit equation.

Extension

Indeed, $a_{n+1} - 1 = \frac{a_n - 1}{3}$, so the error shrinks by a factor of $\frac{1}{3}$.

Source lineage: Original synthesis / diagnosis · monotone convergence theorem / recursive sequence · reference IDs: openstax-v1-2.3-limit-laws, openstax-v2-5.1-sequences. See SOURCES.md for scope and adaptation boundaries; this note does not claim verbatim reproduction.

Advanced Calculation Synthesis Original synthesis / diagnosis

Q075 · A Mixed Exponential-Logarithmic-Trigonometric Limit

Problem

Evaluate $\lim_{x\to 0} \frac{(e^{2x}-1)\tan(3x)}{x\ln(1+4x)}$.

Answer

$\frac{3}{2}$

Knowledge points

- $e^u - 1 \sim u$
- $\tan u \sim u$
- $\ln(1+u) \sim u$
- Tracking constants

Reading and method choice

All four vanishing factors occur in a product-quotient structure, so factorwise substitution is valid. The challenge is preserving the scales 2, 3, and 4.

Detailed derivation

1. $e^{2x} - 1 \sim 2x$.
2. $\tan(3x) \sim 3x$.
3. $\ln(1+4x) \sim 4x$.
4. Thus the expression is equivalent to $\frac{(2x)(3x)}{x(4x)}$.
5. Cancel x^2 to obtain $\frac{6}{4} = \frac{3}{2}$.

Common pitfalls

- Replacing every composite function by x
- Dropping the external x in the denominator
- Canceling an internal x before forming valid ratios

Verification

Factor the expression into three standard ratios and the constant $\frac{3}{2}$; every standard ratio tends to 1.

Method takeaway

For mixed equivalences, pair factors one by one and keep a clear ledger of scale constants.

Extension

Replacing 2,3,4 by a,b,c, with $c \neq 0$, gives the limit $\frac{ab}{c}$.

Source lineage: Original synthesis / diagnosis · mixed equivalent infinitesimals / exponential logarithmic trigonometric · reference IDs: openstax-v1-2.3-limit-laws, mit-18.01sc-session-8-trig-limits, mit-18.01sc-session-19-limit-involving-e. See SOURCES.md for scope and adaptation boundaries; this note does not claim verbatim reproduction.

Advanced Calculation Synthesis Original synthesis / diagnosis

Q076 · Determining Parameters from an Equivalence

Problem

Let $a, b > 0$. If $e^{ax} - 1 \sim b\sin(2x)$ as $x \to 0$ and $a + b = 3$, find a and b.

Answer

$a = 2$ **and** $b = 1$.

Knowledge points

- Ratio definition of equivalence
- Composite scaling
- Parameter equations

Reading and method choice

Reduce both sides to their first-order principal quantities: $e^{ax} - 1 \sim ax$ and $b\sin 2x \sim$ 2bx. Equivalence requires equal leading coefficients.

Detailed derivation

1. From $e^{u} - 1 \sim u$, obtain $e^{ax} - 1 \sim ax$.
2. From $\sin u \sim u$, obtain $b\sin(2x) \sim$ 2bx.
3. Their ratio must tend to 1, so $\frac{a}{2b} = 1$ and $a = 2b$.
4. Combine this with $a + b = 3$ to get $2b + b = 3$.
5. Hence $b = 1$ and $a = 2$.

Common pitfalls

- Using b instead of 2b for $b\sin 2x$
- Stopping at $a = 2b$
- Confusing equivalence with merely having a finite nonzero ratio

Verification

For $a = 2, b = 1, \frac{e^{2x}-1}{\sin 2x} = [\frac{e^{2x}-1}{2x}] \cdot [\frac{2x}{\sin 2x}] \to 1.$

Method takeaway

Parameterized equivalence problems reduce to matching the coefficients of the lowest-order nonzero principal quantities.

Extension

For merely the same order, $\frac{a}{2b}$ need only be finite and nonzero, not necessarily 1.

Source lineage: Original synthesis / diagnosis · parameterized equivalence / exponential function · reference IDs: openstax-v1-2.3-limit-laws, mit-18.01sc-session-19-limit-involving-e. See SOURCES.md for scope and adaptation boundaries; this note does not claim verbatim reproduction.

Advanced Calculation Synthesis Original synthesis / diagnosis

Q077 · Checking a First-Order Remainder in a Difference

Problem

Without Taylor expansions, evaluate $\lim_{x\to 0} \frac{\sqrt{1+x}-1-\frac{x}{2}}{x}$.

Answer

0

Knowledge points

- Rationalization
- Care with difference structures
- Remainder after an infinitesimal equivalence

Reading and method choice

One should not merely replace $\sqrt{1+x}-1$ by $\frac{x}{2}$ inside a difference. Instead, simplify the full quotient exactly.

Detailed derivation

1. Split the expression as $\frac{\sqrt{1+x}-1}{x} - \frac{1}{2}$.
2. Rationalize the first term: $\frac{\sqrt{1+x}-1}{x} = \frac{1}{\sqrt{1+x}+1}$ for $x \neq 0$.
3. Thus the expression is $\frac{1}{\sqrt{1+x}+1} - \frac{1}{2}$.
4. As $x \to 0$, the first term tends to $\frac{1}{1+1} = \frac{1}{2}$.
5. The requested limit is $\frac{1}{2} - \frac{1}{2} = 0$.

Common pitfalls

- Substituting inside a difference without justification
- Dropping $+1$ after rationalization
- Misreading the denominator as x^2

Verification

Exact combination gives $\frac{1-\sqrt{1+x}}{2(\sqrt{1+x}+1)}$, whose numerator tends to 0 and denominator to 4.

Method takeaway

An equivalence may suggest the answer, but differences should be handled by exact identities or regrouping into known ratios.

Extension

With x^2 in the denominator the limit is not 0; higher-order information is needed, as shown in the comprehensive problem.

Source lineage: Original synthesis / diagnosis · difference structure / rationalization · reference IDs: openstax-v1-2.3-limit-laws. See SOURCES.md for scope and adaptation boundaries; this note does not claim verbatim reproduction.

Advanced Calculation Synthesis Open-text method adaptation

Q078 · Three Types of Discontinuity in One Function

Problem

Let $f(x) = -1$ for $x < 0$; $f(x) = \frac{x^2-1}{x-1}$ for $0 \leq x < 2$ with $x \neq 1$; and $f(x) = \frac{1}{x-2}$ for $x > 2$. Find and classify every discontinuity. If one is removable, give the value for a continuous extension.

Answer

$x = 0$ is a jump discontinuity; $x = 1$ is removable with continuous-extension value 2; $x = 2$ is an infinite discontinuity of the second kind.

Knowledge points

- junctions of a piecewise function
- canceling a common factor
- one-sided limits
- first- and second-kind discontinuities

Reading and method choice

The only candidates are the formula junctions $0, 2$ and the zero 1 of the denominator. Every other point lies in the interior of a continuous piece. We classify each candidate from its two one-sided limits.

Detailed derivation

1. For $0 \leq x < 2$ and $x \neq 1$, $\frac{x^2-1}{x-1} = x + 1$. Thus each piece is continuous away from $0, 1, 2$.
2. At $x = 0$, $\lim_{x\to 0^-} f(x) = -1$ while $\lim_{x\to 0^+} f(x) = 1$. The finite one-sided limits are unequal, so 0 is a jump discontinuity.
3. At $x = 1$, both sides use the simplified expression $x + 1$, giving $\lim_{x\to 1} f(x) = 2$. Since $f(1)$ is undefined, 1 is removable.
4. The unique continuous extension at $x = 1$ is obtained by defining $f(1) = 2$.
5. At $x = 2$, $\lim_{x\to 2^-} f(x) = 3$, whereas $\lim_{x\to 2^+} f(x) = +\infty$.
6. Because at least one one-sided limit is not finite, $x = 2$ is an infinite discontinuity of the second kind.
7. Therefore the complete list is $0, 1, 2$, with the classifications stated in the answer.

Common pitfalls

- assuming cancellation makes the original function defined at $x = 1$
- looking only at point values instead of one-sided limits
- misclassifying $x = 2$ as a jump because its left-hand limit is finite

Verification

Every point outside $\{0, 1, 2\}$ is internal to one continuous formula. The one-sided limits at the three candidates exhibit exactly the jump, removable, and infinite patterns, so the list is complete.

Method takeaway

First locate junctions, undefined points, and dangerous points of elementary functions; then classify them with one-sided limits.

Extension

If the third piece is replaced by $\frac{1}{(x-2)^2}$, the right-hand limit as $x \to 2^+$ is still $+\infty$, while the left-hand limit from the middle piece remains 3; thus $x = 2$ is still an infinite discontinuity of the second kind.

Source lineage: Open-text method adaptation · locating discontinuities / jump discontinuity · reference IDs: openstax-v1-2.4-continuity, mit-18.01sc-session-5-discontinuity. See SOURCES.md for scope and adaptation boundaries; this note does not claim verbatim reproduction.

Advanced | Error diagnosis | Synthesis | Original synthesis / diagnosis

Q079 · Must the Sum of Two Discontinuous Functions Be Discontinuous?

Problem

A student claims: "If both f and g are discontinuous at $x = 0$, then $f + g$ must be discontinuous at 0, because only sums of continuous functions are continuous." Identify the logical error and construct and verify a counterexample.

Answer

The claim is false; the theorem "f, g continuous implies $f + g$ continuous" cannot be reversed. Let $f(0) = 0$ and $f(x) = 1$ for $x \neq 0$; let $g(0) = 0$ and $g(x) = -1$ for $x \neq 0$. Both are discontinuous at 0, but $f + g \equiv 0$ is continuous.

Knowledge points

- sufficient conditions and converses
- algebra of continuous functions
- counterexample construction
- single-point removable discontinuity

Reading and method choice

The algebra theorem is a forward sufficient condition: if both summands are continuous, their sum is continuous. It does not say that discontinuity of the summands must survive addition, because their defects can cancel.

Detailed derivation

1. The valid implication is "f, g continuous at $0 \Rightarrow f + g$ continuous at 0." The student has used an invalid converse rather than the actual contrapositive.
2. Define $f(0) = 0$ and $f(x) = 1$ for $x \neq 0$. Then $\lim_{x\to0} f(x) = 1 \neq f(0) = 0$, so f is discontinuous at 0.
3. Define $g(0) = 0$ and $g(x) = -1$ for $x \neq 0$. Then $\lim_{x\to0} g(x) = -1 \neq g(0) = 0$, so g is also discontinuous at 0.
4. For $x \neq 0$, $(f + g)(x) = 1 + (-1) = 0$; at $x = 0$, $(f + g)(0) = 0 + 0 = 0$.
5. Thus $(f + g)(x) = 0$ for every real x, so the sum is a constant function and is continuous at 0.
6. This counterexample disproves the claim and exhibits exact cancellation of two removable defects.

Common pitfalls

- treating a sufficient condition as a necessary one
- failing to verify that both component functions are actually discontinuous
- checking the sum only for $x \neq 0$ and forgetting its value at 0

Verification

Direct checks give limit 1 but value 0 for f, limit -1 but value 0 for g, and both limit and value 0 for the sum. All three conclusions follow from the definition.

Method takeaway

Continuity algebra works forward; it does not guarantee that discontinuity is preserved by algebraic operations.

Extension

Even jump behavior may partly cancel. Always analyze the limit and point value of the resulting function rather than relying only on labels attached to the summands.

Source lineage: Original synthesis / diagnosis · error diagnosis / operations on continuous functions · reference IDs: openstax-v1-2.4-continuity, mit-18.01sc-session-5-discontinuity. See SOURCES.md for scope and adaptation boundaries; this note does not claim verbatim reproduction.

Advanced Proof Synthesis Original synthesis / diagnosis

Q080 · Finding the Unique Junction Value by One-Sided Continuity

Problem

Fix a real number a. Define $f(x) = \frac{\sin(x-a)}{x-a}$ for $x < a$, $f(a) = c$, and $f(x) = \frac{\ln(1+x-a)}{x-a}$ for $x > a$. First prove that a piecewise function is continuous at a exactly when both one-sided limits equal $f(a)$, and then find the unique c making this function continuous.

Answer

$c = 1$

Knowledge points

- equivalence of continuity and one-sided continuity
- standard limits
- joining a piecewise function
- uniqueness of continuous extension

Reading and method choice

The proof uses the equivalence between a two-sided limit and two equal one-sided limits. For the application, set $t = x - a$ on each side and use the two standard limits. Their common finite value is the only possible junction value.

Detailed derivation

1. In general, if f is continuous at a, then $\lim_{x\to a} f(x) = f(a)$. Existence of this two-sided limit implies that both one-sided limits exist and equal $f(a)$.
2. Conversely, if $\lim_{x\to a^-} f(x) = f(a)$ and $\lim_{x\to a^+} f(x) = f(a)$, the two one-sided limits exist and agree. Hence the two-sided limit exists and equals $f(a)$, so f is continuous at a.
3. For the left piece, put $t = x - a$. As $x \to a^-$, $t \to 0^-$, and $\lim_{x\to a^-} \frac{\sin(x-a)}{x-a} = \lim_{t\to 0^-} \frac{\sin t}{t} = 1$.
4. For the right piece, again put $t = x - a$. As $x \to a^+$, $t \to 0^+$, and $\lim_{x\to a^+} \frac{\ln(1+x-a)}{x-a} = \lim_{t\to 0^+} \frac{\ln(1+t)}{t} = 1$.
5. Both one-sided limits are 1. By the criterion just proved, continuity requires and is guaranteed by $f(a) = c = 1$.
6. The value is unique: if $c \neq 1$, the two-sided limit remains 1 but no longer equals the point value.

Common pitfalls

- checking equality of the one-sided limits but not comparing them with c
- reversing a one-sided direction after substitution

- incorrectly assigning limit 0 to $\frac{\ln(1+t)}{t}$

Verification

With $c = 1$, the left-hand limit, right-hand limit, and function value are all 1. Any other c violates the necessary equality between limit and value.

Method takeaway

At a piecewise junction, compute both one-sided limits and compare their common value with the actual point value.

Extension

If the right piece is replaced by $\frac{e^{x-a}-1}{x-a}$, its right-hand limit is still 1, so the unique junction value is unchanged.

Source lineage: Original synthesis / diagnosis · one-sided continuity / standard limits · reference IDs: openstax-v1-2.4-continuity, mit-18.01sc-session-5-discontinuity. See SOURCES.md for scope and adaptation boundaries; this note does not claim verbatim reproduction.

Advanced | Calculation | Synthesis | Original synthesis / diagnosis

Q081 · Finding Closed-Interval Extrema without Derivatives

Problem

Without using derivatives, find the maximum and minimum of $f(x) = x^4 - 2x^2 + 3$ on $[-2, 2]$, together with all points where they are attained.

Answer

The minimum is 2, attained at $x = \pm 1$; the maximum is 11, attained at $x = \pm 2$.

Knowledge points

- extreme value theorem
- completing the square
- range under substitution
- equality conditions

Reading and method choice

Continuity guarantees that extrema are attained, but it does not compute them. Put $t = x^2$ to turn the quartic into a quadratic on $[0, 4]$, then use a completed square to obtain both bounds and their equality cases.

Detailed derivation

1. As a polynomial, f is continuous on the closed interval $[-2, 2]$, so the extreme value theorem guarantees a maximum and a minimum.
2. Set $t = x^2$. Since $x \in [-2, 2]$, the new variable satisfies $t \in [0, 4]$.
3. Rewrite $f = t^2 - 2t + 3 = (t - 1)^2 + 2$, which gives $f \geq 2$.
4. Equality holds exactly when $t = 1$, or $x^2 = 1$. Hence the minimum 2 is attained at $x = \pm 1$.
5. For the upper bound, $t \in [0, 4]$ implies $t - 1 \in [-1, 3]$, and therefore $(t - 1)^2 \leq 9$.
6. The square reaches 9 exactly at $t = 4$, so $f \leq 11$, with $x^2 = 4$ and therefore $x = \pm 2$.
7. Thus the minimum is 2 and the maximum is 11, at the stated points.

Common pitfalls

- writing the range of $t = x^2$ as $[-4, 4]$
- finding only the minimum from the completed square and forgetting the maximum
- failing to check whether equality is attainable in the original interval

Verification

Direct substitution gives $f(\pm 1) = 1 - 2 + 3 = 2$ and $f(\pm 2) = 16 - 8 + 3 = 11$. The substitution interval includes every possible value of x^2, so no interior point is omitted.

Method takeaway

The extreme value theorem guarantees attainment; algebraic transformation computes the values and equality locations.

Extension

Even polynomials of this type can often be reduced by setting $t = x^2$, but the exact range of t must be written first.

Source lineage: Original synthesis / diagnosis · extrema / closed interval · reference IDs: openstax-v1-2.4-continuity, openstax-v1-4.3-extrema. See SOURCES.md for scope and adaptation boundaries; this note does not claim verbatim reproduction.

Advanced Calculation Synthesis Original synthesis / diagnosis

Q082 · Finding a Complete Range from Bounds and the Intermediate Value Property

Problem

Without using derivatives, find the range of $f(x) = x + \frac{4}{x+1}$ on $[0, 3]$, and explain why every number between the two bounds is attained.

Answer

The range is $[3, 4]$; the minimum 3 occurs at $x = 1$, and the maximum 4 occurs at $x = 0$ and $x = 3$.

Knowledge points

- extrema and intermediate values of continuous functions
- substitution
- algebraic inequalities
- range as a closed interval

Reading and method choice

Set $t = x+1$ to transform the interval to $[1, 4]$. A square identity gives the lower bound, and the endpoint signs of a factored expression give the upper bound. Continuity then shows that no values between the extrema are missed.

Detailed derivation

1. On $[0, 3]$, $x + 1 > 0$, so f is continuous. Put $t = x + 1$; then $t \in [1, 4]$ and $f = t - 1 + \frac{4}{t}$.
2. For the lower bound, $t + \frac{4}{t} - 4 = \frac{(t-2)^2}{t} \geq 0$, so $f = t + \frac{4}{t} - 1 \geq 3$.
3. Equality holds exactly at $t = 2$, or $x = 1$, so the minimum is 3.
4. For the upper bound, $4 - f = 5 - t - \frac{4}{t} = \frac{-(t-1)(t-4)}{t}$.
5. On $t \in [1, 4]$, $(t - 1)(t - 4) \leq 0$ and $t > 0$, hence $4 - f \geq 0$ and therefore $f \leq 4$.
6. Equality holds at $t = 1$ or $t = 4$, corresponding to $x = 0$ or $x = 3$, so the maximum is 4.
7. The function is continuous on the connected interval $[0, 3]$ and attains both 3 and 4. By the intermediate value theorem, it attains every value between them.
8. Consequently the complete range is the closed interval $[3, 4]$.

Common pitfalls

- finding only the extrema without justifying the full range
- forgetting $t > 0$ when multiplying or dividing inequalities
- missing one endpoint equality case for the upper bound

Verification

Direct values are $f(1) = 1 + 2 = 3$, $f(0) = 4$, and $f(3) = 3 + 1 = 4$. Continuity rules out gaps inside the range.

Method takeaway

A continuous-function range problem has two stages: establish attained bounds, then use the intermediate value property to fill the interval between them.

Extension

If the domain were a union of separated sets rather than an interval, continuity alone would not guarantee every value between the global extrema.

Source lineage: Original synthesis / diagnosis · range / extreme value theorem · reference IDs: openstax-v1-2.4-continuity, openstax-v1-4.3-extrema. See SOURCES.md for scope and adaptation boundaries; this note does not claim verbatim reproduction.

Hard | Error diagnosis | Synthesis | Original synthesis / diagnosis

Q083 · Diagnose an Endpoint Error in a Domain

Problem

Find the domain of $F(x) = \ln(1 - \sqrt{x-1})$. A student writes: "The radical requires $x \geq 1$. The logarithm requires $1 - \sqrt{x-1} \geq 0$, so $x \leq 2$. Hence the domain is $[1, 2]$." Identify the error and give the correct solution.

Answer

The logarithm requires its argument to be strictly positive, not merely nonnegative. The correct domain is $[1, 2)$.

Knowledge points

- domain of a composite function
- strict positivity of a logarithm argument
- nonnegative radicand
- endpoint checking

Reading and method choice

The radical condition is correct, but the student weakened the logarithmic requirement from strict positivity to nonnegativity, incorrectly retaining $x = 2$ where the logarithm has argument zero.

Detailed derivation

1. The inner radical $\sqrt{x-1}$ requires $x - 1 \geq 0$, so $x \geq 1$.
2. The outer logarithm requires $1 - \sqrt{x-1} > 0$, not ≥ 0.
3. Thus $\sqrt{x-1} < 1$. Both sides are nonnegative, so squaring gives $x - 1 < 1$, or $x < 2$.
4. Combining $x \geq 1$ and $x < 2$ gives the correct domain $[1, 2)$.
5. At $x = 2$, the logarithm argument is $1 - 1 = 0$, and $\ln 0$ is undefined; this is the extra endpoint in the student's answer.

Common pitfalls

- confusing the radical condition ≥ 0 with the logarithmic condition > 0
- solving inequalities without substituting candidate endpoints back into the original expression

Verification

At $x = 1, F(1) = \ln 1 = 0$ is defined. For $1 < x < 2$ the logarithm argument lies in $(0, 1)$. At $x = 2$ it becomes zero and must be excluded.

Method takeaway

Strict inequalities are a major source of domain errors; test every boundary point in the original formula.

Extension

If the outer function were $\sqrt{1 - \sqrt{x-1}}, a$ zero outer radicand would be allowed and the domain would indeed be $[1, 2]$.

Source lineage: Original synthesis / diagnosis · domain / logarithmic composition · reference IDs: openstax-v1-1.1-functions. See SOURCES.md for scope and adaptation boundaries; this note does not claim verbatim reproduction.

Hard Proof Synthesis Original synthesis / diagnosis

Q084 · Inverse of a Composition of Bijections

Problem

Let $f : A \to B$ and $g : B \to C$ be bijections. Prove that $g \circ f : A \to C$ is a bijection and that $(g \circ f)^{-1} = f^{-1} \circ g^{-1}$.

Answer

The composition $g \circ f$ is both injective and surjective, hence bijective. For any $c \in C$, $f^{-1} \circ g^{-1}$ first sends c back to B through g^{-1} and then back to A through f^{-1}. Its compositions with $g \circ f$ in both orders are identity maps, so it equals $(g \circ f)^{-1}$.

Knowledge points

- injectivity and surjectivity
- bijections
- composition
- order of inverse maps

Reading and method choice

First prove injectivity and surjectivity separately. The inverse order reverses because one must undo the last-applied map g before undoing the first-applied map f.

Detailed derivation

1. Injectivity: if $(g \circ f)(a_1) = (g \circ f)(a_2)$, then $g(f(a_1)) = g(f(a_2))$. Injectivity of g gives $f(a_1) = f(a_2)$, and injectivity of f gives $a_1 = a_2$.
2. Surjectivity: take any $c \in C$. Since g is surjective, some $b \in B$ satisfies $g(b) = c$. Since f is surjective, some $a \in A$ satisfies $f(a) = b$. Thus $(g \circ f)(a) = c$.
3. Therefore $g \circ f$ is both injective and surjective, so it is a bijection from A to C and has an inverse.
4. Let $H = f^{-1} \circ g^{-1}$. For every $c \in C, (g \circ f)(H(c)) = g(f(f^{-1}(g^{-1}(c)))) = g(g^{-1}(c)) = c$.
5. For every $a \in A, H((g \circ f)(a)) = f^{-1}(g^{-1}(g(f(a)))) = f^{-1}(f(a)) = a$.
6. Since H composes with $g \circ f$ in both orders to give the relevant identity maps, $H = (g \circ f)^{-1}$; hence $(g \circ f)^{-1} = f^{-1} \circ g^{-1}$.

Common pitfalls

- writing the inverse in the unreversed order $g^{-1} \circ f^{-1}$
- claiming invertibility without proving injectivity and surjectivity or checking both compositions

Verification

The set flow is $A - -f \to B - -g \to C$. Reversing it must give $C - -g^{-1} \to B - -f^{-1} \to A$, exactly the composition $f^{-1} \circ g^{-1}$.

Method takeaway

To invert a composition, undo the maps in reverse order: the last operation is undone first.

Extension

Likewise, for three bijections, $(h \circ g \circ f)^{-1} = f^{-1} \circ g^{-1} \circ h^{-1}$.

Source lineage: Original synthesis / diagnosis · bijection / composition · reference IDs: openstax-v1-1.1-functions, openstax-v1-1.4-inverse-functions. See SOURCES.md for scope and adaptation boundaries; this note does not claim verbatim reproduction.

Hard | Error diagnosis | Synthesis | Original synthesis / diagnosis

Q085 · A Limit Equation Cannot Replace a Convergence Proof

Problem

Let $a_1 = 2$ and $a_{n+1} = \sqrt{3 + a_n}$. A student writes: "Assume $a_n \to L$. Then $L = \sqrt{3 + L}$, so $L = \frac{1+\sqrt{13}}{2}$. Therefore the sequence converges to this value." Identify the logical error and complete a rigorous proof.

Answer

The work shows only that, if the sequence converges, its limit must be $\frac{1+\sqrt{13}}{2}$; it does not prove convergence. Let $L = \frac{1+\sqrt{13}}{2}$. Induction gives $0 < a_n < L$, and one can prove $a_{n+1} > a_n$. Thus the sequence is increasing and bounded above by L, so it converges. Passing to the limit in the recurrence then gives the limit L.

Knowledge points

- limits of recursive sequences
- monotone bounded principle
- mathematical induction
- a limit equation supplies only candidates

Reading and method choice

Passing to a limit assumes that a limit exists. The correct order is to use the positive fixed point as an upper bound, prove monotonicity, invoke monotone bounded convergence, and only then determine the limit.

Detailed derivation

1. The gap is treating a necessary condition under an assumed convergence as proof of convergence. Let $L = \frac{1+\sqrt{13}}{2}$; then $L > 2$ and $L^2 = L + 3$.
2. Upper bound: $a_1 = 2 < L$. If $0 < a_n < L$, then $a_{n+1} = \sqrt{3 + a_n} < \sqrt{3 + L} = \sqrt{L^2} = L$. Induction gives $0 < a_n < L$.
3. Monotonicity: because $0 < a_n < L$ and $t^2 - t - 3 < 0$ for $0 < t < L$, we have $a_n^2 < a_n + 3$.
4. Both sides are positive, so taking square roots yields $a_n < \sqrt{a_n + 3} = a_{n+1}$. Hence the sequence is strictly increasing.
5. The sequence is increasing and bounded above by L, so the monotone bounded principle guarantees convergence. Denote its limit by ℓ.
6. Now pass to the limit in the recurrence: $\ell = \sqrt{3 + \ell}$, so $\ell^2 - \ell - 3 = 0$. Since $\ell > 0$, $\ell = \frac{1+\sqrt{13}}{2} = L$.

Common pitfalls

- passing to a limit in a recurrence before proving that the limit exists
- solving the fixed-point equation without rejecting a root incompatible with the signs of the sequence

Verification

The first terms are $2, \sqrt{5} \approx 2.236$, and $\sqrt{3+\sqrt{5}} \approx 2.288$; they increase and remain below $L \approx 2.303$. Also $L^2 = L + 3$ verifies the fixed point.

Method takeaway

For a recursive sequence, prove convergence first and determine the limit second; a fixed-point equation only identifies candidates.

Extension

For recurrences $a_{n+1} = \sqrt{c + a_n}$, first find a positive fixed point and test whether it creates an invariant interval and a useful upper bound.

Source lineage: Original synthesis / diagnosis · recursive sequence / monotone bounded principle · reference IDs: openstax-v2-5.1-sequences. See SOURCES.md for scope and adaptation boundaries; this note does not claim verbatim reproduction.

Hard Proof Synthesis Original synthesis / diagnosis

Q086 · Prove Uniqueness of a Function Limit

Problem

Let a be an accumulation point of the domain of f. If $\lim_{x\to a} f(x) = A$ and $\lim_{x\to a} f(x) = B$, prove that $A = B$.

Answer

Assume $A \neq B$ and choose $\varepsilon = \frac{|A-B|}{3}$. The two limit definitions imply that, in one sufficiently small deleted neighborhood, both $|f(x) - A| < \varepsilon$ and $|f(x) - B| < \varepsilon$ hold. Then $|A - B| \le |A - f(x)| + |f(x) - B| < \frac{2|A-B|}{3}$, a contradiction. Hence $A = B$.

Knowledge points

- uniqueness of function limits
- epsilon-delta definition
- taking the minimum of two deltas
- triangle inequality

Reading and method choice

If A and B differ, they have a positive separation. Forcing $f(x)$ into neighborhoods of both centers whose radii total less than that separation creates an impossible overlap.

Detailed derivation

1. Assume for contradiction that $A \neq B$, and set $d = |A - B| > 0$. Choose $\varepsilon = \frac{d}{3}$.
2. From $\lim_{x\to a} f(x) = A$, there exists $\delta_1 > 0$ such that $0 < |x - a| < \delta_1$ implies $|f(x) - A| < \frac{d}{3}$.
3. From $\lim_{x\to a} f(x) = B$, there exists $\delta_2 > 0$ such that $0 < |x - a| < \delta_2$ implies $|f(x) - B| < \frac{d}{3}$.
4. Let $\delta = \min\{\delta_1, \delta_2\}$. Because a is an accumulation point, choose a domain point x with $0 < |x - a| < \delta$; both inequalities then hold simultaneously.
5. By the triangle inequality, $d = |A - B| \le |A - f(x)| + |f(x) - B| < \frac{d}{3} + \frac{d}{3} = \frac{2d}{3}$.
6. This gives $d < \frac{2d}{3}$, contradicting $d > 0$. Therefore the assumption is false and $A = B$.

Common pitfalls

- obtaining δ_1 and δ_2 but failing to take their minimum, so the estimates are not guaranteed simultaneously
- forgetting why the accumulation-point condition supplies a domain point in the deleted neighborhood

Verification

Geometrically, the intervals centered at A and B with radius $\frac{d}{3}$ still have a gap of $\frac{d}{3}$, so they cannot contain the same value $f(x)$, matching the algebraic contradiction.

Method takeaway

For uniqueness, place both limit estimates in one common deleted neighborhood and compare the two candidate limits with the triangle inequality.

Extension

The same proof establishes uniqueness of a sequence limit by combining two thresholds with max$\{N_1, N_2\}$.

Source lineage: Original synthesis / diagnosis · uniqueness of a function limit / epsilon-delta proof · reference IDs: openstax-v1-2.2-function-limits, openstax-v1-2.5-precise-limit. See SOURCES.md for scope and adaptation boundaries; this note does not claim verbatim reproduction.

Hard Proof Synthesis Original synthesis / diagnosis

Q087 · A Bounded Function Times an Infinitesimal

Problem

Suppose $\alpha(x)$ is infinitesimal as $x \to a$ and $g(x)$ is bounded in a deleted neighborhood of a. Using definitions only, prove that $\alpha(x)g(x)$ is also infinitesimal, treating separately the case where the bound may be 0.

Answer

$\alpha(x)g(x) \to 0$. If $|g(x)| \le M$ with $M > 0$, require $|\alpha(x)| < \frac{\varepsilon}{M}$. If $M = 0$, then $g(x) = 0$ throughout that neighborhood and the product is identically 0 there.

Knowledge points

- Epsilon-delta definition of an infinitesimal
- Local boundedness
- Product estimates
- Boundary-case analysis

Reading and method choice

The target is to make $|\alpha g|$ smaller than ε. Boundedness supplies a fixed multiplier M, and the case $M = 0$ must be handled before dividing by M.

Detailed derivation

1. Since g is locally bounded, there exist $\delta_0 > 0$ and $M \ge 0$ such that $|g(x)| \le M$ whenever $0 < |x - a| < \delta_0$.
2. If $M = 0$, then $|g(x)| \le 0$ implies $g(x) = 0$ in that neighborhood, so $\alpha(x)g(x) = 0$ and the result is immediate.
3. Now assume $M > 0$. Given $\varepsilon > 0, \alpha(x) \to 0$ provides $\delta_1 > 0$ such that $|\alpha(x)| < \frac{\varepsilon}{M}$ whenever $0 < |x - a| < \delta_1$.
4. Choose $\delta = \min(\delta_0, \delta_1)$. Both estimates then hold whenever $0 < |x - a| < \delta$.
5. Therefore $|\alpha(x)g(x)| = |\alpha(x)||g(x)| \le M|\alpha(x)| < \frac{M \cdot \varepsilon}{M} = \varepsilon$.
6. By the definition of an infinitesimal, $\lim_{x \to a} \alpha(x)g(x) = 0$.

Common pitfalls

- Failing to intersect the two neighborhoods
- Dividing by M when $M = 0$
- Citing the conclusion without an epsilon estimate

Verification

For every ε, the proof explicitly selects $\delta = \min(\delta_0, \delta_1)$ and derives $|\alpha g| < \varepsilon$, with exactly the quantifier order required by the definition.

Method takeaway

For a bounded multiplier, obtain a uniform bound first and scale the epsilon target for the infinitesimal by that bound.

Extension

If $g(x) \to B$, then g is locally bounded, so this result becomes a key ingredient in proving the product limit law.

Source lineage: Original synthesis / diagnosis · property of infinitesimals / bounded function · reference IDs: openstax-v1-2.2-function-limits, openstax-v1-2.5-precise-limit. See SOURCES.md for scope and adaptation boundaries; this note does not claim verbatim reproduction.

Hard | Error diagnosis | Synthesis | Original synthesis / diagnosis

Q088 · Diagnosing Errors in $\frac{0}{0}$ and Cancellation

Problem

A student evaluates $\lim_{x\to1}\frac{x^2-1}{x-1}$ as follows: first, substitution gives $\frac{0}{0}=0$; second, canceling $x-1$ gives $x+1$, so the limit is also 2; hence the answer is both 0 and 2. Identify the logical error in each step and give the complete correct solution.

Answer

The form $\frac{0}{0}$ has no value, so the first step is invalid. Cancellation holds only for $x\neq1$, but that is exactly enough for a limit. The correct limit is 2.

Knowledge points

- Meaning of an indeterminate form
- Point values versus limits
- Equality in a deleted neighborhood
- Factorization

Reading and method choice

The contradiction comes from assigning a numerical value to $\frac{0}{0}$. The cancellation itself is valid, provided one states that it holds for $x\neq1$, which is sufficient for a limit.

Detailed derivation

1. Direct substitution produces only the form $\frac{0}{0}$; this expression is undefined and cannot be set equal to 0.
2. Since $x^2-1=(x-1)(x+1)$, for every $x\neq1$ the quotient equals $x+1$.
3. The original function may be undefined at $x=1$, but a limit uses points with $0<|x-1|<\delta$, so deleted-neighborhood equality is sufficient.
4. Thus the original limit equals $\lim_{x\to1}(x+1)$.
5. Direct substitution into the polynomial gives $1+1=2$.
6. The unique correct answer is 2; the value 0 arose from an illegal assignment to $\frac{0}{0}$.

Common pitfalls

- Treating an indeterminate form as a number
- Demanding equality at $x = 1$ after cancellation
- Assuming an undefined point prevents a limit

Verification

At $x = 0.99$ and $x = 1.01$, the original quotient is 1.99 and 2.01, respectively, close to 2 rather than 0.

Method takeaway

The form $\frac{0}{0}$ is a signal to transform the expression; a limit only requires equality in a deleted neighborhood.

Extension

If two functions agree in some deleted neighborhood of a, then their limits as $x \to a$ exist together and are equal, regardless of their values at a.

Source lineage: Original synthesis / diagnosis · error diagnosis / $\frac{0}{0}$ form · reference IDs: openstax-v1-2.3-limit-laws. See SOURCES.md for scope and adaptation boundaries; this note does not claim verbatim reproduction.

Hard | Parameter / synthesis / application | Synthesis | Original synthesis / diagnosis

Q089 · A Variable-Exponent Limit at Infinity

Problem

Evaluate $\lim_{x\to+\infty}(1+\frac{3}{x})^{2x-\sqrt{x}}$, explicitly identifying the 1^∞ form and the effective standardized exponent.

Answer

e^6. With $t=\frac{3}{x}$, the base is $1+t$ and the effective exponent is $t(2x-\sqrt{x})=6-\frac{3}{\sqrt{x}}\to 6$.

Knowledge points

- Recognition of 1^∞
- Second fundamental limit
- Variable-exponent standardization
- Composite limits

Reading and method choice

The base tends to 1 while the exponent tends to $+\infty$, so the answer is not obtained by direct substitution. The product of the base increment and exponent tends to 6.

Detailed derivation

1. As $x\to+\infty, 1+\frac{3}{x}\to 1$ and $2x-\sqrt{x}\to+\infty$, so the form is 1^∞.
2. Set $t=\frac{3}{x}$. Then $t\to 0^+$ and $(1+t)^{2x-\sqrt{x}}=[(1+t)^{\frac{1}{t}}]^{t(2x-\sqrt{x})}$.
3. The second fundamental limit gives $(1+t)^{\frac{1}{t}}\to e$.
4. The effective exponent is $t(2x-\sqrt{x})=(\frac{3}{x})(2x-\sqrt{x})=6-\frac{3}{\sqrt{x}}\to 6$.
5. Continuity of the power expression yields the full limit e^6.
6. This also explains why the base limit 1 alone is insufficient.

Common pitfalls

- Treating 1^∞ as 1
- Ignoring the $-\sqrt{x}$ correction
- Computing $(\frac{3}{x})\sqrt{x}$ as $3\sqrt{x}$

Verification

At $x = 10^4$ the effective exponent is $6 - 0.03 = 5.97$, and the expression is close to $e^{5.97}$, moving toward e^6.

Method takeaway

A general 1^∞ limit is controlled by the product of the base increment and the exponent.

Extension

Replacing the exponent by $2x - c\sqrt{x}$ for any constant c leaves the limit e^6 because the correction times $\frac{3}{x}$ tends to 0.

Source lineage: Original synthesis / diagnosis · second fundamental limit / infinity · reference IDs: openstax-v1-2.3-limit-laws, mit-18.01sc-session-19-limit-involving-e. See SOURCES.md for scope and adaptation boundaries; this note does not claim verbatim reproduction.

Hard Proof Synthesis Open-text method adaptation

Q090 · A Geometric Proof of the First Fundamental Limit

Problem

Assuming radian measure, start from $\sin x < x < \tan x$ for $0 < x < \frac{\pi}{2}$ and prove $\lim_{x\to 0} \frac{\sin x}{x} = 1$, covering both $x \to 0^+$ and $x \to 0^-$.

Answer

The inequalities give $\cos x < \frac{\sin x}{x} < 1$, so the right-hand limit is 1. Since $\frac{\sin(-x)}{-x} = \frac{\sin x}{x}$, the left-hand limit is the same.

Knowledge points

- Unit-circle geometric inequality
- Squeeze theorem
- Even functions
- One-sided limits

Reading and method choice

The right-hand limit follows from area comparison and squeezing. The left-hand limit must be handled explicitly by the evenness of $\frac{\sin x}{x}$.

Detailed derivation

1. For $0 < x < \frac{\pi}{2}$, comparison of areas in the unit circle gives $\sin x < x < \tan x$.
2. Because $\sin x$ and $\cos x$ are positive, $\sin x < x$ gives $\frac{\sin x}{x} < 1$, while $x < \tan x = \frac{\sin x}{\cos x}$ gives $\cos x < \frac{\sin x}{x}$.
3. As $x \to 0^+$, $\cos x \to 1$, so both bounds tend to 1.
4. The squeeze theorem yields $\lim_{x\to 0^+} \frac{\sin x}{x} = 1$.
5. For $x < 0$, put $u = -x > 0$; then $\frac{\sin x}{x} = \frac{\sin(-u)}{-u} = \frac{\sin u}{u}$.
6. Thus the left-hand limit also equals 1, and the two-sided limit is 1.

Common pitfalls

- Not stating that angles are in radians
- Dividing inequalities without checking positivity
- Proving only the right-hand limit

- Calling $\frac{\sin x}{x}$ odd instead of even

Verification

The squeeze can be written $0 < 1 - \frac{\sin x}{x} < 1 - \cos x$, whose right side tends to 0; evenness transfers the result to the negative side.

Method takeaway

A rigorous proof of the fundamental limit combines geometry, squeezing, and both one-sided limits.

Extension

It immediately implies $\lim_{x \to 0} \frac{\tan x}{x} = 1$ and $\lim_{x \to 0} \frac{1 - \cos x}{x^2} = \frac{1}{2}$.

Source lineage: Open-text method adaptation · first fundamental limit / geometric squeeze · reference IDs: openstax-v1-2.3-limit-laws, mit-18.01sc-session-8-trig-limits. See SOURCES.md for scope and adaptation boundaries; this note does not claim verbatim reproduction.

Hard Proof Synthesis Original synthesis / diagnosis

Q091 · Equivalence Characterized by Relative Error

Problem

Let $\alpha(x)$ and $\beta(x)$ both be infinitesimal as $x \to a$, with $\beta(x)$ nonzero in a deleted neighborhood. Prove that $\alpha(x) \sim \beta(x)$ if and only if $\alpha(x) - \beta(x) = o(\beta(x))$.

Answer

Since $\frac{\alpha-\beta}{\beta} = \frac{\alpha}{\beta} - 1$, the conditions $\frac{\alpha}{\beta} \to 1$ and $\frac{\alpha-\beta}{\beta} \to 0$ are exactly equivalent.

Knowledge points

- Definition of equivalent infinitesimals
- Little-o notation
- Biconditional proof
- Relative error

Reading and method choice

An exact identity connects the two statements. Both directions must be shown, and eventual nonvanishing of β ensures that every ratio is defined.

Detailed derivation

1. Where $\beta(x) \neq 0$, $\frac{\alpha(x)-\beta(x)}{\beta(x)} = \frac{\alpha(x)}{\beta(x)} - 1$.
2. If $\alpha \sim \beta$, then $\frac{\alpha}{\beta} \to 1$ by definition.
3. Hence $\frac{\alpha-\beta}{\beta} \to 1 - 1 = 0$, so $\alpha - \beta = o(\beta)$.
4. Conversely, if $\alpha - \beta = o(\beta)$, then $\frac{\alpha-\beta}{\beta} \to 0$.
5. Using $\frac{\alpha}{\beta} = 1 + \frac{\alpha-\beta}{\beta}$ gives $\frac{\alpha}{\beta} \to 1$.
6. Therefore $\alpha \sim \beta$, proving both directions.

Common pitfalls

- Proving only one direction
- Forgetting eventual nonvanishing of β

- Replacing $o(\beta)$ by merely tending to 0

Verification

Both implications come from the same reversible identity, with the two limits differing exactly by the constant 1.

Method takeaway

Equivalent infinitesimals have a difference negligible relative to β, not merely an absolute difference tending to 0.

Extension

This criterion explains the common notation $\alpha = \beta + o(\beta)$ for $\alpha \sim \beta$.

Source lineage: Original synthesis / diagnosis · criterion for equivalent infinitesimals / higher-order infinitesimal · reference IDs: openstax-v1-2.3-limit-laws. See SOURCES.md for scope and adaptation boundaries; this note does not claim verbatim reproduction.

Hard Proof Synthesis Original synthesis / diagnosis

Q092 · Proving Safe Replacement in a Quotient

Problem

Suppose $\alpha \sim \alpha_1$ and $\beta \sim \beta_1$ as $x \to a$, with all relevant quotients defined in a deleted neighborhood. Prove $\frac{\alpha}{\beta} \sim \frac{\alpha_1}{\beta_1}$, and explain why this supports equivalent substitution in products and quotients.

Answer

$$[\frac{\frac{\alpha}{\beta}}{\frac{\alpha_1}{\beta_1}}] = (\frac{\alpha}{\alpha_1})(\frac{\beta_1}{\beta}) \to 1 \cdot 1 = 1.$$

Knowledge points

- Ratio definition of equivalence
- Limits of reciprocals
- Transfer of equivalence through quotients
- Domain conditions

Reading and method choice

To compare two quotients for equivalence, take their ratio. It factors into two quantities already known to tend to 1.

Detailed derivation

1. From $\alpha \sim \alpha_1$, $\frac{\alpha}{\alpha_1} \to 1$; from $\beta \sim \beta_1$, $\frac{\beta}{\beta_1} \to 1$.
2. Since the latter limit is $1 \neq 0$, its reciprocal satisfies $\frac{\beta_1}{\beta} \to 1$.
3. Where all quotients are defined, $\frac{\frac{\alpha}{\beta}}{\frac{\alpha_1}{\beta_1}} = (\frac{\alpha}{\alpha_1})(\frac{\beta_1}{\beta})$.
4. The two factors on the right tend to 1 and 1.
5. Their product tends to 1, so $\frac{\alpha}{\beta} \sim \frac{\alpha_1}{\beta_1}$.
6. Thus factorwise replacement in a product or quotient preserves the final relative ratio.

Common pitfalls

- Not checking eventual nonvanishing of denominators
- Giving $\frac{\beta_1}{\beta}$ the wrong limit

- Extending the theorem without conditions to sums or differences

Verification

The proof directly checks that the ratio required by the definition of equivalence tends to 1.

Method takeaway

Equivalent substitution is safe in products and quotients because relative errors multiply as factors tending to 1.

Extension

The same result for any finite number of product and quotient factors follows by induction.

Source lineage: Original synthesis / diagnosis · equivalent substitution theorem / quotient structure · reference IDs: openstax-v1-2.3-limit-laws. See SOURCES.md for scope and adaptation boundaries; this note does not claim verbatim reproduction.

Hard | Proof | Synthesis | Original synthesis / diagnosis

Q093 · A Counterexample to Replacement inside a Sum

Problem

Construct and verify a counterexample showing that $\alpha \sim \alpha_1$ and $\beta \sim \beta_1$ need not imply $\alpha + \beta \sim \alpha_1 + \beta_1$. Require both sums to be nonzero in a deleted neighborhood.

Answer

Take $\alpha = x + x^2, \alpha_1 = x, \beta = -x + x^3$, and $\beta_1 = -x + x^3$. Then $\alpha \sim \alpha_1$ and $\beta \sim \beta_1$, but $\frac{\alpha+\beta}{\alpha_1+\beta_1} = \frac{x^2+x^3}{x^3} = \frac{1}{x} + 1$, which does not tend to 1.

Knowledge points

- Counterexample construction
- Cancellation of principal terms
- Failure of additive stability
- Ratio verification

Reading and method choice

Arrange for first-order terms x and $-x$ to cancel in the sums. The previously negligible higher-order terms then determine the result.

Detailed derivation

1. Let $\alpha = x + x^2$ and $\alpha_1 = x$. Then $\frac{\alpha}{\alpha_1} = 1 + x \to 1$, so $\alpha \sim \alpha_1$.
2. Let $\beta = \beta_1 = -x + x^3$. They are nonzero in a sufficiently small deleted neighborhood, and $\frac{\beta}{\beta_1} = 1$.
3. Compute $\alpha + \beta = (x + x^2) + (-x + x^3) = x^2 + x^3 = x^2(1 + x)$.
4. Compute $\alpha_1 + \beta_1 = x + (-x + x^3) = x^3$, nonzero for $x \neq 0$.
5. Their ratio is $\frac{x^2(1+x)}{x^3} = \frac{1+x}{x} = \frac{1}{x} + 1$.
6. This ratio does not tend to 1, so the sums are not equivalent.

Common pitfalls

- Using a counterexample whose comparison denominator is identically zero
- Mentioning cancellation without checking the ratio
- Assuming higher-order terms are always disposable

Verification

The two premise ratios tend to 1, while the conclusion ratio is unbounded; the premises hold and the conclusion fails.

Method takeaway

Addition or subtraction may cancel principal terms, allowing formerly negligible terms to become dominant.

Extension

Replacement in a sum may become safe under an extra condition preventing cancellation and keeping the comparison sum at the original principal order.

Source lineage: Original synthesis / diagnosis · forbidden zone of equivalent substitution / counterexample proof · reference IDs: openstax-v1-2.3-limit-laws. See SOURCES.md for scope and adaptation boundaries; this note does not claim verbatim reproduction.

Hard | Parameter / synthesis / application | Synthesis | Original synthesis / diagnosis

Q094 · Can a Power Factor Suppress Oscillation?

Problem

For a real parameter p, define $f_p(0) = 0$ and $f_p(x) = |x|^p \sin(\frac{1}{x})$ for $x \neq 0$. Discuss continuity at $x = 0$. When it is discontinuous, classify the discontinuity and decide whether the function is bounded in some neighborhood of 0.

Answer

For $p > 0$, it is continuous at 0 and locally bounded; for $p = 0$, it has a bounded oscillatory discontinuity of the second kind; for $p < 0$, it has an unbounded oscillatory discontinuity of the second kind.

Knowledge points

- squeeze theorem
- oscillatory functions
- parameter case analysis
- sequential test for limits
- local boundedness

Reading and method choice

The decisive quantity is the amplitude $|x|^p$. For $p > 0$ it tends to zero and suppresses the sine oscillation; for $p = 0$ it stays fixed; for $p < 0$ it grows without bound. In the last two cases, explicit sequences rigorously disprove the existence of a limit.

Detailed derivation

1. If $p > 0$, then $|f_p(x)| = |x|^p|\sin(\frac{1}{x})| \leq |x|^p \to 0$. The squeeze theorem gives $\lim_{x\to 0} f_p(x) = 0 = f_p(0)$, so the function is continuous.
2. Still for $p > 0$, if $|x| < 1$, then $|f_p(x)| \leq |x|^p < 1$, so the function is locally bounded.
3. If $p = 0$, then $f_0(x) = \sin(\frac{1}{x})$. For $x_n = \frac{1}{\frac{\pi}{2}+2\pi n}$, $f_0(x_n) = 1$; for $y_n = \frac{1}{\frac{3\pi}{2}+2\pi n}$, $f_0(y_n) = -1$.
4. Since $x_n, y_n \to 0$ but the function values approach different numbers, the limit fails to exist. This is an oscillatory discontinuity of the second kind, while $|f_0(x)| \leq 1$ proves local boundedness.
5. If $p < 0$, use $x_n = \frac{1}{\frac{\pi}{2}+2\pi n}$ again. Then $f_p(x_n) = |x_n|^p \to +\infty$.
6. Now take $z_n = \frac{1}{\pi n}$. Since $\sin(\frac{1}{z_n}) = 0$, we have $f_p(z_n) = 0$. Two sequences approaching 0 produce infinite and zero behavior, so no single limit exists.

7. The divergence $f_p(x_n) \to +\infty$ also proves unboundedness in every neighborhood of 0. Thus $p < 0$ gives an unbounded oscillatory discontinuity of the second kind.

8. Combining the three parameter ranges yields the complete classification.

Common pitfalls

- assuming $|\sin(\frac{1}{x})| \leq 1$ forces the product to zero for every p
- writing the limit as infinity for $p < 0$ while ignoring zeros and sign changes of the sine factor
- misclassifying bounded oscillation as a first-kind discontinuity

Verification

The boundary case $p = 0$ is treated separately. A uniform bound proves the $p > 0$ result, while explicit sequences for $p = 0$ and $p < 0$ both disprove the limit and establish the stated boundedness behavior.

Method takeaway

A vanishing amplitude times a bounded oscillation is handled by squeezing; if the amplitude does not vanish, reveal the oscillation with carefully chosen sequences.

Extension

If $\sin(\frac{1}{x})$ is replaced by any bounded function $h(\frac{1}{x})$, the conclusion for $p > 0$ remains true; for $p \leq 0$, the behavior depends on the actual oscillation of h.

Source lineage: Original synthesis / diagnosis · parameter analysis / squeeze theorem · reference IDs: openstax-v1-2.4-continuity, mit-18.01sc-session-5-discontinuity. See SOURCES.md for scope and adaptation boundaries; this note does not claim verbatim reproduction.

Hard Proof Synthesis Original synthesis / diagnosis

Q095 · Equal Values Half an Interval Apart

Problem

Suppose f is continuous on $[0,1]$ and $f(0)=f(1)$. Prove that there is at least one $\xi \in [0,\frac{1}{2}]$ such that $f(\xi)=f(\xi+\frac{1}{2})$.

Answer

Define $g(x)=f(x)-f(x+\frac{1}{2})$ for $0\le x\le \frac{1}{2}$. Then $g(\frac{1}{2})=-g(0)$. If an endpoint value is zero, the result is immediate; otherwise the endpoint values have opposite signs and the zero theorem supplies the required ξ.

Knowledge points

- construction of an auxiliary function
- difference of continuous functions
- zero theorem
- opposite endpoint values

Reading and method choice

The desired equation compares values of the same function at points separated by $\frac{1}{2}$. Moving the two sides into a difference converts the problem into finding a zero. The condition $f(0)=f(1)$ makes the endpoint values of this difference exact negatives.

Detailed derivation

1. Define $g(x)=f(x)-f(x+\frac{1}{2})$ on $[0,\frac{1}{2}]$.
2. For $x\in[0,\frac{1}{2}]$, the shifted point $x+\frac{1}{2}$ belongs to $[\frac{1}{2},1]$. Since f is continuous, both $f(x)$ and $f(x+\frac{1}{2})$ are continuous, so g is continuous.
3. At the left endpoint, $g(0)=f(0)-f(\frac{1}{2})$.
4. At the right endpoint, $g(\frac{1}{2})=f(\frac{1}{2})-f(1)=f(\frac{1}{2})-f(0)=-g(0)$, using $f(0)=f(1)$.
5. If $g(0)=0$, choose $\xi=0$ and obtain $f(0)=f(\frac{1}{2})$; in this case $g(\frac{1}{2})$ is also zero.
6. If $g(0)\neq 0$, then $g(0)$ and $g(\frac{1}{2})=-g(0)$ have opposite signs.
7. The zero theorem gives some $\xi\in(0,\frac{1}{2})$ with $g(\xi)=0$.
8. In either case, some $\xi\in[0,\frac{1}{2}]$ satisfies $f(\xi)-f(\xi+\frac{1}{2})=0$, which is the desired equality.

Common pitfalls

- applying the zero theorem directly to f without constructing a difference
- failing to check that $x + \frac{1}{2}$ stays in the domain of f
- handling only strict opposite signs and omitting the endpoint-zero case

Verification

Continuity of the auxiliary function, its endpoint relation, and every hypothesis of the zero theorem have been checked; the equation $g(\xi) = 0$ is exactly equivalent to the target statement.

Method takeaway

For equal-value existence problems, form the difference of the two sides and seek a sign relation at the endpoints.

Extension

For an integer $n \geq 2$, one can similarly study differences $f(x) - f(x + \frac{1}{n})$ and use cyclic relations among subdivision values.

Source lineage: Original synthesis / diagnosis · intermediate value theorem / auxiliary function · reference IDs: openstax-v1-2.4-continuity. See SOURCES.md for scope and adaptation boundaries; this note does not claim verbatim reproduction.

Hard | Parameter / synthesis / application | Synthesis | Original synthesis / diagnosis

Q096 · Optional: Comparing Uniform Continuity on an Unbounded Interval

Problem

On $[0, +\infty)$, determine whether $f(x) = \sqrt{x}$ and $g(x) = x^2$ are uniformly continuous. Prove both conclusions directly from the definition. For the function that is not uniformly continuous, give a specific ε_0 and a contradiction construction.

Answer

$f(x) = \sqrt{x}$ is uniformly continuous on $[0, +\infty)$ with $\delta = \varepsilon^2$; $g(x) = x^2$ is not uniformly continuous there, as shown by $\varepsilon_0 = 1$ and $x = n, y = n + \frac{1}{n}$.

Knowledge points

- definition of uniform continuity
- estimate for differences of square roots
- negation of uniform continuity
- unbounded interval

Reading and method choice

Uniform continuity requires one δ to work for every pair of points in the interval. The global estimate $|\sqrt{x} - \sqrt{y}| \le \sqrt{|x-y|}$ provides uniform control. The square function becomes increasingly steep far out, allowing pairs whose input distance tends to zero while the output difference stays above a fixed number.

Detailed derivation

1. First prove the square-root estimate. For $x, y \ge 0$, assume without loss of generality that $x \ge y$. Then $x-y = (\sqrt{x}-\sqrt{y})(\sqrt{x}+\sqrt{y}) \ge (\sqrt{x}-\sqrt{y})^2$.
2. Hence for all $x, y \ge 0$, $|\sqrt{x} - \sqrt{y}| \le \sqrt{|x-y|}$.
3. Given $\varepsilon > 0$, choose $\delta = \varepsilon^2$. If $|x-y| < \delta$, then $|f(x) - f(y)| \le \sqrt{|x-y|} < \sqrt{\delta} = \varepsilon$.
4. This δ depends only on ε, not on the particular points, so $\sqrt{x}$ is uniformly continuous on $[0, +\infty)$.
5. Now consider $g(x) = x^2$. To negate uniform continuity, fix $\varepsilon_0 = 1$.
6. Given any $\delta > 0$, choose a positive integer $n > \frac{1}{\delta}$, and set $x = n$ and $y = n + \frac{1}{n}$. Then $|x-y| = \frac{1}{n} < \delta$.
7. However, $|g(y) - g(x)| = (n + \frac{1}{n})^2 - n^2 = 2 + \frac{1}{n^2} > 1 = \varepsilon_0$.

8. Thus every proposed δ is defeated by a pair closer than δ, so x^2 is not uniformly continuous on $[0, +\infty)$.

Common pitfalls

- using a point-dependent delta from ordinary continuity as though it were uniform
- concluding from an unbounded derivative instead of following the requested definition
- allowing epsilon to vary with n when negating uniform continuity

Verification

For the square root, $\delta = \varepsilon^2$ works uniformly over the whole interval. For the square, the same fixed $\varepsilon_0 = 1$ defeats every δ through an explicit pair, exactly matching the quantified negation.

Method takeaway

Uniform continuity is global uniform control; on an unbounded interval, ordinary continuity does not automatically imply it.

Extension

A similar estimate proves that x^α is uniformly continuous on $[0, +\infty)$ for $0 < \alpha \leq 1$, while the square-function construction adapts when $\alpha > 1$.

Source lineage: Original synthesis / diagnosis · optional / uniform continuity · reference IDs: openstax-v1-2.4-continuity. See SOURCES.md for scope and adaptation boundaries; this note does not claim verbatim reproduction.

Challenge | Parameter / synthesis / application | Challenge | Original synthesis / diagnosis

Q097 · Characterize Parameter Convergence Using Even and Odd Subsequences

Problem

For real p and q, define $a_{2k} = p + \frac{1}{k}$ and $a_{2k-1} = q - \frac{1}{k}$ for $k = 1, 2, \cdots$. Find a necessary and sufficient condition for $\{a_n\}$ to converge. In the convergent case, state the limit and prove it using the ε-N definition.

Answer

$\{a_n\}$ converges if and only if $p = q$; in that case its limit is the common value $p = q$.

Knowledge points

- even and odd subsequences
- subsequences inherit the original limit
- proof of an if-and-only-if statement
- uniform control of both index classes

Reading and method choice

Necessity follows because the two natural subsequences of a convergent sequence must have the same limit. For sufficiency, one should provide a single N that controls both even and odd terms.

Detailed derivation

1. Necessity: if $\{a_n\}$ converges to A, then both its even and odd subsequences must also converge to A.
2. Since $a_{2k} = p + \frac{1}{k}$, $\lim_{k\to\infty} a_{2k} = p$. Since $a_{2k-1} = q - \frac{1}{k}$, $\lim_{k\to\infty} a_{2k-1} = q$. Thus $p = q = A$ is necessary.
3. Sufficiency: suppose $p = q = r$. Given $\varepsilon > 0$, choose a positive integer $K > \frac{1}{\varepsilon}$ and set $N = 2K$.
4. If $n > N$ is even, write $n = 2k$. Then $k > K$, so $|a_n - r| = \frac{1}{k} < \varepsilon$.
5. If $n > N$ is odd, write $n = 2k - 1$. From $2k - 1 > 2K$ we get $k > K + \frac{1}{2}$, hence $k > K$ and $|a_n - r| = \frac{1}{k} < \varepsilon$.
6. Thus $|a_n - r| < \varepsilon$ for every $n > N$, regardless of parity. The ε-N definition gives $a_n \to r$. Therefore convergence occurs exactly when $p = q$, with that common value as the limit.

Common pitfalls

- computing the two subsequence limits without stating why convergence of the original forces them to agree
- choosing separate thresholds for even and odd terms without combining them into one N

Verification

If $p \neq q$, choose $\varepsilon = \frac{|p-q|}{4}$. Sufficiently large even terms lie near p and odd terms near q; these neighborhoods are disjoint, so the full sequence cannot approach one limit.

Method takeaway

For a parity-defined sequence, use subsequences to obtain necessary parameter conditions, then prove sufficiency with one tail estimate covering both parities.

Extension

If $\frac{1}{k}$ is replaced by any positive sequence $b_k \to 0$, the same condition $p = q$ remains valid; use $b_k \to 0$ in place of the explicit reciprocal estimate.

Source lineage: Original synthesis / diagnosis · piecewise sequence / subsequences · reference IDs: openstax-v2-5.1-sequences. See SOURCES.md for scope and adaptation boundaries; this note does not claim verbatim reproduction.

Challenge | Parameter / synthesis / application | Challenge | Original synthesis / diagnosis

Q098 · A Complete Convergence Chain for a Radical Recurrence

Problem

Let $0 < a_1 < 2$ and $a_{n+1} = \sqrt{2 + a_n}$. Prove: $(1) 0 < a_n < 2; (2) a_n$ is increasing; (3) it converges; and (4) find its limit. Do not guess a closed form or merely solve a limit equation.

Answer

For every $n, 0 < a_n < 2$ and $a_{n+1} > a_n$. The sequence is increasing and bounded above by 2, so it converges to 2.

Knowledge points

- Invariant interval for a recurrence
- Mathematical induction
- Comparison by squaring
- Monotone convergence theorem
- Filtering roots of a limit equation

Reading and method choice

First prove that $(0, 2)$ is invariant under the recurrence. Inside this interval, $a_{n+1} > a_n$ is equivalent to $2 + a_n > a_n^2$, whose sign factors cleanly.

Detailed derivation

1. The base case has $0 < a_1 < 2$. If $0 < a_n < 2$, then $2 < 2 + a_n < 4$, hence $\sqrt{2} < a_{n+1} < 2$; induction gives $0 < a_n < 2$ for all n.
2. Both a_n and a_{n+1} are positive, so their order may be compared by squaring.
3. Compute $a_{n+1}^2 - a_n^2 = 2 + a_n - a_n^2 = (2 - a_n)(a_n + 1) > 0$.
4. Thus $a_{n+1} > a_n$. The sequence is increasing and, from the first step, bounded above by 2.
5. The monotone convergence theorem gives convergence. Let the limit be L, with $0 < L \leq 2$.
6. Passing to the limit gives $L = \sqrt{2 + L}$, hence $L^2 - L - 2 = 0$ and $L = 2$ or $L = -1$.
7. Since $L > 0$, reject -1 and obtain $L = 2$.

Common pitfalls

- Solving $L = \sqrt{2+L}$ without first proving convergence
- Squaring without confirming nonnegativity
- Keeping both algebraic roots
- Proving only the upper bound in the induction

Verification

For $a_1 = 1$, the terms $1, \sqrt{3}, \sqrt{2+\sqrt{3}}$ remain in $(0, 2)$ and increase; the limit 2 satisfies the recurrence equation.

Method takeaway

For nonlinear recurrences: invariant interval, monotonicity, boundedness, existence, limit equation, and root filtering.

Extension

If $a_1 > 2$, one can instead prove the sequence decreases and is bounded below by 2, again converging to 2.

Source lineage: Original synthesis / diagnosis · monotone and bounded / nonlinear recurrence · reference IDs: openstax-v1-2.3-limit-laws, openstax-v2-5.1-sequences. See SOURCES.md for scope and adaptation boundaries; this note does not claim verbatim reproduction.

Challenge | Parameter / synthesis / application | Challenge | Original synthesis / diagnosis

Q099 · Safe and Unsafe Zones for Equivalent Substitution

Problem

Complete the following without derivatives, L'Hôpital's rule, or Taylor expansions. (1) Evaluate $A = \lim_{x\to 0} \frac{\sin(2x)}{e^x - 1}$. (2) A student uses $\sqrt{1+x} - 1 \sim \frac{x}{2}$ to replace $\sqrt{1+x} - 1$ inside a difference and claims $B = \lim_{x\to 0} \frac{\sqrt{1+x}-1-\frac{x}{2}}{x^2} = 0$. Judge the method and conclusion, then find B by exact algebra. (3) State one rule describing when direct equivalent-infinitesimal substitution is valid.

Answer

$A = 2$. The student's method is invalid and the conclusion is false; $B = -\frac{1}{8}$. Direct substitution is safe for multiplicative and divisive factors, but not mechanically inside sums or differences where principal terms may cancel.

Knowledge points

- Equivalent substitution in products and quotients
- Cancellation in differences
- Exact radical rationalization
- Method-validity audit

Reading and method choice

Part (1) has a safe quotient structure. Part (2) magnifies the second-order error discarded by a first-order equivalence, so the exact remainder must be retained.

Detailed derivation

1. For A, $\sin(2x) \sim 2x$ and $e^x - 1 \sim x$. These are quotient factors, so $A = \lim \frac{2x}{x} = 2$.
2. For B, replacing one term inside a difference makes equal first-order terms cancel; the equivalence contains no information about the remaining second-order term, so the method is invalid.
3. Set $r = \sqrt{1+x}$. Then $x = r^2 - 1 = (r-1)(r+1)$, and $r \to 1$.
4. The numerator is $r - 1 - \frac{r^2-1}{2} = (r-1)[1 - \frac{r+1}{2}] = -\frac{(r-1)^2}{2}$.
5. Since $x^2 = (r-1)^2(r+1)^2$, cancellation for $x \neq 0$ gives $-\frac{1}{2(r+1)^2}$.
6. Letting $r \to 1$ yields $B = -\frac{1}{2\cdot(2)^2} = -\frac{1}{8}$, so the numerical claim is also false.
7. Rule: replace equivalent infinitesimals factorwise in products and quotients; in sums or differences with possible cancellation, use exact algebra or higher-order remainder information.

Common pitfalls

- Treating equivalence as identity
- Ignoring cancellation of principal terms
- Using Taylor expansions outside the allowed chapter methods
- Losing a square during rationalization

Verification

A factors as $[\frac{\sin 2x}{2x}] \cdot [\frac{x}{e^x-1}] \cdot 2 \to 2$. For B, the exact expression $-\frac{1}{2(\sqrt{1+x}+1)^2}$ tends to $-\frac{1}{8}$.

Method takeaway

Equivalence preserves first-order relative information; once subtraction cancels that first-order term, exact algebra or higher-order information is required.

Extension

The same danger appears in $\infty - \infty$ forms: subtracting separate leading equivalents may erase the information that determines the limit.

Source lineage: Original synthesis / diagnosis · equivalent-infinitesimal synthesis / difference prohibition · reference IDs: openstax-v1-2.3-limit-laws. See SOURCES.md for scope and adaptation boundaries; this note does not claim verbatim reproduction.

Challenge | Parameter / synthesis / application | Challenge | Original synthesis / diagnosis

Q100 · Optional: The Uniform-Continuity Boundary for Oscillatory Functions

Problem

On $(0,1]$, define $F(x) = x\sin(\frac{1}{x})$ and $G(x) = \sin(\frac{1}{x})$. Determine whether each is uniformly continuous. For F, use a continuous extension and the uniform-continuity theorem on a closed interval. For G, construct two sequences whose distance tends to 0 while their function values differ by exactly 2.

Answer

F is uniformly continuous on $(0,1]$ because defining $F(0) = 0$ gives a continuous function on $[0,1]$. G is not uniformly continuous; take $x_n = \frac{1}{\frac{\pi}{2}+2\pi n}$ and $y_n = \frac{1}{\frac{3\pi}{2}+2\pi n}$.

Knowledge points

- continuous extension
- uniform continuity of a continuous function on a closed interval
- squeeze theorem
- counterexample sequences for uniform continuity
- rapid oscillation

Reading and method choice

Both functions oscillate increasingly rapidly, but the amplitude x in F tends to zero and permits a continuous extension at 0. The closed-interval theorem then gives uniform continuity. The amplitude of G does not decay, so increasingly close peaks and troughs remain two units apart in value.

Detailed derivation

1. For F, $|x\sin(\frac{1}{x})| \le x$. As $x \to 0^+$, the right-hand side tends to 0, so the squeeze theorem gives $\lim_{x\to 0^+} F(x) = 0$.
2. Define $\widetilde{F}(0) = 0$ and $\widetilde{F}(x) = F(x)$ on $(0,1]$. The original expression is continuous at positive points, and the previous limit proves right-continuity at 0.
3. Thus $\widetilde{F}$ is continuous on the closed interval $[0,1]$. By the theorem that a continuous function on a closed interval is uniformly continuous, $\widetilde{F}$ is uniformly continuous on $[0,1]$.
4. A restriction of a uniformly continuous function to any subset is uniformly continuous, so F is uniformly continuous on $(0,1]$.
5. For G, set $x_n = \frac{1}{\frac{\pi}{2}+2\pi n}$ and $y_n = \frac{1}{\frac{3\pi}{2}+2\pi n}$. For all sufficiently large n, both lie in $(0,1]$, and both tend to 0.

6. We have $G(x_n) = 1$ and $G(y_n) = -1$, so $|G(x_n) - G(y_n)| = 2$ for every such n.

7. Meanwhile, $|x_n - y_n| = \frac{\pi}{(\frac{\pi}{2}+2\pi n)(\frac{3\pi}{2}+2\pi n)} \to 0$.

8. If G were uniformly continuous, choosing $\varepsilon = 1$ would give a $\delta > 0$ forcing every pair less than δ apart to have value difference below 1. Large n violate this with distance below δ but value difference 2.

9. Therefore G is not uniformly continuous. The distinction is whether the oscillation amplitude decays enough to allow a continuous endpoint extension.

Common pitfalls

- declaring both functions non-uniformly continuous merely because they oscillate rapidly
- proving only pointwise continuity of F without obtaining a uniform result
- choosing two sequences without checking both mutual distance and a fixed output separation

Verification

The extension of F satisfies every hypothesis of the closed-interval theorem. For G, the point-distance formula tends to 0 while the value difference stays exactly 2, directly contradicting uniform continuity with fixed $\varepsilon = 1$.

Method takeaway

Infinite oscillation alone need not destroy uniform continuity; decay of amplitude and continuous extendability at the missing endpoint are decisive.

Extension

For $x^p \sin(\frac{1}{x})$ on $(0, 1]$, every $p > 0$ allows a continuous extension to $[0, 1]$ and hence uniform continuity, while $p = 0$ gives the present function G.

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